

Filtering and Affordability along the Housing Ladder

Arthur Adam* Francisco M. Cocco†
Princeton University Princeton University

September 30, 2026

Abstract

Are housing policies progressive? We study this question in an equilibrium model where low-income households sort into old units at the bottom of the housing ladder. We show that price changes along the ladder are a sufficient statistic for welfare changes across the income distribution. Policy propagates to prices through the affordability effect, which moves demand along the ladder, and the tear-down effect, which shifts supply at the bottom. The progressivity of specific policies depends on their loadings on each effect. Policy can also cause construction peaks and troughs, inducing vintage effects which further shape progressivity along a transition. We take the model to the data, calibrating to a representative US city. Our findings suggest that construction subsidies are regressive, especially in the short run. Maintenance subsidies emerge as a progressive alternative.

*Princeton University, aa7032@princeton.edu.

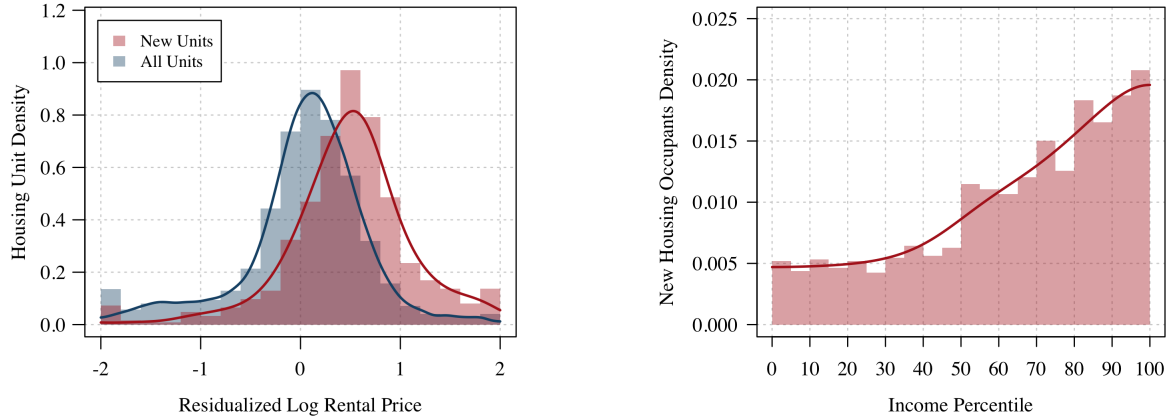
†Princeton University, fl5293@princeton.edu.

We are especially grateful to Eduardo Morales, Karthik Sastry, and Gianluca Violante for their guidance. We thank Mark Aguiar, John Sturm Becko, Markus Brunnermeier, Leonardo D’Amico, Morris Davis, John Grigsby, Veronica Guerrieri, Moritz Lenel, Stanimira Milcheva (discussant), Daniel O’Connor, Gary Painter, Richard Rogerson, Ayşegül Şahin, Evan Soltas, Motohiro Yogo, and seminar participants at Wharton, the 2026 Conference on Low-Income Housing Supply and Housing Affordability, the 2026 University of Chicago Macro Finance Summer Session for Young Scholars, the Princeton Finance Student Workshop, the Princeton IES Student Workshop, and the Princeton Student Macro Lunch.

1 Introduction

A central feature of housing markets is that new units command higher prices than old ones. As a result, low-income households predominantly live in old housing. Figure 1 illustrates this pattern in the United States (US). The left panel shows the log rental price distribution of all units (in blue) and new units (in red), residualized from metropolitan statistical area (MSA) $\times$ year fixed effects.¹ The median rental price of new units lies at the 82nd percentile of the rental price distribution. The right panel shows the income percentile distribution of new housing occupants. The share of households in new units is four times lower in the bottom 10% than in the top 10% of the income distribution. Understanding the distributional implications of recent policy initiatives, ranging from streamlining permitting (e.g. ROAD Act Title 2; California AB 130) to public housing expansion (e.g. NYC Block by Block), therefore requires understanding how they shape the equilibrium prices of old units through maintenance, tear-down, and construction incentives.

Figure 1: Prices and Sorting into Old vs. New Housing.



Note: Left panel: log rental price distributions of all units and new units (age < 2). Right panel: income percentile distribution of new housing occupants. Rental prices are residualized from MSA $\times$ year fixed effects. Source: 2001–2023 American Housing Survey.

In this paper, we study a model of the housing market where housing units go down the housing ladder as they age. Consistent with Figure 1, the model generates sorting of low-income households into old units. We derive a theory of the progressivity of housing policy. First, we show that in response to changes in policy, changes in welfare across the income distribution are captured by changes in prices along the housing ladder. Second, we show that the long-run impact on prices of any policy can be interpreted through the lens of

¹Figure A1 shows that the log housing value distributions of new owner-occupied units and existing owner-occupied units are similarly separated.

two effects, which we label *affordability* and *tear-down*. The progressivity of specific policy instruments can then be understood through their (potentially negative) loadings on these effects. Third, we study transition dynamics and show how *vintage* effects further shape prices and the progressivity of policy.

We take the model to the data. The calibrated model matches the empirical relationship between rental prices and housing age, and its variation across MSAs. Counterfactual policy experiments suggest that policies which act as a subsidy to construction are regressive, and even more so in the short run. Public housing construction is progressive, but ineffective at lowering prices of market-rate housing. Housing maintenance subsidies are both progressive and effective.

While it is well established that low-income households have been hardest hit by rising housing costs in the US, little is known about the distributional implications of policy initiatives attempting to improve housing affordability. Our paper provides a framework to think through the forces that shape the progressivity of housing policy and suggests that among the wide array of envisioned policies, often overlooked maintenance subsidy programs (e.g. Whole-Home Repairs Act) may deliver the best outcomes for low-income households.

Housing Equilibrium Model with Filtering. In our model, housing units are differentiated along a single quality dimension. The model features *filtering*: units lose quality as they age (Sweeney 1974). This generates a housing ladder with new units at the top and old units at the bottom. Over time, any given unit sees its price and the income of its occupants decline, consistent with the findings of Rosenthal (2014).

Households with heterogeneous income make up the demand side of the housing market. For tractability, we model all households as renters. Households sort into housing units based on the rental price schedule. On the supply side, landlords make maintenance, tear-down, and construction decisions. Absent a tear-down tax, landlords choose to maintain housing units whose rental price net of maintenance costs exceeds an opportunity cost captured by land value. As housing units age, their rental price declines and their maintenance costs rise. Landlords eventually cease to maintain them and tear them down. Landlords construct new units when present discounted future rent net of maintenance and opportunity costs outweigh building costs.

A wide array of policies shape the housing market. First, the government sets a minimal quality occupancy standard. Second, the government can build public housing and implement rent control for low-quality housing. Third, the government sets taxes and subsidies on maintenance, tear-downs, and construction. Last, the government controls a land use regulation wedge. Equilibrium is then characterized by four objects: an assignment function

mapping housing quality to the income of its occupants, a slope for the rental price schedule derived from household optimality conditions, a maintenance policy derived from landlord optimality conditions, and a housing construction free-entry condition. We establish steady-state equilibrium existence and uniqueness.

The Progressivity of Housing Policy. Our model is tractable and thus well suited for theoretical analysis. We first establish a sufficient statistic result: in response to a policy perturbation, the consumption-equivalent welfare change of any household occupying market-rate housing is given by the rental price change at their initial housing quality. The result follows from an envelope condition and implies that progressivity in welfare changes across the income distribution is equivalent to progressivity in rental price changes along the housing ladder. We move on to study the latter object.

We consider the comparative statics induced by a policy perturbation. Its impact on prices along the housing ladder is governed by two propagation channels: the affordability effect and the tear-down effect. Specific policy instruments load on each effect differently. The affordability effect captures how an increase in the price of new units propagates down the ladder, holding maintenance, tear-downs, and construction fixed. We show that the affordability effect is *amplified* going down the ladder. The reason is that higher (lower) prices at the top move demand down (up) the ladder. Hence, a policy that induces a positive affordability effect (a rise in the price of new units) produces a regressive force, while one that induces a negative affordability effect (a decline in the price of new units) produces a progressive force. The tear-down effect captures how old unit tear-downs propagate to prices. Because old units are at the bottom of the ladder, the tear-down effect is *dampened* going up. A policy that induces a positive tear-down effect produces a regressive force, while one that induces a negative tear-down effect produces a progressive force.

We consider the transition dynamics induced by a permanent policy perturbation. Its impact on prices is governed by the affordability and tear-down effects as well as transitional vintage effects, which capture policy-induced construction peaks and troughs. Construction peaks shift supply up the housing ladder, which raises prices at the bottom and produces a regressive force. Construction troughs have the opposite effect.

What do our theoretical results reveal about specific housing policy instruments? Throughout the paper, we focus our discussion on three instruments: construction subsidies, public housing construction, and maintenance subsidies. We note that construction subsidies in the model can be interpreted both as direct subsidies (e.g. tax credits) and as indirect subsidies reflecting administrative costs (e.g. ROAD Act Title 2; California AB 130).

Our comparative statics shed light on the long run. Construction subsidies cause a

progressive affordability effect: the decrease in prices of new units induced by construction subsidies is amplified going down the housing ladder. However, construction subsidies also push landlords to tear down old units in favor of building new, causing a regressive tear-down effect. Public housing construction frees up housing at the bottom of the ladder, which behaves as a progressive tear-down effect. Public housing construction also raises private building costs through congestion, leading to an increase in the price of new units and hence a regressive affordability effect. Maintenance subsidies incentivize landlords to further maintain old housing units, causing a progressive tear-down effect. The long-run affordability effect from maintenance subsidies is ambiguous.

Our transition dynamics shed light on the short run. We compare construction and maintenance subsidies. Construction subsidies load on the affordability and tear-down effects with the same sign in the short run as in the long run. In addition, construction subsidies cause a (regressive) construction peak. Maintenance subsidies, on the other hand, are unambiguously progressive in the short run. Indeed, they cause progressive affordability and tear-down effects as well as a construction trough. The short-run effects of public housing construction are ambiguous and depend on the timing of private construction crowding out induced by public housing.

Taking the Model to the Data. Towards taking the model to the data, we enrich the technology governing maintenance and construction. We then calibrate to a representative US city. To validate the calibrated model, we compare the relationship between rental prices and housing age in the model and in the data. The relationship is captured by the *rent-age* gradient, which reflects the strength of filtering. The calibrated model matches the empirical rent-age gradient, as well as its variation across MSAs.

We use the calibrated model to conduct quantitative counterfactual policy experiments. We compare three equally costly policy changes corresponding to a construction subsidy, public housing construction, and a maintenance subsidy. Discounting the lifetime consumption-equivalent welfare changes induced by the policies, we find the construction subsidy to be regressive, and the maintenance subsidy to be progressive. 90th-percentile households prefer the construction subsidy, from which they gain 1.6 consumption-equivalent units per unit spent by the government, to the maintenance subsidy, from which they gain only 1.1 units. 10th-percentile households prefer the maintenance policy, from which they gain 1.6 consumption-equivalent units per unit spent by the government, to the construction subsidy, from which they gain only 0.9 units. Public housing construction is progressive, but lowers welfare for almost all households living in market-rate housing.

We interpret our quantitative results through the lens of our theory of the progressiv-

ity of housing policy. First, consistent with our sufficient statistic result, welfare changes across the income distribution are captured by rental price changes across the housing ladder. Our quantitative results allow us to go one step further and to decompose welfare gains between changes in rent paid and changes in housing quality. Welfare gains from the construction subsidy are split between both, while welfare gains from the maintenance subsidy come from large decreases in rent paid. Second, the progressivity of each policy can be understood through the affordability, tear-down, and vintage effects they induce. The construction subsidy is regressive because its progressive affordability effect is overpowered by its regressive tear-down effect. It is particularly regressive in the short run, as it causes a major construction peak. Public housing construction is progressive because its progressive tear-down effect overpowers its regressive affordability effect. The maintenance subsidy is progressive because it produces progressive affordability and tear-down effects, in addition to a construction trough.

Related Literature. At the core of our paper is assignment of households to housing units of varying quality as in Määttänen and Terviö (2014), Landvoigt et al. (2015), Abramson and Landvoigt (2025), and Nikolakoudis (2025). Contrary to these papers, we fully endogenize housing supply through maintenance, tear-downs, and construction. This allows us to study the equilibrium impact of housing policy changes.

To endogenize supply of housing of varying quality, we build upon early theoretical work on housing filtering. Sweeney (1974) departs from canonical housing *quantity* depreciation by studying a model where housing units transition from higher to lower qualities over time. Ohls (1975) performs numerical simulations using this model. Braid (1984) introduces filtering along a continuum of housing qualities and Arnott and Braid (1997) consider a housing maintenance technology which can interrupt filtering. We expand this literature in three ways. First, our tractable model delivers a general theory of the progressivity of housing policy which we apply to specific policy instruments. Second, while previous analyses were limited to either the very short run (with housing held fixed) or the long run (where housing has reached a steady state), our model lends itself to the study of transition dynamics. Because the housing stock is slow to adjust, transition dynamics are essential to housing policy analysis. Third, we take our model to the data. It matches the empirical relationship between rental prices and housing age – a key outcome of filtering – lending credibility to the theory. We use the calibrated model to quantify policy counterfactuals.

Our paper also connects to empirical work studying how a given housing unit is occupied by lower-income households over time. This phenomenon is often also referred to as filtering. For clarity, we label it *income filtering* and reserve the word filtering for the physical process

through which a given housing unit loses quality over time. Rosenthal (2014) documents income filtering to be a robust feature of housing markets, with average yearly income filtering rates of 2.5% for rental housing. Liu et al. (2022) document substantial heterogeneity in income filtering rates across space and show that they are lower in housing markets with constrained supply. Spader (2025) documents a recent decline in income filtering, coinciding with declining housing affordability. We interpret these findings through the lens of our model when we take it to the data.

More broadly, our paper is connected to a literature studying the effect of housing supply on housing prices along the ladder. A first strand of this literature treats housing supply as given and studies exogenous increases in new construction. Exploiting weather shocks, Mense (2025) provides empirical evidence of price declines along the ladder. The result is attributed to moving chains: households moving into added units free up existing units, whose prices then fall. Nathanson (2025) studies moving chains in a model calibrated to Los Angeles and highlights the limiting role of external migration. Hauck and Kluser (2025), Bratu et al. (2023), and Mast (2023) provide direct evidence of moving chains. Our paper fits into a second strand of this literature which seeks to endogenize housing supply to study the distributional implications of policy changes. Diamond et al. (2025) and Ma (2025) study land use reforms. Favilukis et al. (2023) study an array of policies in a dynamic, stochastic, and spatial environment. Most related to our work, Hong and Macek (2025) study a tear-down tax in Chicago using a quantitative model which features filtering. Our general theory based on filtering distinguishes our paper. Because our model is tractable, the theory is transparent with respect to the forces that shape progressivity. Our theoretical and quantitative results reveal that maintenance subsidies, a policy instrument absent from the literature, may deliver the best outcomes for low-income households.

Outline. The rest of the paper is organized as follows. In Section 2, we introduce our model of a housing market with filtering. In Section 3, we develop our theory of the progressivity of housing policy. In Section 4, we extend and calibrate our model to take it to the data. We conclude in Section 5.

2 A Housing Equilibrium Model with Filtering

This section introduces our model. Time is continuous and indexed by $t \geq 0$. Households are endowed with heterogeneous income and rent housing units. Landlords choose whether to build new housing units and whether to maintain existing ones. Maintained units gradually filter down the housing ladder. In Section 3, we use the model to study the progressivity of

housing policy. We study a quantitative version of the model in Section 4.

2.1 Environment

Households. A unit mass of households inhabits the economy. Households are heterogeneous in their income endowment w_t , distributed according to the CDF $F(\cdot)$ with density $f(\cdot)$ and support $[\underline{w}, \bar{w}]$. Households spend their income on consumption of a numéraire good and rental of a housing unit. We make no assumptions on the income process faced by each household and abstract from household saving and homeownership.

Housing units are differentiated along a single quality dimension which we denote by $q \in [\underline{q}, 1]$, where $\underline{q} > 0$ is a minimal quality occupancy standard. All households must rent a housing unit and household flow utility over consumption c and housing quality q is given by

$$u(c, q) = cq.$$

We note that equal Cobb-Douglas exponents on consumption and housing quality are without loss of generality up to a re-scaling of quality.² Scaling quality simply scales utility, leaving the model's results unchanged.

The economy features market-rate (MR) and non-market-rate (non-MR) housing units. Non-MR units are allocated by the government to households with income $w_t < w_t^{MR}$, where w_t^{MR} is the MR housing cutoff. Households with $w_t \geq w_t^{MR}$ choose consumption and housing quality to solve

$$\max_{c \geq 0, q \in \mathcal{Q}_t^{MR}} u(c, q) \quad \text{subject to} \quad c + p_t(q) = w_t, \quad (1)$$

where $p_t(\cdot)$ is the rental price schedule and $\mathcal{Q}_t^{MR}$ is the MR housing choice set.

So far, we have implicitly imposed that all households agree on orderings of housing units by assuming the existence of a quality index, but have remained agnostic as to what constitutes housing quality.³ We now introduce the housing technology, which pins down the quality of a housing unit based on its quality at construction, its age, and its maintenance history. Our model abstracts from housing size and location and therefore from households' ability to substitute across housing units based on those characteristics.

²To see this, consider the more general formulation $\tilde{u}(c, q) = c^\alpha q^\beta$, yielding the marginal rate of substitution of housing quality for consumption of $(\beta/\alpha)(c/q) = c/\tilde{q}$ for $\tilde{q} = (\beta/\alpha)q$.

³While limiting, the assumption of a quality index is crucial for tractability in housing assignment models (Määtänen and Terviö 2014).

Housing Technology. New housing units are built at the top of the housing ladder, with quality $q = 1$. This is in line with Figure 1, where the median rental price of new units lies at the 82nd percentile of the rental price distribution. In Section 4, we extend the model to allow for construction at many qualities. Given aggregate construction C_t^A , building is priced competitively at the marginal cost $\kappa(C_t^A) \geq 0$. We assume $\kappa'(C_t^A) > 0$ to capture congestion in construction.

Filtering is endogenous in our model. Given an amount of maintenance spending m , a housing unit of quality q_t filters down the housing ladder at a rate

$$\dot{q}_t = -\mathcal{D}(m, q_t).$$

We specify $\mathcal{D}(m, q)$ to capture two properties. First, filtering is decreasing in maintenance spending: better maintained housing units filter down the housing ladder at a slower rate. Second, the maintenance spending required for a given filtering rate is decreasing in q : as housing units go down the housing ladder, more maintenance is required to slow down filtering. Specifically, we impose

$$\mathcal{D}(m, q) = \begin{cases} \infty & \text{if } m < \mathcal{M}(q) \\ \delta & \text{if } m \geq \mathcal{M}(q) \end{cases}, \quad \mathcal{M}(q) = \gamma \frac{1-q}{q}, \quad (2)$$

where $\mathcal{M}(q)$ captures maintenance costs, with $\gamma > 0$. Our choice for $\mathcal{D}(m, q)$ entails that improperly maintained housing units lose their quality instantly, while maintained units progressively lose their quality at the rate $\delta > 0$. As becomes clear upon introducing the landlord problem, this generates an extensive margin maintenance decision, with landlords deciding whether or not to maintain units of different qualities, but abstracts from any intensive margin maintenance decision allowing landlords to further slow down or interrupt filtering. We extend the model in this direction in Section 4. Our choice for $\mathcal{M}(q)$ entails that new units at the top of the housing ladder require no maintenance and that the cost of maintaining a unit with quality approaching 0 becomes prohibitive.

Landlords. Landlords are endowed with a mass $L > 1$ of land. We normalize the size of a land unit so it fits one housing unit. Land can be used for housing or for a commercial purpose, reflected by the exogenous outside option value P_0 .

We let $P_t(q)$ denote the value of a land unit with quality q housing and $\underline{P}_t$ denote the value of a vacant land unit. Landlords must tear down housing units that reach the minimal quality occupancy standard $q = \underline{q}$, subject to a tear-down tax $\tau^{TD} \geq 0$. The maintenance

policy function $m_t : [\underline{q}, 1] \rightarrow \mathbb{R}^+$ solves the Hamilton–Jacobi–Bellman (HJB) equation for housing value

$$\begin{aligned} rP_t(q) = \max_{m \geq 0} & \mathbf{1}_{m < \mathcal{M}(q)} \left(r(\underline{P}_t - \tau^{TD}) - (1 - \tau^m)m \right) \\ & + \mathbf{1}_{m \geq \mathcal{M}(q)} \left(p_t(q) - (1 - \tau^m)m + \dot{P}_t(q) - \delta P'_t(q) \right), \end{aligned} \quad (3)$$

where $r > 0$ denotes the interest rate at which landlords discount future profits and $\tau^m \in (-1, 1)$ is a subsidy (or a tax if $\tau^m < 0$) on maintenance. The HJB reflects the filtering process $\mathcal{D}(m, q)$. The first term captures the value of improperly maintained units. They lose their quality instantly and are torn down, yielding the value of vacant land. The second term captures the value of maintained units. Maintained units provide landlords with flow profits $p_t(q) - (1 - \tau^m)m$ and capital gains $\dot{P}_t(q)$ and filter down the housing ladder at the rate δ .

We assume landlord free entry into building. Because the economy has sufficient land to house the unit mass of households, free entry entails

$$\underline{P}_t = (1 + \tau^{P_0})P_0 \geq P_t(1) - (1 - \tau^C)\kappa(C_t^A) \quad (4)$$

where τ^{P_0} introduces a wedge between the value of using land for housing and for a commercial purpose, capturing land use regulation, and where $\tau^C \in (-1, 1)$ is a subsidy (or a tax if $\tau^C < 0$) on private construction. (4) must hold with equality whenever there is private construction, defined as $C_t \equiv C_t^A - C^{PH}$.

Housing Policy. We now detail the instruments of housing policy. First, the government sets the minimal quality occupancy standard $\underline{q}$. Second, the government provides two types of non-MR housing: public and rent-controlled. Public housing construction C^{PH} directly provides non-MR units. We assume that the government maintains public housing units following the maintenance policy of landlords. Rent control indirectly provides private non-MR housing. Specifically, the government imposes a minimal quality q^{RC} such that landlords must rent units with $q < q^{RC}$ at the price $p_t(q) = \min\{p_t(q^{RC}), p^{RC}\}$. All non-MR housing is assigned to the lowest-income households in the economy. For now, we take no further stance on non-MR housing. We have abstracted from households rejecting non-MR units but note that this would never be optimal for sufficiently low public housing rental prices and p^{RC} .

The remaining instruments of housing policy are the taxes and subsidies on housing tear-downs (τ^{TD}), maintenance (τ^m), land use (τ^{P_0}), and private construction (τ^C). We define

housing policy as

$$\mathcal{I} \equiv \{\underline{q}, C^{PH}, q^{RC}, p^{RC}, \tau^{TD}, \tau^m, \tau^C, \tau^{P_0}\}.$$

Because our aim is to understand the progressivity of housing policy and not to study its normative optimality (or lack thereof), we do not specify a government budget constraint. As a result, our theoretical analysis in Section 3 abstracts from the financing of government policy and its own implications for progressivity. In Section 4, we ensure fair quantitative comparisons by considering equally costly policy changes in our counterfactual experiments.

2.2 Equilibrium

Housing Market Clearing. The private housing mass function, which we denote by $h_t(\cdot)$, evolves according to the Kolmogorov Forward Equation (KFE)

$$\dot{h}_t(q) = \delta \frac{\partial h_t(q)}{\partial q} \quad (5)$$

wherever $m_t(q) \geq \mathcal{M}(q)$, together with the boundary conditions

$$h_t(q) = 0 \quad \text{for} \quad m_t(q) < \mathcal{M}(q), \quad h_t(1) = \frac{C_t}{\delta}.$$

Consider an allocation of consumption and housing $\{(c_{wt}, q_{wt})\}_w$. Housing market clearing for MR units requires that for any quality $q \in \mathcal{Q}_t^{MR}$,

$$\int_{w_t^{MR}}^{\bar{w}} \mathbf{1}_{q_{wt} \geq q} dF(w) = \int_q^1 h_t(x) dx. \quad (6)$$

Market clearing for non-MR units requires

$$F(w_t^{MR}) = \int_{\underline{q}}^1 h_t^{PH}(q) dq + \int_{\underline{q}}^{q^{RC}} h_t(q) dq, \quad (7)$$

where $h_t^{PH}(\cdot)$ is the public housing mass function, defined identically to $h_t(q)$ but with the boundary condition $h_t^{PH}(1) = C^{PH}/\delta$. The first term of the RHS of (7) is the mass of non-MR housing directly supplied by the government through public housing while the second is the mass supplied privately by landlords in the presence of rent control.

Assumptions and Definition. We make three assumptions. First, we assume that public housing construction is insufficient to cover the housing needs of all households, given by the rate δ and the range of housing quality $1 - \underline{q}$. This ensures the economy requires private units.

Assumption 1. *Public housing construction satisfies $C^{PH} < \frac{\delta}{1-\underline{q}}$.*

Second, we assume that the lowest-income household earns sufficient income to cover maintenance costs and the flow opportunity cost of using land for housing, net of delaying tear-down tax payment. This ensures all households can afford housing.⁴

Assumption 2. *The lowest-income household satisfies*

$$\underline{w} > (1 - \tau^m)\mathcal{M}(\underline{q}) + r(1 + \tau^{P_0})P_0 - r\tau^{TD}.$$

Third, we assume income in excess of $\underline{w}$ is sufficient to cover building and tear-down costs.

Assumption 3. *The CDF $F(w)$ satisfies*

$$\int_0^{\frac{1-\underline{q}^{RC}}{\delta}} e^{-rt} \int_{q^{RC}}^{1-\delta t} \left(F^{-1} \left(\frac{x - q^{RC}}{1 - q^{RC}} \right) - \underline{w} \right) dx dt > (1 - \tau^C)\kappa \left(\frac{\delta}{1 - q^{RC}} \right) + \tau^{TD}. \quad (8)$$

To make sense of Assumption 3, consider the scenario where landlords maintain housing units from when they are built until they reach the rent-control threshold q^{RC} . Suppose landlords rent the lowest-quality units to the lowest-income households for their entire income $\underline{w}$, and rent higher-quality units to higher-income households. The LHS of (8) represents income in excess of $\underline{w}$ landlords can extract over the lifetime of a new unit. Building costs in the RHS are evaluated at $\delta/(1 - q^{RC})$, the required aggregate construction in this scenario. Together, Assumptions 2 and 3 ensure sufficient income in the economy for landlords to break even on housing development. We now define an equilibrium of the economy.⁵

Definition 1 (Equilibrium). *Given housing policy $\mathcal{I}$ and initial housing mass functions $h_0(q)$ and $h_0^{PH}(q)$, an equilibrium is characterized by allocations $\{(c_{wt}, q_{wt})\}_w\}_{t \geq 0}$, maintenance policies $\{m_t(q)\}_{t \geq 0}$, private and aggregate construction $\{C_t\}_{t \geq 0}$ and $\{C_t^A\}_{t \geq 0}$, MR housing cutoffs $\{w_t^{MR}\}_{t \geq 0}$, MR housing choice sets $\{\mathcal{Q}_t^{MR}\}_{t \geq 0}$, housing mass functions $\{h_t(q)\}_{t > 0}$ and $\{h_t^{PH}(q)\}_{t > 0}$, rental price schedules $\{p_t(q)\}_{t \geq 0}$, and housing value functions $\{P_t(q)\}_{t \geq 0}$ such that:*

⁴In doing so, we abstract away from homelessness, which we view as beyond the scope of our analysis. O’Flaherty (1995) studies the determinants of homelessness in a model with housing filtering.

⁵We abstract from numéraire good market clearing, which holds by Walras’ law.

1. Given $\mathcal{Q}_t^{MR}$ and $p_t(q)$, households with income $w_t \geq w_t^{MR}$ choose (c_{wt}, q_{wt}) to maximize utility for all $t \geq 0$.
2. Given $p_t(q)$, landlords choose $m_t(q)$ according to the HJB for $P_t(q)$.
3. Free entry into building holds for all $t \geq 0$, with equality when $C_t = C_t^A - C^{PH} > 0$.
4. $h_t(q)$ and $h_t^{PH}(q)$ evolve according to the KFE and boundary conditions for private and public housing.
5. MR housing choice sets are $\mathcal{Q}_t^{MR} = \{q \geq q^{RC} : \int_{q-\epsilon}^{q+\epsilon} h_t(x)dx > 0 \text{ for all } \epsilon > 0\}$.
6. Housing market clearing holds consistently with w_t^{MR} for all $t \geq 0$.

We now define a regular equilibrium. We restrict our attention to continuous rental price schedules.⁶ We also restrict our attention to rental prices and hence to maintenance policies and housing values for qualities with non-zero housing, with the understanding that any equilibrium can be supported by a range of prices for qualities with zero housing. Last, we restrict our attention to allocations of consumption and housing quality for households in MR housing, as we have not taken a stance on the government's allocation of non-MR housing. Throughout the remainder of the paper, we study regular equilibria.

Definition 2 (Regular Equilibrium). *Given housing policy $\mathcal{I}$, a regular equilibrium consists of the objects of an equilibrium, with $p_t(q)$ continuous for $q \in \mathcal{Q}_t^{MR}$, $m_t(q)$, $p_t(q)$, and $P_t(q)$ defined only over housing qualities $\{q : \int_{q-\epsilon}^{q+\epsilon} h_t(x)dx > 0 \text{ for all } \epsilon > 0\}$, and $\{(c_{wt}, q_{wt})\}_w$ defined only over $w \geq w_t^{MR}$.*

Characterization. First, we consider the assignment of MR housing units to households. We define the assignment function $\tilde{w}_t : \mathcal{Q}_t^{MR} \rightarrow [\underline{w}, \bar{w}]$ mapping MR housing quality to the income of its occupants:

$$\tilde{w}(q_{wt}) = w.$$

Starting from any initial quality, a marginal quality increase is worth less in terms of consumption to low-income households. This results in positive assortative matching, corresponding to an increasing $\tilde{w}_t(\cdot)$ that assigns higher-quality units to higher-income households. Lemma 1 follows.

⁶We do so because starting from any equilibrium featuring a continuous rental price schedule, the price schedule can be raised discontinuously at a finite number of qualities without affecting equilibrium conditions. This is the case because market clearing is imposed as an integral condition, which cannot be affected by measure 0 re-assignments of households.

Lemma 1. *There is positive assortative matching of MR units with*

$$\tilde{w}_t(q) = F^{-1} \left(\int_{\underline{q}}^1 h_t^{PH}(x) dx + \int_{\underline{q}}^q h_t(x) dx \right)$$

for $q \in \mathcal{Q}_t^{MR}$.

Proof. See Appendix B.1. □

Second, we characterize the slope of the rental price schedule. Lemma 2 follows from the first-order conditions of the household problem (1).

Lemma 2. *The rental price schedule is increasing and differentiable with slope*

$$p'_t(q) = \frac{u_q}{u_c} = \frac{\tilde{w}_t(q) - p_t(q)}{q}$$

for $q \in \mathcal{Q}_t^{MR}$.

Proof. See Appendix B.2. □

In equilibrium, the slope of the rental price schedule for MR units is equal to the marginal rate of substitution of housing quality for consumption of its occupants. Third, we turn to the landlord problem. Landlords maintain housing units when

$$\Phi_t(q) \equiv \underbrace{p_t(q) - (1 - \tau^m)\mathcal{M}(q)}_{\text{Flow profits}} - \underbrace{r(1 + \tau^{P_0})P_0}_{\text{Opportunity cost}} + \underbrace{r\tau^{TD}}_{\text{Postponing tear-down}} \geq 0. \quad (9)$$

For those units, flow profits and the value of postponing payment of the tear-down tax outweigh the flow opportunity cost of using land for housing. Non-decreasing rental prices and decreasing maintenance costs ensure $\Phi_t(q)$ is increasing. This entails a tear-down cutoff $q_{min,t}$ where landlords cease to maintain and tear down housing. Lemma 3 characterizes the maintenance policy.

Lemma 3. *The maintenance policy satisfies*

$$m_t(q) = \begin{cases} \mathcal{M}(q) & \text{for } q \geq q_{min,t} \\ 0 & \text{for } q < q_{min,t} \end{cases}, \quad q_{min,t} = \inf\{q : \Phi_t(q) \geq 0\}.$$

Proof. See Appendix B.3. □

Let $T_t \equiv (1 - q_{min,t})/\delta$ denote the age of the oldest housing unit at time t . Combining the maintenance policy with the HJB (3) allows us to re-express the free-entry condition (4) as

$$\begin{aligned} P_t(1) &= \int_0^{\bar{s}_t} e^{-rs} (p_{t+s}(1 - \delta s) - (1 - \tau^m)\mathcal{M}(1 - \delta s)) ds + e^{-r\bar{s}_t} ((1 + \tau^{P_0})P_0 - \tau^{TD}) \\ &\leq (1 - \tau^C)\kappa(C_t^A) + (1 + \tau^{P_0})P_0, \end{aligned} \quad (10)$$

where $\bar{s}_t = \inf\{s : t + s = T_t\}$ is the age at which a housing unit built at time t will be torn down. When considering housing development, landlords weigh the present discounted value of flow profits and residual land value after payment of the tear-down tax against building costs and the land opportunity cost.

Together, the assignment function, the slope of the rental price schedule, the maintenance policy, and the free-entry condition characterize equilibrium. Theorem 1 establishes existence and uniqueness of the steady-state equilibrium.

Theorem 1. *There exists a unique steady-state regular equilibrium.*

Proof. See Appendix B.4. □

We distinguish steady-state objects from their dynamic counterparts by omitting t subscripts. Changing variables from s to q allows us to write $P(1)$ as

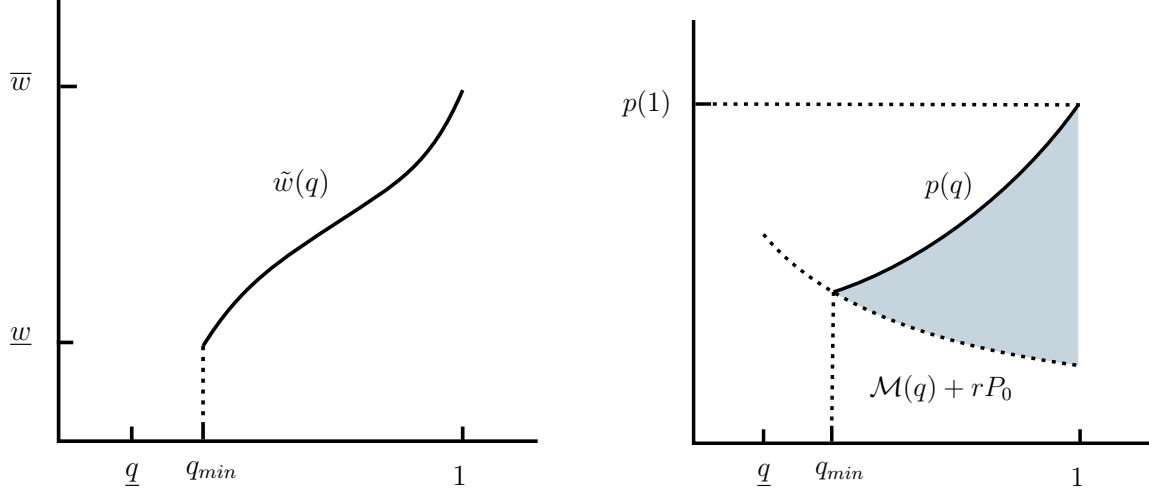
$$\int_{q_{min}}^1 e^{-r\frac{1-q}{\delta}} \frac{p(q) - (1 - \tau^m)\mathcal{M}(q) - r((1 + \tau^{P_0})P_0 - \tau^{TD})}{\delta} dq + (1 + \tau^{P_0})P_0 - \tau^{TD}. \quad (11)$$

The value of a new unit depends on flow profits over the entire housing ladder, net of flow opportunity cost minus value from postponing tear-down. Net flow profits are discounted based on the rate δ , which determines the speed at which a newly built unit will reach the ladder's many echelons. The filtering rate also appears as a denominator to net flow profits, with a slower rate leading to profits at any q being collected for longer.

We represent a steady-state equilibrium graphically in Figure 2. For simplicity, we consider a baseline housing policy without a binding minimal quality occupancy standard ($\underline{q} \rightarrow 0$), non-MR housing ($C^{PH} = 0$, $q^{RC} = \underline{q}$), and taxes ($\tau^{TD} = \tau^m = \tau^{P_0} = \tau^C = 0$). The left panel shows the assignment function $\tilde{w}(q)$. It is increasing, consistent with positive assortative matching. Households with the lowest income $\underline{w}$ occupy housing units of quality q_{min} : the oldest and lowest-quality units in the economy. The right panel shows the rental price schedule. The rental price is highest for new units of quality $q = 1$. Rental prices fall as units age and filter down the housing ladder. $p(q)$ intersects the sum of maintenance costs

$\mathcal{M}(q)$ and of the flow opportunity cost of using land for housing rP_0 at q_{min} . The shaded area represents landlord net flow profits, which once discounted as in (11), exactly offset building costs $\kappa(C)$.

Figure 2: Steady-State Equilibrium.



Note: Left panel: assignment function. Right panel: rental price schedule. The figure has $\underline{q} \rightarrow 0$, $C^{PH} = 0$, $q^{RC} = \underline{q}$, and $\tau^{TD} = \tau^m = \tau^{P_0} = \tau^C = 0$.

3 The Progressivity of Housing Policy

In this section, we derive a theory of the progressivity of housing policy. We first show that changes in rental prices along the housing ladder provide a sufficient statistic for welfare changes across the income distribution. Second, we study comparative statics, shedding light on the progressive and regressive effects of policy in the long run. Third, we consider transitional dynamics and highlight the distributional implications of policy-induced construction peaks and troughs.

3.1 A Sufficient Statistic for Welfare

Throughout this section, we consider a permanent perturbation di to a housing policy instrument $i \in \mathcal{I}$, starting from a steady-state equilibrium. We define the indirect utility function $v_t(w) \equiv u(c_{wt}, q_{wt})$. Differentiating $v_t(w)$ yields

$$\frac{dv_t(w)}{di} = - \underbrace{u_c \frac{dp_t(q_{wt})}{di}}_{\text{Rental price change}} + \underbrace{u_q \frac{dq_{wt}}{di}}_{\text{Housing quality change}} - \underbrace{u_c p'_t(q) \frac{dq_{wt}}{di}}_{\text{Cost of quality change}}. \quad (12)$$

The policy change di impacts welfare in three ways. First, it changes the rental price faced by households at their initial housing quality. Second, it changes the housing quality that they rent. Third, it changes the rent they pay as a result of the quality change. We let

$$\xi_t^i(w) = \frac{1}{u_c(c_{wt}, q_{wt})} \frac{dv_t(w)}{di}$$

denote the consumption-equivalent welfare change of a household with income w at time t . Theorem 2 establishes that for households in MR housing, the first term in (12) is a sufficient statistic for welfare changes.

Theorem 2. *To first order, rental price changes along the housing ladder provide a sufficient statistic for welfare changes across the income distribution:*

$$\xi_t^i(\tilde{w}_t(q)) = \frac{-dp_t(q)}{di}$$

for $q \in \mathcal{Q}_t^{MR}$.

Proof. The proof follows from an envelope condition. (12) rewrites as

$$\xi_t^i(\tilde{w}_t(q)) = \frac{-dp_t(q)}{di} + \left(\frac{u_q}{u_c} - \cancel{p'_t(q)} \right) \frac{dq_{\tilde{w}_t(q)t}}{di}.$$

The second term cancels out by the household optimality conditions in Lemma 2. □

Theorem 2 establishes that the change in rental price at a given quality characterizes the change in welfare of households occupying that quality. Combined with positive assortative matching, this implies an equivalence between progressivity in welfare changes across the income distribution and in rental price changes along the housing ladder. We state this result in Corollary 1.

Corollary 1. *Progressivity in welfare changes across the income distribution is equivalent to progressivity in rental price changes along the housing ladder:*

$$\frac{d\xi_t^i(w)}{dw} < 0 \text{ for all } w \in [w^{MR}, \bar{w}] \iff \frac{d}{dq} \left(\frac{dp_t(q)}{di} \right) > 0 \text{ for all } q \in \mathcal{Q}_t^{MR}.$$

Proof. Starting from Theorem 2, we derive

$$\frac{d\xi_t^i(\tilde{w}(q))}{dw} \tilde{w}'_t(q) = \frac{d}{dq} \left(\frac{-dp_t(q)}{di} \right).$$

We observe that $\tilde{w}'_t(q) > 0$ corresponds to positive assortative matching, established in Lemma 1. $\square$

Having established that the distributional implications of housing policy are captured by its propagation along the housing ladder, we move on to study how di shapes the rental price schedule $p_t(q)$.

3.2 Comparative Statics: Affordability and Tear-Down Effects

We first consider the comparative statics induced by the policy change di , starting from a steady-state equilibrium without a binding minimal quality occupancy standard ($\underline{q} \rightarrow 0$) and non-MR housing ($C^{PH} = 0$, $q^{RC} = \underline{q}$). Theorem 3 establishes two channels for the propagation of di to $p(q)$.

Theorem 3. *The rental price schedule admits a representation $p(q) = \tilde{p}(q; p(1), q_{min})$ such that for any policy instrument $i \in \mathcal{I}$*

$$\frac{dp(q)}{di} = \underbrace{\alpha^i \frac{\partial \tilde{p}(q)}{\partial p(1)}}_{\text{Affordability effect}} + \underbrace{\beta^i \frac{\partial \tilde{p}(q)}{\partial q_{min}}}_{\text{Tear-down effect}}$$

where α^i and β^i are policy dependent loadings.

Proof. See Appendix B.5. $\square$

The key insight from this result is that *any* policy $i \in \mathcal{I}$ propagates along the housing ladder through the same two channels, each corresponding to a partial equilibrium perturbation. The (potentially negative) loadings α^i and β^i on these channels are what distinguish specific policies. The first propagation channel is the affordability effect, which corresponds to a partial equilibrium increase in the price of new units $p(1)$, holding q_{min} and the underlying construction and maintenance policy fixed. The second propagation channel is the tear-down effect, which corresponds to a partial equilibrium increase in the tear-down cutoff q_{min} , holding $p(1)$ fixed.

The decomposition of $dp(q)/di$ in Theorem 3 is a step towards understanding the progressivity of housing policy because the affordability and tear-down effects can be characterized as follows in Proposition 1.

Proposition 1. *The affordability effect is regressive if $\alpha^i > 0$ and progressive if $\alpha^i < 0$:*

$$\frac{\partial \tilde{p}(q)}{\partial p(1)} \geq 0, \quad \frac{d}{dq} \left(\frac{\partial \tilde{p}(q)}{\partial p(1)} \right) < 0.$$

The tear-down effect is regressive if $\beta^i > 0$ and progressive if $\beta^i < 0$:

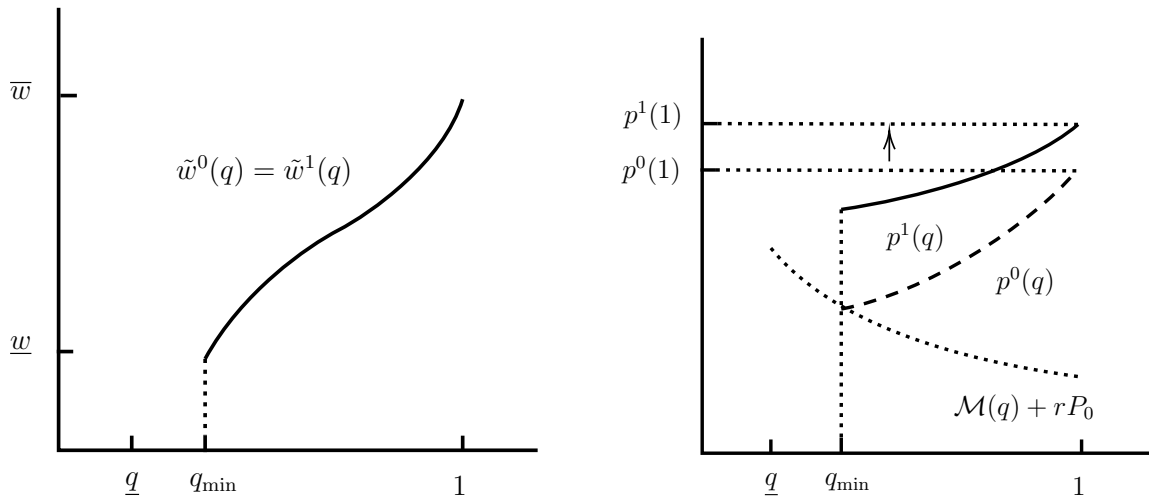
$$\frac{\partial \tilde{p}(q)}{\partial q_{\min}} \geq 0, \quad \frac{d}{dq} \left(\frac{\partial \tilde{p}(q)}{\partial q_{\min}} \right) < 0.$$

Proof. See Appendix B.6. □

To understand this result, we now discuss the propagation channels one by one. Suppose $\alpha^i > 0$. An increase in the price of new units $p(1)$ increases rental prices in every part of the housing ladder. This is because old (low-quality) units are substitutes for new (high-quality) units. By how much do rental prices of low-quality units increase? An intuitive view is as follows: because units are differentiated in quality along the ladder, the increase in $p(1)$ should be dampened going down. The regressivity of the affordability effect reveals that this logic fails. The reason is that unaffordable housing lowers consumption and hence households' willingness to pay for housing quality, pushing demand down the ladder. Holding construction and maintenance fixed, households compete for a now scarce supply of old units, causing their prices to increase more than $p(1)$.

We represent an affordability effect with $\alpha^i > 0$ in Figure 3. Raising $p(1)$ from $p^0(1)$ to $p^1(1)$ while holding $q_{\min}$ fixed induces no re-assignment of households to housing units, as shown in the left panel. Lower housing affordability lowers households' consumption and hence their marginal rate of substitution of housing quality for consumption. By Lemma 2, this flattens the rental price schedule. Thus, in the right panel, price increases are highest at the bottom of the housing ladder, consistent with a regressive affordability effect.

Figure 3: Affordability Effect.

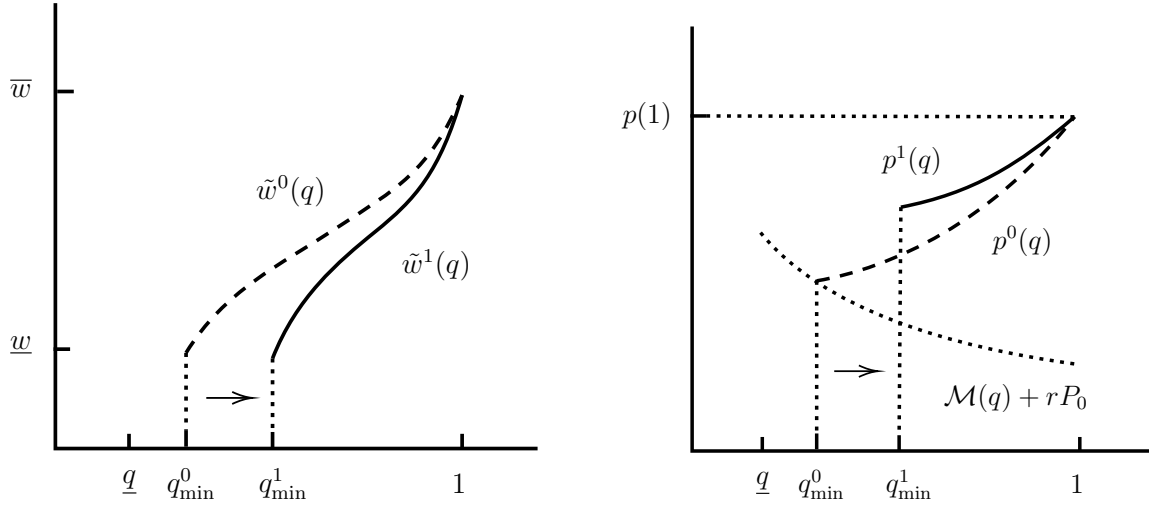


Note: Effect of a partial equilibrium increase in $p(1)$. Left panel: assignment function. Right panel: rental price schedule. The figure has $\underline{q} \rightarrow 0$, $C^{PH} = 0$, $q^{RC} = \underline{q}$, and $\tau^{TD} = \tau^m = \tau^{P_0} = \tau^C = 0$.

Now suppose $\beta^i > 0$. An increase in the tear-down cutoff q_{min} pushes households out of the lowest-quality units in the economy. As they are re-assigned to more expensive units, households' willingness to pay for housing quality falls. Hence, supply and demand move in opposite directions along the housing ladder, causing prices to rise at the bottom. The tear-down effect fades away toward the top of the ladder, where households occupy new units and are unaffected by the tear-down of old ones.

We represent a tear-down effect with $\beta^i > 0$ in Figure 4. Raising q_{min} from q_{min}^0 to q_{min}^1 pushes households at the bottom of the income distribution into higher-quality units. As old units are torn down, their scarcity rises and the rental-price schedule contracts and flattens, as shown in the right panel. The increase in prices is highest at the bottom of the housing ladder, consistent with a regressive tear-down effect.

Figure 4: Tear-Down Effect.



Note: Effect of a partial equilibrium increase from q_{min}^0 to q_{min}^1 . Left panel: assignment function. Right panel: rental price schedule. The figure has $\underline{q} \rightarrow 0$, $C^{PH} = 0$, $q^{RC} = \underline{q}$, and $\tau^{TD} = \tau^m = \tau^{P_0} = \tau^C = 0$.

Taken together, Theorem 3 and Proposition 1 imply that we can understand the progressivity of any policy instrument $i \in \mathcal{I}$ through its loadings α^i and β^i . Proposition 2 signs these loadings for policies of interest.

Proposition 2. *Construction subsidies τ^C cause a progressive affordability effect ($\alpha^{\tau^C} < 0$) but a regressive tear-down effect ($\beta^{\tau^C} > 0$). Public housing construction C^{PH} causes a progressive tear-down effect ($\beta^{C^{PH}} < 0$) but a regressive affordability effect ($\alpha^{C^{PH}} > 0$). Maintenance subsidies τ^m cause a progressive tear-down effect ($\beta^{\tau^m} < 0$) and an ambiguous affordability effect ($\alpha^{\tau^m} \geq 0$).*

Proof. See Appendix B.7. □

Construction subsidies – whether direct or indirect through reductions in administrative costs – and public housing construction are at the forefront of the housing affordability policy debate. Construction subsidies have been criticized as a solution to housing affordability challenges on the grounds that their effects on prices are unlikely to propagate to the old units which low-income households occupy. The logic of the affordability effect contradicts this claim: lower prices for new units push demand up the housing ladder, causing larger decreases in prices at the bottom. The progressivity of construction subsidies is however limited by the tear-down effect. Landlords respond to lower building costs by tearing down old housing units in favor of building new, causing a regressive increase in prices at the bottom of the housing ladder.

The proponents of public housing construction claim it is better suited to improve housing affordability for low-income households because it targets the bottom of the housing ladder, not the top. Indeed, when assigned to the lowest-income households, public housing effectively frees up supply of the lowest-quality MR units. As a result of this targeting, it is argued public housing will induce progressive rental price decreases for households who remain in MR units. This force is captured by $\beta^{C^{PH}} < 0$: the re-assignment of households into the low-quality units vacated by the households to whom public housing is assigned behaves identically to a lower q_{min} that decreases prices, and more so at the bottom of the ladder. However, public housing construction raises aggregate construction and hence building costs. The rise in costs is absorbed in part by crowding out of private construction, with $dC/dC^{PH} < 0$ and in part by an increase in $p(1)$. This causes a regressive affordability effect which raises prices the most for low-income households in MR housing.

Maintenance subsidies receive little attention in comparison to construction subsidies and public housing. They cause a progressive tear-down effect as they incentivize landlords to further maintain old housing units, lowering q_{min} . While lower maintenance costs push $p(1)$ down, they also lower private construction and hence the supply of new units, pushing $p(1)$ up. The resulting affordability effect is ambiguous.

3.3 Transition Dynamics and Vintage Effects

Conditional on convergence of the price changes induced by the permanent policy change di , our comparative statics have informed the long run. However, housing units are highly durable, with lifespans of many decades. Thus, the housing stock is slow to adjust to policy changes and the long run may be very far down the road from $t = 0$. Furthermore, as a result of filtering, the relative abundance of low- and high-quality housing depends on the history of construction in the economy, suggesting that dynamics may matter not only for

the scale of the price changes induced by di , but also for their progressivity.

To study transition dynamics, we define a stable equilibrium. Consider the aggregate construction supply elasticity $\varepsilon^{C^A} = \partial \log C^A / \partial \log P(1)$ and $|d \log P(1) / d \log C^A|$, the absolute value of the total elasticity of the value of a new unit with respect to aggregate construction. The product of both objects bounds the feedback to past and future construction of changes in construction at a given time. When this feedback is small, the transition induced by di can be characterized using a contraction mapping for $\{dC_t^A/di\}_{t \geq 0}$.

Definition 3 (Stable Equilibrium). *A stable equilibrium is a steady-state regular equilibrium satisfying*

$$\varepsilon^{C^A} \left| \frac{d \log P(1)}{d \log C^A} \right| < \frac{1}{2}.$$

To reflect the adjustment of the housing stock to di , we introduce the notion of housing vintages. We define the vintage s share of historical construction, $V_{s,t}$, as

$$V_{s,t} = \frac{C_{t-s}}{\hat{C}_t}, \quad \hat{C}_t = \frac{\int_0^{T_t} C_{t-s} ds}{T_t}. \quad (13)$$

We extend the comparative statics decomposition of rental price changes in Theorem 3 to the full transition in Theorem 4.

Theorem 4. *The rental price schedule admits a representation $p_t(q) = \tilde{p}^{TD}(q; p_t(1), q_{min,t}, \{V_{s,t}\}_{s \in [0, T_t]})$ such that for any policy instrument $i \in \mathcal{I}$*

$$\frac{dp_t(q)}{di} = \alpha_t^i \frac{\partial \tilde{p}^{TD}(q)}{\partial p(1)} + \beta_t^i \frac{\partial \tilde{p}^{TD}(q)}{\partial q_{min}} + \underbrace{\int_0^T \eta_{s,t}^i \frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} ds}_{\text{Vintage Effects}},$$

where α_t^i , β_t^i , and $\{\eta_{s,t}^i\}_{s \in [0, T]}$ are policy and time-dependent loadings with $\int_0^T \eta_{s,t}^i ds = 0$. In addition, $\partial \tilde{p}^{TD} / \partial p(1) = \partial \tilde{p} / \partial p(1)$, $\partial \tilde{p}^{TD} / \partial q_{min} = \partial \tilde{p} / \partial q_{min}$, and starting from a stable equilibrium,

$$\alpha_t^i \rightarrow \alpha^i, \quad \beta_t^i \rightarrow \beta^i, \quad \eta_{s,t}^i \rightarrow 0$$

as $t \rightarrow \infty$.

Proof. See Appendix B.8. □

Theorem 4 establishes that along the transition, di propagates through affordability,

tear-down, and vintage effects. The transitional affordability and tear-down effects behave identically to their comparative statics counterparts and thus inherit the properties detailed in Proposition 1. The policy loadings α_t^i and β_t^i are time-dependent and converge to their comparative statics counterparts, formalizing our interpretation of α^i and β^i as long-run. Proposition 3 signs these loadings in the short run for policies of interest. Vintage effects, which we then turn to, capture how the history of housing construction shapes prices along the housing ladder.

Proposition 3. *Suppose the economy is in a stable equilibrium. In a short-run window $t \in (0, \epsilon)$ for $\epsilon > 0$, construction subsidies τ^C cause a progressive affordability effect ($\alpha_t^{\tau^C} < 0$) but a regressive tear-down effect ($\beta_t^{\tau^C} > 0$). Public housing construction C^{PH} causes ambiguous affordability and tear-down effects ($\beta_t^{C^{PH}} \gtrless 0$, $\alpha_t^{C^{PH}} \gtrless 0$). Maintenance subsidies τ^m cause a progressive affordability effect ($\alpha_t^{\tau^m} < 0$) and a progressive tear-down effect ($\beta_t^{\tau^m} < 0$).*

Proof. See Appendix B.9. □

Construction subsidies cause a progressive affordability effect but a regressive tear-down effect in the short run, as they do in the long run. The loadings of public housing construction are ambiguous in the short run. This is because public housing is built only gradually, so that the short run captures the effect of anticipated – not built – public housing. Anticipation may in the short run crowd out private housing faster than public housing is added to the economy, flipping the signs of $\alpha_t^{C^{PH}}$ and $\beta_t^{C^{PH}}$ relative to their comparative statics counterparts. Maintenance subsidies cause a progressive tear-down effect in the short run, as in the long run. They also cause a progressive affordability effect in the short run. This is because scarcity of new units, the force pushing $p_t(1)$ up in the long run, builds up only gradually through the housing stock, while lower flow maintenance costs, the force pushing $p_t(1)$ down, act immediately.

A high share $V_{s,t}$ of recent vintages (low s) captures a recent rise in construction. A low share of recent vintages instead captures a recent drop in construction. We formalize this idea by defining construction peaks and troughs.

Definition 4 (Construction Peaks and Troughs). *The economy is in a construction peak (trough) at time t if there exists $z \in (0, T_t)$, such that $V_{s,t} \geq 1$ ($V_{s,t} \leq 1$) for $s \leq z$ and $V_{s,t} < 1$ ($V_{s,t} > 1$) for $s > z$.*

Whether di induces a construction peak or trough is then captured by the loadings $\{\eta_{s,t}^i\}_{s \in [0, T]}$.

Corollary 2. *A perturbation to any policy instrument $i \in \mathcal{I}$ induces a construction peak (trough) a time t if and only if there exists $z \in (0, T)$ such that $\eta_{s,t}^i \geq 0$ ($\eta_{s,t}^i \leq 0$) for $s \leq z$ and $\eta_{s,t}^i < 0$ ($\eta_{s,t}^i > 0$) for $s > z$.*

Proposition 4 characterizes vintage effects during construction peaks and troughs.

Proposition 4. *In a construction peak, vintage effects $\{\eta_{s,t}^i\}_{s \in [0,T]}$ are regressive:*

$$\int_0^T \eta_{s,t}^i \frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} ds \geq 0, \quad \frac{d}{dq} \left(\int_0^T \eta_{s,t}^i \frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} ds \right) < 0.$$

In a construction trough, they are progressive:

$$\int_0^T \eta_{s,t}^i \frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} ds \leq 0, \quad \frac{d}{dq} \left(\int_0^T \eta_{s,t}^i \frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} ds \right) > 0.$$

Proof. See Appendix B.10. □

Suppose di induces a construction peak. Market clearing requires households to upgrade the quality of their housing unit. Their willingness to pay for housing quality falls, causing prices to rise towards the bottom of the housing ladder. Contrary to the tear-down effect, construction peaks do not induce re-assignment of the lowest-income households in the economy, who remain in quality q_{min} . Nevertheless, the largest price increase resulting from a construction peak is at $p(q_{min})$. This is because the higher-income households re-assigned to new units exert downstream pressure on prices of old (low-quality) units.

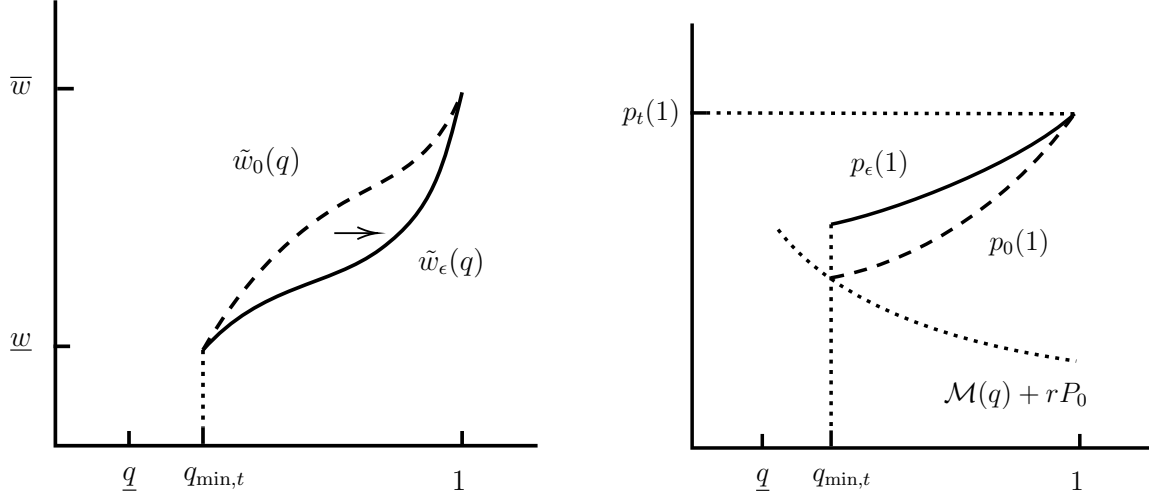
We represent a construction peak graphically in Figure 5. At time $t = 0$, the economy is in steady state, but at time $\epsilon > 0$, a policy change has caused construction to surge. The left panel shows that households other than the lowest and highest-income are re-assigned to higher-quality units. For markets to clear, re-assigned households must face a flatter rental price schedule, consistent with regressive vintage effects in the right panel.

Theorem 4 established that vintage effects, which reflect adjustment of the housing stock, fade away in the long run. Understanding short-run vintage effects is important because the long run is a distant future for housing policy. Proposition 5 characterizes them for policies of interest.

Proposition 5. *Suppose the economy is in stable equilibrium. In a short-run window $t \in (0, \epsilon)$ for $\epsilon > 0$, construction subsidies τ^C cause a construction peak. Public housing construction C^{PH} causes ambiguous vintage effects. Maintenance subsidies τ^m cause a construction trough.*

Proof. See Appendix B.11. □

Figure 5: Vintage Effects in a Construction Peak.



Note: Vintage effects in a construction peak at $t = \epsilon$: $\eta_{s,\epsilon}^i \geq 0$ for $s \leq z$ and $\eta_{s,\epsilon}^i < 0$ for $s > z$ for $z \in (0, T)$. Left panel: assignment function. Right panel: rental price schedule. The figure has $\underline{q} \rightarrow 0$, $C^{PH} = 0$, $q^{RC} = \underline{q}$, and $\tau^{TD} = \tau^m = \tau^{P_0} = \tau^C = 0$.

Construction subsidies cause landlords to tear down old units in favor of building new. The resulting shift in supply from the bottom to the top of the housing ladder is captured by the tear-down effect in the long run. But this shift in supply is especially severe in the short run, before the new units being built have had time to filter down the housing ladder. This is captured by construction peak vintage effects, which cause prices to increase at the bottom of the ladder. As a result, construction subsidies may be more regressive in the short run than they are in the long run. The vintage effects induced by public housing construction cannot be characterized. Again, this is because we cannot pin down short-run private housing crowding out in anticipation of public housing.

Maintenance subsidies cause landlords to further maintain old housing units. For markets to clear with more units at the bottom of the ladder, there must be fewer units in other parts. In the short run, the increase in low-quality units is compensated by a drop in construction, resulting in construction trough vintage effects. Thus, maintenance subsidies are unambiguously progressive in the short run.

4 Taking the Model to the Data

In this section, we take our model to the data. First, we extend the model. Second, we calibrate the model to a representative US city and conduct validation exercises. Third, we study counterfactual policy experiments.

4.1 Extending the Model

We first extend the technology governing construction. In Section 2, we assumed that all new units are built with quality $q = 1$. This assumption is consistent with new units commanding higher prices than old ones and lends tractability to our theory. Yet, it entails a single price for new units, at odds with the data.⁷ Towards taking the model to the data, we thus allow for construction to occur at any quality.

We then extend the technology governing the relationship between maintenance and the filtering rate. In Section 2, we assumed a constant filtering rate δ for sufficiently maintained units. The data however suggests that higher maintenance can bring this rate further down. Indeed, a repeat-sales analysis of the relationship between housing value depreciation and maintenance expenditure which we discuss in further detail upon calibrating the model reveals that a 1% increase in maintenance expenditure lowers a housing unit's annual value depreciation by 0.02 percentage points. Thus, we include an intensive margin maintenance decision.

Extending the Construction Technology. We denote aggregate construction of quality q housing by $C_t^A(q)$. The cost of $C_t^A(q)$ is $\kappa(C_t^A(q), q) \geq 0$, with $\kappa_C(C_t^A(q), q) > 0$. We note that while this extension allows construction at any quality, it does not impose construction at every quality. Indeed, landlords do not build at any q where the building cost of the first unit, $\kappa(0, q)$, is too high relative to $P_t(q)$. We also note that congestion in construction is quality specific. We make congestion quality specific to generate dispersion in new unit prices. Indeed, absent quality-specific congestion, for a given housing value function $P_t(q)$, all construction tends to concentrate at a single quality, preventing dispersion in new unit prices (see Baum-Snow and Duranton 2025 for a discussion). Taking $\kappa(0, q) \rightarrow \infty$ for $q < 1$, our extended construction technology behaves identically to that in Section 2, with all construction at $q = 1$.

Extending the Maintenance Technology. As in Section 2, a housing unit's filtering rate is decreasing in maintenance spending and the maintenance spending required for a given filtering rate is decreasing in quality. We again assume that insufficiently maintained units lose their quality instantly, generating an extensive margin maintenance decision. We now also assume that sufficiently maintained units filter down the housing ladder at a rate

⁷The fact that there is a range of prices for new units is visible in Figure 1.

which depends continuously on maintenance. Specifically, the filtering rate now satisfies

$$\mathcal{D}(m_t, q) = \begin{cases} \infty & \text{if } m_t < \mathcal{M}(q) \\ \delta e^{-\phi(m_t - \mathcal{M}(q))} & \text{if } m_t \geq \mathcal{M}(q) \end{cases},$$

where $\phi \geq 0$ controls the effectiveness of maintenance in slowing down filtering beyond δ and generates an intensive margin maintenance decision. The first-order condition corresponding to this decision is

$$1 \geq -(1 - \tau^m) \frac{\partial \mathcal{D}(m_t(q), q)}{\partial m} \frac{\partial P_t(q)}{\partial q} = (1 - \tau^m) \phi \mathcal{D}(m_t(q), q) \frac{\partial P_t(q)}{\partial q}, \quad (14)$$

and characterizes the maintenance policy $m_t(q)$ for sufficiently maintained units. The inequality captures that optimal maintenance may be $\mathcal{M}(q)$. When optimal maintenance exceeds $\mathcal{M}(q)$, it is increasing in the housing value gradient $\partial P_t(q)/\partial q$, which captures the marginal value of quality for landlords.⁸ Because $\lim_{m \rightarrow \infty} \mathcal{D}(m, q) = 0$, filtering can be brought to a crawl. As before, however, filtering cannot be reversed. Through the lens of the model, large-scale renovations are thus captured by the combination of a tear-down and a new build. This generates a challenge for mapping the model to the data, which we discuss upon calibrating.

We have allowed for congestion in construction but not in maintenance. Our micro-foundation for construction costs in Appendix C.1 attributes congestion to limited arrival of development opportunities. We think it unlikely that maintenance costs would be subject to this force. However, alternative drivers of congestion in construction, such as an inelastic supply of specialized labor, could affect maintenance. Because our data provides us with no straightforward way to estimate a maintenance supply elasticity, we choose not to model congestion in maintenance and leave to future work a more in-depth exploration of the structure of maintenance costs.

4.2 Calibration

We calibrate the model steady-state to a representative US city at an annual frequency. Our main data source is the biennial waves of the American Housing Survey (AHS) from 2001 to 2023. We restrict our analysis to the 15 MSAs with geographic identification in all AHS waves.⁹ As additional sources of data, we rely on Davis et al. (2021) for land value shares,

⁸To see this, note that $\mathcal{D}(m, q)$ is decreasing in m for any q .

⁹These are Atlanta, Boston, Chicago, Dallas, Detroit, Houston, Los Angeles, Miami, New York, Philadelphia, Phoenix, Riverside, San Francisco, Seattle, and Washington.

on Baum-Snow and Han (2024) for housing supply elasticities, on RSMeans for a breakdown of construction costs, and on the Building Permits Survey (BPS) for building permit counts. We provide additional detail on our data in Appendix C.2.

Calibration requires us to bridge two gaps in interpretation between the model and the data. First, because the model abstracts from homeownership, it treats owner-occupants as renters. Taking the model to the data therefore requires imputing rental prices for owner-occupants in the data, which we do following the procedure outlined by Diamond and Diamond (2024). We omit imputed rental prices from regressions which include characteristics used for imputation as explanatory variables. Second, the model captures large-scale renovations which effectively reset the quality of a housing unit as the combination of a tear-down and a new build. Thus, large-scale renovations reset the age of housing units in the model, but not in the data. This results in *effectively* new units recorded as old in the data. To account for the presence of these units, we avoid targeting moments of the housing unit age distribution in the calibration. Furthermore, throughout the calibration and our validation exercises, we restrict regressions which include housing unit age as an explanatory variable to units aged less than 35 years. This choice is based on the observation that the relationship between rental prices and housing age flattens sharply beyond age 35, consistent with the presence of effectively new units recorded as old. Figure A2 shows this pattern.

We set the income distribution to a truncated log normal distribution whose mean and standard deviation we back out from post-tax income data.¹⁰ We set $r = 0.05$ externally. As a functional form for building costs, we use

$$\kappa(C_t^A(q), q) = q \left(\mu + \zeta (C_t^A(q))^\lambda \right).$$

We microfound this construction technology as resulting from an arrival process for development opportunities in Appendix C.1. The parameters δ , P_0 , μ , ζ , ϕ , λ and γ are calibrated by minimizing the distance between moments computed in the data and in the model. We compute data moments for the representative city as the weighted median across MSAs of within-MSA data moments. Instead of attempting to explicitly measure the taxes and subsidies on maintenance (τ^m), land use (τ^{P_0}), and private construction (τ^C) in place in the baseline equilibrium, we incorporate them into the parameters γ , μ , ζ , and P_0 . Our calibration does however abstract from a binding minimal quality occupancy standard ($\underline{q}$), public housing construction (C^{PH}), rent control (q^{RC} , p^{RC}), and tear-down taxes (τ^{TD}) in the baseline equilibrium. This is consistent with limited use of these policy instruments in

¹⁰We estimate post-tax income as $\tilde{y} = \mu y^{1-\tau}$ where y is pre-tax income, τ is 0.181, and μ is chosen to match an average tax rate of 0.2, based on Heathcote et al. (2017).

the representative US city. Table 1 summarizes our calibration.

Table 1: Calibration Summary.

Parameter	Value	Moment	Model	Data	Source
r	0.050	—	—	—	—
δ	0.0037	Median Rent-to-Income	0.302	0.291	AHS
P_0	4.98	Median Land-to-Value	0.464	0.443	Davis et al. (2021)
μ	5.51	Median New Unit Rent to Median Income	0.448	0.473	AHS
ζ	2,140.19	Share of Materials in Construction Costs	0.527	0.525	RSMeans
λ	2.23	Housing Supply Elasticity	0.189	0.199	Baum-Snow and Han (2024)
γ	1.88	Maintenance on Age Regression Coefficient	0.023	0.023	AHS
ϕ	1.76	Average Maintenance Effectiveness	0.019	0.019	AHS

Note: Summary of calibration with parameters, calibrated parameter values, moments used to inform parameters, model moment values, data moment values, and data source. Horizontal lines group parameters with the moments used to inform them.

We target the median rent-to-income ratio to inform δ . Rent-to-income informs δ because all else equal, a higher δ accelerates filtering and thus causes housing units to be torn down and rebuilt more frequently, raising the rental prices required for landlords to break even on housing development. We target the median ratio of land price to housing value to inform the outside option value of land P_0 . In the data, the median ratio of land price to housing value corresponds to the median land value share, which we take directly from Davis et al. (2021) for each MSA in our sample.

Construction costs consist of both material and labor costs. Consistent with our construction costs microfoundation in Appendix C.1, we attribute congestion to labor costs. Thus, μ parameterizes material costs, and ζ and λ parameterize labor costs. All three parameters are jointly informed by three moments. First, to capture overall construction costs, we target the ratio of median rental price of new units to median income. We use the rental price of new units as an indirect measure of construction costs instead of RSMeans construction costs directly because the latter exclude some cost components (e.g. regulatory and financing costs). Second, we target the share of materials in construction costs, measured at 52.5% in RSMeans.¹¹ Third, we target housing supply elasticity.¹²

The parameters governing the maintenance technology, γ and ϕ , are informed by two moments. First, we target the coefficient from a regression of log maintenance spending on

¹¹Our method implicitly assumes that the share of materials for cost components excluded from RSMeans is equal to the share of materials for cost components included in RSMeans.

¹²To compute housing supply elasticity in the model, we permanently perturb the housing value function $P_t(q)$ by 1% and compute the resulting percent increase in the total housing stock after 10 years, holding rental prices (and thus the tear-down cutoff $q_{min,t}$) fixed. Computing the 10-year change aligns the supply elasticity computed in the model with those measured in the data by Baum-Snow and Han (2024).

housing unit age, to capture the degree to which maintenance costs rise as housing quality declines.¹³ Second, to capture the effectiveness of maintenance in slowing down filtering, we target the relationship between housing value depreciation and maintenance expenditure. We uncover this relationship in the data through a repeat-sales estimation of α_1 in the regression specification

$$\frac{\log P_{h,c,t} - \log P_{h,c,t-s}}{s} = \alpha_0 + \alpha_1 \log(\text{maintenance}_{h,c,t,s}) + \Gamma_{c,t} + \varepsilon_{h,c,t},$$

where $P_{h,c,t}$ is the value of a housing unit h in MSA c sold at year t , $\text{maintenance}_{h,c,t,s}$ is average yearly maintenance per square foot between $t - s$ and t and $\Gamma_{c,t}$ are MSA $\times$ year fixed effects.¹⁴ The counterpart to α_1 in the model is

$$\int \frac{d}{d \log m} \left(\frac{d \log P(q)}{dt} \right) h(q) dq.$$

Figure A3 presents a calibration sensitivity analysis which shows the effect of small perturbations to calibrated parameter values on model moment values. Model moment values are sensitive to the parameters informed by their data counterparts.

Untargeted Moments. We motivated our paper by observing that new housing units command higher prices than old ones, resulting in low-income households sorting away from new housing (Figure 1). Our parsimonious calibration imposed neither pattern. Therefore, we begin assessing our model’s fit of the data by reproducing Figure 1. The left panel of Figure 6 compares the distributions of rental prices generated by the calibrated model for new and existing units with their empirical counterparts.¹⁵ The model successfully matches the separation between rental prices of new and existing units. The median rental price of new units lies at the 90th percentile of the rental price distribution in the model, compared to the 87th percentile in the data.¹⁶ The model matches price dispersion within new units and within existing units reasonably well. The right panel of Figure 6 compares the income percentile distribution of new housing occupants in the model and in the data. The model fits the data reasonably well, with disproportionately higher income households occupying

¹³In the data, we control for year, condo status, and number of bedrooms fixed effects.

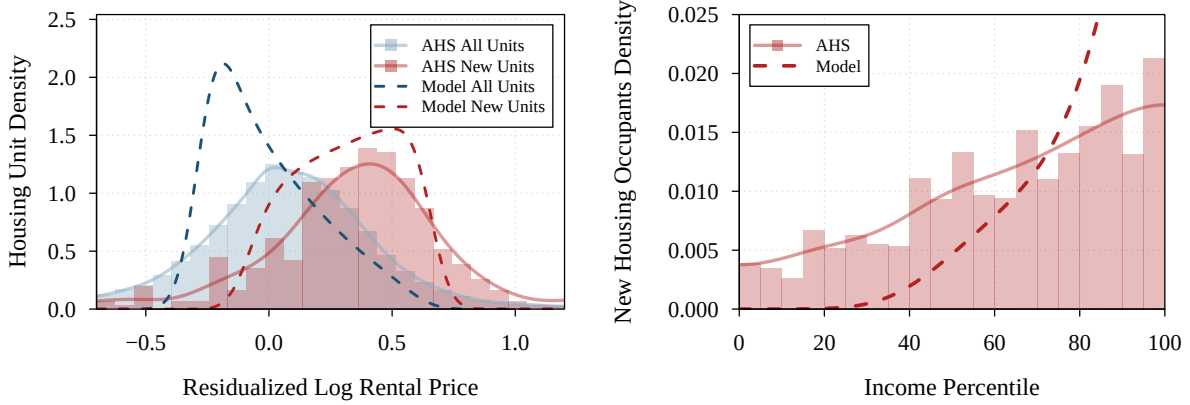
¹⁴Because of limited repeat-sales observations, we pool this regression across MSAs instead of estimating it separately within each MSA in a first step, as we do for all other data moments.

¹⁵The data version of the distributions in Figure 6 differ from the distributions in Figure 1 in two ways. First, the sample is different, as Figure 1 is generated using the full AHS dataset, before applying any data restrictions. Second, log rental prices in Figure 6 are residualized from square footage fixed effects, as price variation due to housing size is absent from the model.

¹⁶The 87th-percentile value reported here is different from the 82nd-percentile value from Figure 1 because it is computed on the calibration dataset.

new housing units.

Figure 6: Prices and Sorting into Old vs. New Housing: Model and Data.



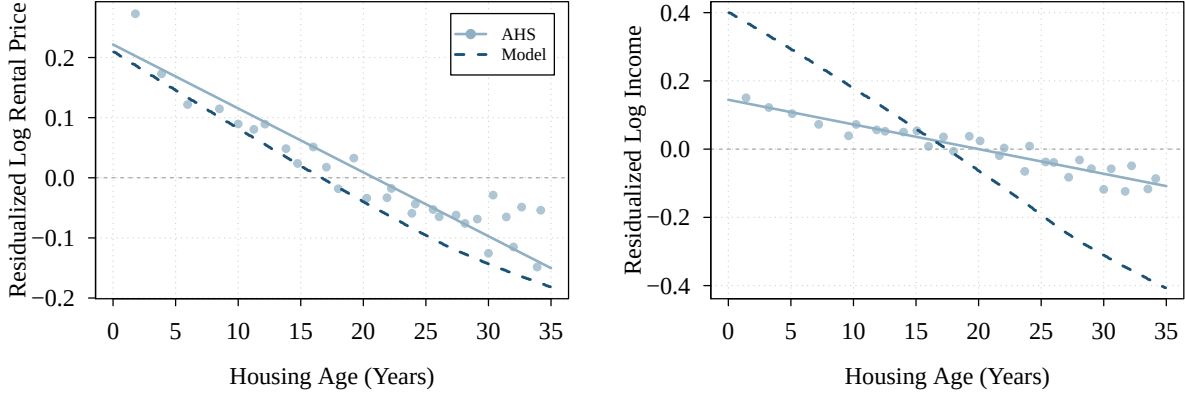
Note: Left panel: log rental price distributions of all units and new units. Right panel: income percentile distribution of new housing occupants. Rental prices are residualized from $MSA \times year$ and square footage fixed effects. Source: AHS.

In Appendix D, we benchmark our model against a calibrated model *without* filtering, where housing depreciates in quantity instead of quality. The benchmark model assumes that maintenance costs rise as housing units age and therefore also generates maintenance, tear-down, and construction decisions. However, Figure A4 shows that it fails to reproduce the qualitative pattern from Figure 1. Without filtering, old units are *more* expensive than new ones due to survivorship bias: landlords only find old units with high rental prices profitable to maintain. This generates sorting in the opposite direction of the data, with low-income households disproportionately occupying new units. Upon studying counterfactual policies, we show that this failure can flip the benchmark model’s assessment of policy progressivity.

Central to our theory of the progressivity of housing policy is the relationship between rental prices and housing quality, captured by the rental price schedule $p(q)$, and household sorting into housing units, captured by the assignment function $\tilde{w}(q)$. An ideal assessment of our calibrated model would therefore compare these objects in the model and in the data. Because housing quality q is unobservable in the data, this comparison is infeasible. Filtering however implies a tight relationship between housing quality and housing age, which suggests using age as a negative proxy for quality. The left panel of Figure 7 shows the relationship between log rental prices and housing age – the rent-age gradient – in the model and in the data. The model closely matches the empirical rent-age gradient. The right panel shows the relationship between log household income and housing age – the income-age gradient. The model overshoots the income-age gradient. This is due to steeper positive assortative matching in the model than in the data, where assortative matching is

dampened by forces such as idiosyncratic tastes and moving frictions. Adding these forces would make solving dynamic equilibria numerically infeasible.¹⁷ Because model and data agree on positive assortative matching, we expect it would leave the progressivity patterns we document in our counterfactuals unchanged.

Figure 7: Rent-Age and Income-Age Gradients.



Note: Left panel: rent-age gradient. Right panel: income-age gradient. Rental prices and income are residualized from MSA $\times$ year fixed effects. Source: AHS.

We assess the model's fit of other key untargeted moments. Our calibration targeted the median rent-to-income ratio, but not its dispersion across the income distribution. The model generates a rent-to-income ratio 1.61 times greater for 25th-percentile households relative to 75th-percentile households, compared to 1.63 in the data. We compare housing unit ages in the model and in the data. Due to the presence of effectively new units recorded as old in the data, we expect median housing age to be lower in the model. This is indeed the case, with a median housing age of 30 years in the model, compared to 46 in the data. Accordingly, the model generates a construction rate higher than its empirical counterpart (0.016 compared to 0.010). The lifespan of units in the model ranges from 26 to 86 years, with a median of 62 years. We summarize the model's fit of untargeted moments in Table 2.

Cross-MSA Validation. To validate our model, we test its ability to replicate variation in rent-age and income-age gradients across MSAs. Studying these objects in the data reveals a salient pattern: MSAs with high construction costs feature flatter rent-age gradients and

¹⁷Solving the model numerically is facilitated by the assignment function established in Lemma 1, which, combined with Lemma 2, yields a differential equation for the rental price schedule that can be solved analytically. Adding idiosyncratic tastes or moving frictions would break the assignment function, forcing us to solve for the rental price schedule – an infinite dimensional object – numerically. Along a transition, this proves infeasible.

Table 2: Untargeted Moments.

Moment	Model	Data	Source
Median New Unit Rent Percentile	90th	87th	AHS
Rent-Age Gradient	-0.012	-0.011	AHS
Income-Age Gradient	-0.023	-0.007	AHS
25th-percentile / 75th-percentile Rent-to-Income	1.61	1.63	AHS
Median Housing Age	30	46	AHS
Construction Rate	0.016	0.010	BPS

Note: Summary of untargeted moments fit. The empirical construction rate is measured using the number of permitted housing units recorded in the BPS. Source: AHS and BPS.

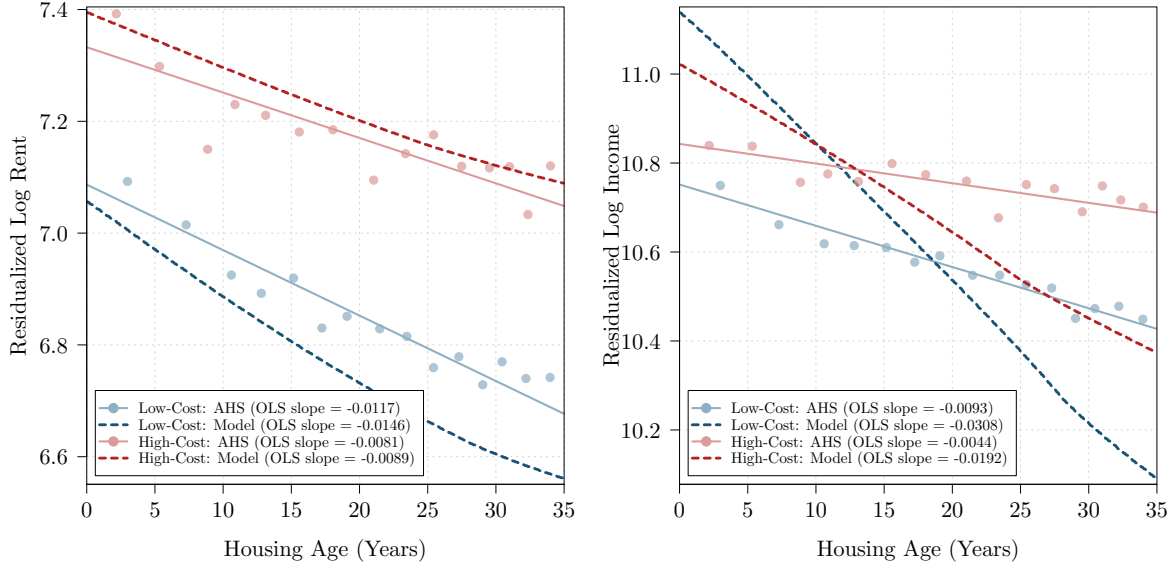
flatter income-age gradients.¹⁸ The left panel of Figure 8 shows that a housing unit’s rental price falls on average by 0.8% every year in the top 25% MSAs by construction costs (high-cost MSAs), compared to 1.2% in the bottom 25% (low-cost MSAs). The right panel reveals that the income of this housing unit’s occupants falls on average by 0.4% every year in high-cost MSAs, compared to 0.9% in low-cost MSAs. Can the model account for this pattern?

Our first approach to simulating high- and low-cost MSAs in the model is to solve for a steady-state under different values of ζ , capturing different construction labor costs, holding other parameters fixed. Our choice of ζ for this experiment is grounded in the fact that 97% of variation in construction costs across MSAs is attributed by RSMeans to labor costs. We then compare high-cost MSAs in the data to a high-cost calibration of the model which matches their average building costs, and proceed accordingly for low-cost MSAs. In the model, a lower $p(1)$ in the low-cost calibration steepens $p(q)$ through an affordability effect. Lower costs also raise q_{min} , producing a counteracting tear-down effect which pushes low-income households into higher-quality units, steepening $\tilde{w}(q)$. Quantitatively, the combined effects result in a moderately steeper rent-age gradient and a steeper income-age gradient in the low-cost calibration of the model. Figure 8 overlays model results on empirical results, with the model successfully tracking the data.

Acknowledging that ζ may be confounded with income distributions, land prices, construction material costs, housing supply elasticities, and maintenance costs, our second approach re-calibrates each of these objects for each MSA in our sample. Specifically, we estimate MSA-specific income distributions and calibrate P_0 , μ , ζ , λ , ϕ , and γ at the MSA level to match median rent-to-income ratios, median land-to-value ratios, material shares

¹⁸We use RSMeans construction costs as a proxy for building costs. In our sample, construction costs are 86% higher in the most expensive MSA relative to the least expensive MSA.

Figure 8: Rent-Age and Income-Age Gradients in High-Cost and Low-Cost MSAs.



Note: Left panel: rent-age gradient for high-cost MSAs (top 25% population-weighted MSAs by RSMeans construction costs) and low-cost MSAs (bottom 25%). Right panel: income-age gradient for high- and low-cost MSAs. Model overlay is generated by adjusting ζ to match construction costs in high- and low-cost MSAs. Rental prices and income are residualized from MSA $\times$ year fixed effects. OLS stands for ordinary least squares. Source: RSMeans and AHS.

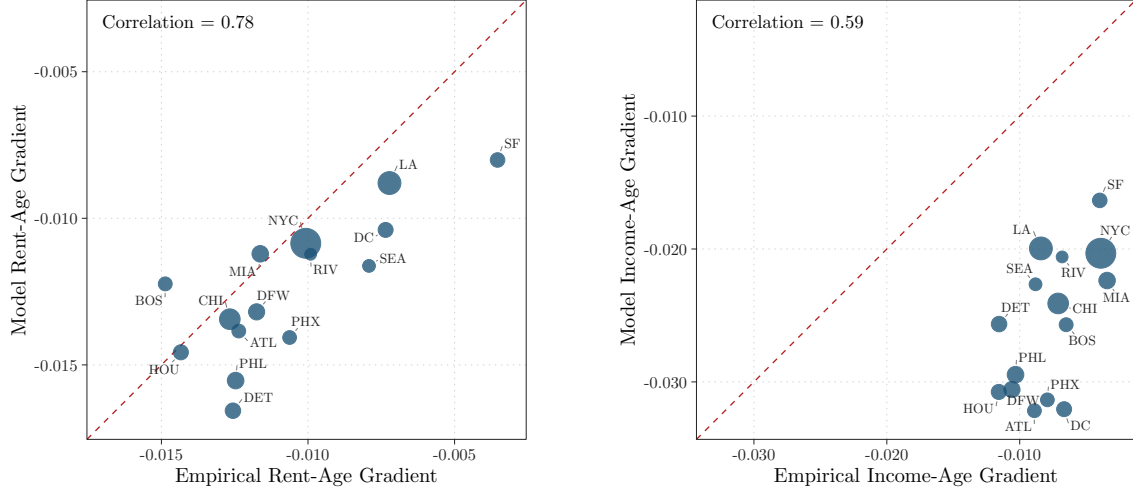
in construction costs, housing supply elasticities, and maintenance spending.¹⁹ Figure A5 compares targeted moments in the model and in the data for each MSA.

Calibrated to each MSA in our sample, the model reproduces the empirical pattern of flatter rent-age and income-age gradients in high-cost MSAs: San Francisco, Los Angeles, and (to a lesser extent) New York City than in low-cost MSAs: Detroit, Houston, and (to a lesser extent) Philadelphia. Figure 9 compares model moments on the y -axis to their empirical counterparts on the x -axis for all 15 MSAs. The correlation between model and data moments is 0.78 for the rent-age gradient and 0.59 for the income-age gradient.

The results from our cross-MSA validation connect to observations made in prior work. First, our exercises show that high-cost cities face disproportionately higher prices for old units. This finding is connected to the observation by Malpezzi and Green (1996) of higher prices for low-quality units relative to high-quality units in areas with costly housing regulation. Molloy (2020) cites this observation as conflicting with survey evidence by Emrath (2016) that regulation places a higher burden on high-quality construction. Our model resolves this tension: the direct effect of regulation may be on the price of high-quality units,

¹⁹Variation in maintenance costs is captured by scaling γ by some k and ϕ by $1/k$. k is informed by maintenance spending at the MSA level. Specifically, we target the ratio of MSA-specific maintenance expenditure for units aged less than 35 to representative city maintenance expenditure for units aged less than 35.

Figure 9: Rent-Age and Income-Age Gradients across MSAs.



Note: Left panel: rent-age gradients in model (MSA-specific calibrations) and data. Right panel: income-age gradients in model and data. Markers are scaled by MSA population. Rental prices and income are residualized from year fixed effects. Source: AHS.

but the affordability effect amplifies price increases going down the housing ladder. Second, our exercises highlight a slower occupant income decline – i.e. slower income filtering (Rosenthal 2014) – for housing units in high-cost cities. This is consistent with a negative correlation between costly regulation and the speed of income filtering, a fact documented by Liu et al. (2022). Our model attributes this pattern to the tear-down effect. Similarly, Spader (2025) documents a recent slowdown in income filtering for which our model offers the coinciding rise in construction costs (Baum-Snow and Han 2024) as an explanation.

4.3 Counterfactual Policy Experiments

Despite a parsimonious calibration, our model successfully matches key within- and cross-MSA moments. We now use it for policy experiments. Starting from our baseline equilibrium, we consider three policy perturbations, corresponding to a construction subsidy, public housing construction, and a maintenance subsidy.

Construction Subsidy (τ^C). We consider the unanticipated and permanent implementation of $\tau^C = 0.01$, corresponding to a 1% decrease in the building costs faced by landlords. We compute the cost of the policy perturbation by discounting the resulting path of government spending, using $r = 0.05$.

Public Housing Construction (C^{PH}). We consider an unanticipated and permanent increase in public housing construction C^{PH} . We assume that the government builds public housing at the median quality of private new builds. This is in line with the *Unit Con-*

construction Cost Guideline of the US Department of Housing and Urban Development (HUD), which sets RSMeans average quality costs as a target for public housing development.²⁰ We assume that the government charges public housing tenants 30% of their income as rent. This is in line with Section 42 of the US Housing Act, which establishes guidelines for funding of public housing agencies.²¹ Last, we assume that the government spends exactly $\mathcal{M}(q)$ on maintenance of a quality q unit and tears units down when they reach the baseline equilibrium cutoff q_{min} . We set C^{PH} such that discounted government spending, net of rental income, equals the cost of the construction subsidy. This results in a yearly public housing construction rate of 1 unit per 15,000 households.

Maintenance Subsidy (τ^m). We consider an unanticipated and permanent implementation of $\tau^m > 0$. We set τ^m such that discounted government spending equals the cost of the construction subsidy. This results in a maintenance subsidy covering 0.4% of maintenance costs.

Welfare Analysis. Table 3 shows the discounted lifetime welfare changes from all three policies across the income distribution. Welfare changes are expressed in consumption-equivalent terms and normalized by government spending. The discount rate is $\rho = r = 0.05$.²² The construction subsidy is regressive: its consumption-equivalent transfer is close to twice as large for households in the 90th percentile of the income distribution as for households in the 10th percentile. Public housing construction is progressive, but fails to improve welfare for households who remain in MR housing. The gains for households assigned a public housing unit are very large, but even in the long run, these are only 0.4% of households.²³ The maintenance subsidy is progressive: its consumption-equivalent transfer is close to 40% larger for 10th-percentile households than for 90th-percentile households. Overall, close to the bottom half of households prefers the maintenance subsidy to the construction subsidy. Only the households assigned a public housing unit prefer public housing construction.

Welfare changes may occur either through changes in rent paid (first and third terms in (12)) or through changes in housing quality (second term in (12)). Figure 10 shows, for each policy, the contribution of both sources to lifetime welfare changes, with household income percentile on the x -axis. The construction subsidy lowers rent paid. However, it generates much of its welfare gains through quality improvements. Low-income households, whose

²⁰Source: HUD, Public Housing Capital Fund Program; Final Rule.

²¹Source: United States Code, 42 U.S.C. § 1437f.

²²Table A1 shows that results are similar when discounting at the rate $\rho = 0.02$.

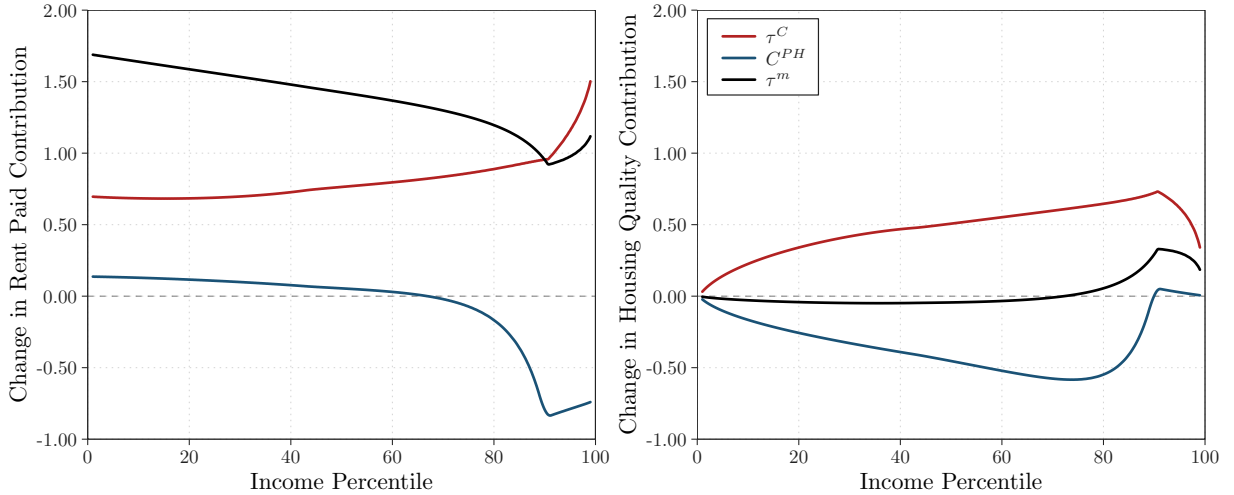
²³To compute welfare changes for households assigned a public housing unit, we assume random assignment of public housing units to public housing tenants.

Table 3: Lifetime Welfare Changes from τ^C , C^{PH} , and τ^m .

Income Percentile	Construction Subsidy (τ^C)	Public Housing Construction (C^{PH})	Maintenance Subsidy (τ^m)
0.2nd	0.663	17.792	1.647
10th	0.868	-0.080	1.569
35th	1.105	-0.325	1.405
65th	1.325	-0.608	1.246
90th	1.569	-0.901	1.136

Note: Summary of discounted lifetime welfare changes from equally costly τ^C , C^{PH} , and τ^m policies. Welfare changes are expressed in consumption-equivalent terms, normalized by government spending, and discounted at the rate $\rho = 0.05$. The 0.2nd percentile of the income distribution corresponds to the median income assigned a public housing unit in the long run under the C^{PH} policy.

rent-to-income ratio is high – implying a low marginal rate of substitution of housing quality for consumption – value housing quality improvements less than reductions in rent paid. Hence they prefer the maintenance subsidy, whose welfare gains result from large reductions in rent paid. Public housing construction does reduce rent paid by low-income households in MR housing.²⁴ However, the reduction is modest and, being driven by re-assignment into the low-quality units vacated by new public housing tenants, comes at the expense of housing quality.

Figure 10: Sources of Lifetime Welfare Changes from τ^C , C^{PH} , and τ^m .

Note: Sources of discounted lifetime welfare changes from equally costly τ^C , C^{PH} , and τ^m policies. Welfare changes are expressed in consumption-equivalent terms, normalized by government spending, and discounted at the rate $\rho = 0.05$. Public housing tenants are omitted.

²⁴Figure 10 does not decompose welfare changes from the C^{PH} policy for the 0.4% of households assigned a public housing unit. As highlighted in Table 3, these households experience large welfare gains. These are driven by reductions in rent paid and improvements in quality.

We note that both the construction and maintenance subsidies induce aggregate household welfare gains that outweigh government spending. That is not to say that they are Pareto improving: being unanticipated, they depreciate housing value on impact as present and future rental prices fall, making landlords worse off.²⁵ In the presence of homeownership, unanticipated construction and maintenance subsidies would thus affect homeowners differently than renters. Because homeowners are disproportionately high-income households, this suggests an additional progressivity channel to unanticipated housing policy which lies outside of the scope of our analysis.

To think through the forces underlying the results in Table 3, we turn to our theory of the progressivity of housing policy. So far, we have called policies *progressive* and *regressive* when their welfare gains are respectively decreasing and increasing in income. It is now useful to introduce a progressivity index for any policy i at any time t . Following Kakwani (1977), we use $\Omega_t^i \equiv -2 \text{cov}[\xi_t^i(w), F(w)]/E[|\xi_t^i(w)|]$.²⁶ We can derive

$$\Omega_t^i = \frac{1}{E[|\xi_t^i(w)|]} \int_{\underline{w}}^{\bar{w}} \frac{-d\xi_t^i(w)}{dw} F(w)(1 - F(w))dw.$$

Thus, Ω_t^i is a weighted average of minus the derivative of time t welfare changes with respect to income ($-d\xi_t^i/dw$), normalized by the size of welfare changes $E[|\xi_t^i(w)|]$. This implies two desirable properties. First, any time- t progressive policy ($d\xi_t^i/dw < 0$ for all w) satisfies $\Omega_t^i > 0$, while any time- t regressive policy ($d\xi_t^i/dw > 0$ for all w) satisfies $\Omega_t^i < 0$. Second, Ω_t^i is invariant to scaling policy perturbations and thus is well suited to compare the progressivity of our equally costly policy perturbations at different time horizons.²⁷

Theorem 2 established that $\xi_t^i(w) = -dp_t(q)/di$. Following our analysis from Section 3, we exploit this sufficient statistic by relying on changes in prices along the housing ladder to shed light on changes in welfare across the income distribution. We first consider the long run, relying on our theoretical results on comparative statics. We then consider the short run, relying on our theoretical results on transition dynamics.

The Long Run: Affordability and Tear-Down Effects. Theorem 3 established that policy shapes long-run prices through the affordability and tear-down effects. Based on the

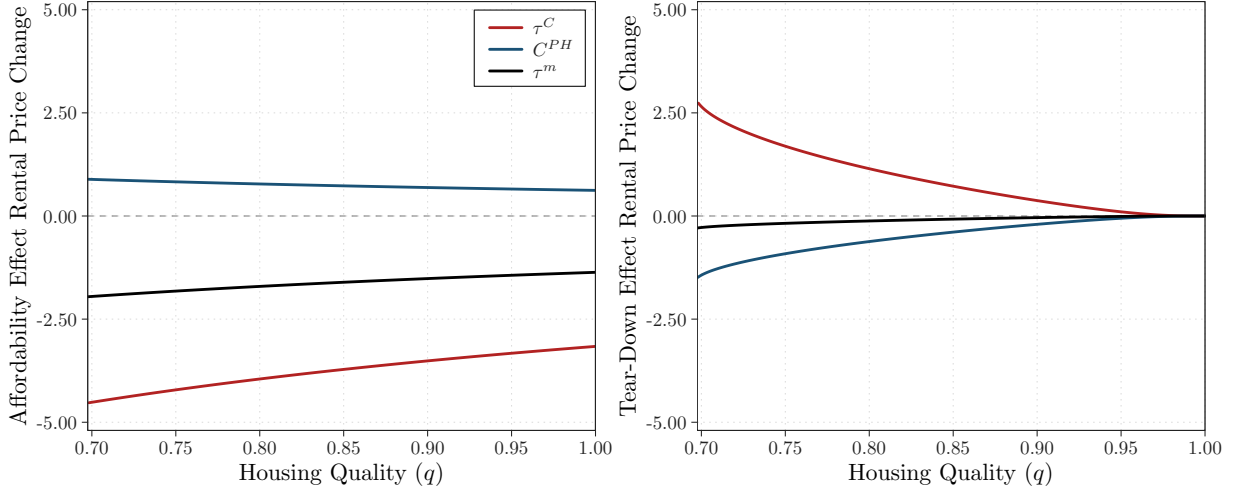
²⁵In fact, because our model includes no source of inefficiency, policy interventions are not Pareto improving.

²⁶When $\xi_t^i(w) > 0$ for all w , the progressivity index Ω_t^i corresponds to twice the area between the concentration curve $L_t^i(x)$, defined as the share of $\int \xi_t^i(w)dF(w)$ going to households with income w below percentile x ($F(w) < x$), and the 45° line. Indeed, we then have $\Omega_t^i = 2 \int_0^1 L_t^i(x)dx - 1$.

²⁷To see this, note that both the numerator (through $\xi_t^i(w)$) and denominator (through $E[|\xi_t^i(w)|]$) scale with di .

formula for the policy loadings α^i and β^i derived in the proof of Theorem 3 (see Appendix B.5), we compute the affordability effect as the contribution to changes in $p(q)$ of changes in new unit prices $p(1)$ and the tear-down effect as the contribution to changes in $p(q)$ of changes in the tear-down cutoff q_{min} , net of household re-assignment into public housing. Figure 11 shows how each effect shapes long-run rental price changes.

Figure 11: Affordability and Tear-Down Effects in the Long Run.



Note: Left panel: long-run affordability effect on rental prices from equally costly τ^C , C^{PH} , and τ^m policies. Right panel: tear-down effect. Public housing tenants are omitted.

Consistent with Proposition 1, affordability and tear-down effects are strongest at the bottom of the housing ladder. Mapping rental price changes to welfare changes using Theorem 2, we decompose long-run progressivity indexes Ω^i between affordability and tear-down effect contributions. Results track Proposition 2. The construction subsidy induces a progressive affordability effect by lowering the price of new units $p(1)$. This contributes positively to the progressivity index Ω^{τ^C} . However, landlords respond to the construction subsidy by tearing down old units in favor of building new, resulting in a regressive tear-down effect which overpowers the affordability effect.²⁸ The progressivity of public housing is explained by its progressive tear-down effect, which overpowers its regressive affordability effect. Proposition 2 derived an ambiguous long-run affordability effect for maintenance subsidies. Numerically, our maintenance subsidy experiment causes a progressive affordability effect, in addition to its progressive tear-down effect. A lower price for new units and further maintenance of old units amplify the decrease in rental prices resulting from τ^m at the bottom of the housing ladder.

²⁸ $\xi_t^i(w)$ is ill-defined for the 0.4% of households assigned a public housing unit. Thus, they are omitted from $\Omega_t^{C^{PH}}$. As clear from Table 3, their large welfare gains are consistent with progressive public housing construction.

Table 4: Long-Run Progressivity Index (Ω^i) Decomposition.

	Construction Subsidy (τ^C)	Public Housing Construction (C^{PH})	Maintenance Subsidy (τ^m)
Affordability Effect Contribution	0.075	-0.095	0.064
Tear-Down Effect Contribution	-0.149	0.520	0.031
Cumulative Progressivity Index	-0.074	0.424	0.095
Total Progressivity Index (Ω^i)	-0.083	0.372	0.038

Note: Decomposition of Ω^i between affordability (row 1) and tear-down (row 2) effects. The cumulative progressivity index (row 3) sums contributions of affordability and tear-down effects. Public housing tenants are omitted.

The third row of Table 4 computes a cumulative progressivity index as the sum of affordability and tear-down effect contributions to Ω^i . Long-run gaps between cumulative and total progressivity indexes result from our two model extensions: construction at many qualities and an intensive margin for maintenance. Indeed, adjustments in the quality mix of housing construction and in the intensity with which landlords maintain housing units impact policy progressivity beyond the effects studied in Section 3. As clear from Table 4, however, their contribution is small for the construction subsidy and public housing construction. This is not the case for the maintenance subsidy, whose progressivity is considerably dampened by the presence of an intensive margin for maintenance. The first-order condition for maintenance intensity (14) illustrates why. All else equal, subsidizing maintenance incentivizes landlords to further maintain *all* housing units. Thus, the maintenance subsidy expands housing supply at the bottom of the ladder through a progressive tear-down effect, but also restricts it, as greater maintenance intensity slows down filtering at the top of the ladder.

We contrast our baseline maintenance subsidy with an equally costly maintenance subsidy targeted to housing units with below-median rental prices in Table 5. The targeted maintenance subsidy achieves higher progressivity for two reasons. First, due to its narrower scope, it is able to cover 0.5% of maintenance costs for eligible units, a 0.1 percentage point increase. As a result, it provides a stronger incentive for further maintenance to landlords with old housing units, increasing the tear-down effect contribution to Ω^m . Second, it does not increase maintenance intensity at the top of the housing ladder. The regressive impact of intensive margin maintenance fades, which shrinks the gap between cumulative and total progressivity indexes.

The Short Run and Vintage Effects. Housing units are highly durable, with lifespans of up to 86 years in our calibrated model. Thus, the impact of our three policy perturbations may differ substantially in the short and long run. In particular, welfare changes may

Table 5: Long-Run Progressivity Index (Ω^i): Untargeted vs. Targeted τ^m .

	Baseline Maintenance Subsidy	Targeted Maintenance Subsidy
Affordability Effect Contribution	0.064	0.042
Tear-Down Effect Contribution	0.031	0.073
Cumulative Progressivity Index	0.095	0.115
Total Progressivity Index (Ω^i)	0.038	0.114

Note: Decomposition of Ω^i between affordability (row 1) and tear-down (row 2) effects for baseline and targeted maintenance subsidies. The targeted maintenance subsidy is restricted to below-median units. The cumulative progressivity index (row 3) sums contributions of affordability and tear-down effects.

materialize at different horizons for low-income and high-income households, resulting in time-varying policy progressivity. We now study the path of Ω_t^i and its underlying forces.

Theorem 4 established that policy shapes rental prices along the transition through time-dependent affordability and tear-down effects as well as vintage effects capturing construction peaks and troughs. Based on the formula for the policy loadings α_t^i , β_t^i , and $\{\eta_{s,t}^i\}_{s \in [0,T]}$ derived in the proof of Theorem 4 (see Appendix B.8), we compute the contribution to changes in rental prices of each type of effect. As we did for long-run progressivity indexes, we then rely on Theorem 2 to decompose time-dependent progressivity indexes Ω_t^i between affordability, tear-down, and vintage effects contributions. Table 6 presents our results at 5-, 25-, and 50-year time horizons.

Consistent with Theorem 4, over time, the contributions of affordability and tear-down effects approach their long-run values, shown in Table 4, while the contributions of vintage effects converge to zero. Gaps between cumulative progressivity indexes – computed by summing the contributions of affordability, tear-down, and vintage effects – and total progressivity indexes are small in the short run. Comparing the progressivity of each policy at different time horizons, we note two stark patterns. First, the construction subsidy is substantially more regressive in the short than in the long run, with $\Omega_5^{\tau^C}$ more than 70% lower than $\Omega_{50}^{\tau^C}$. The welfare gains resulting from τ^C are not only lower but also slower to materialize for low-income households, compared to high-income households.

This is driven by vintage effects: Table 6 reveals that net of their contribution, the construction subsidy would be *progressive* in the short run. As the construction subsidy causes landlords to tear down in favor of building new, housing supply shifts from the bottom to the top of the housing ladder. The shift is sharpest in the short run, before new units have had time to filter down the ladder. Figure 12 shows a heat map of vintage shares with time t on the x -axis and vintages s on the y -axis. The 45° degree line traces the path of the time $t = 0$ vintage. The short-run increase in recent vintage shares induced by the

Table 6: Time-Dependent Progressivity Index (Ω_t^i) Decomposition.

	Construction Subsidy (τ^C)	Public Housing Construction (C^{PH})	Maintenance Subsidy (τ^m)
<i>5 Years</i>			
Affordability Effect Contribution	0.108	-0.153	0.051
Tear-down Effect Contribution	-0.125	0.219	0.011
Vintage Effects Contribution	-0.168	0.253	0.010
Cumulative Progressivity Index	-0.185	0.319	0.072
Total Progressivity Index (Ω_5^i)	-0.156	0.345	0.061
<i>25 Years</i>			
Affordability Effect Contribution	0.083	-0.124	0.050
Tear-down Effect Contribution	-0.135	0.260	0.041
Vintage Effects Contribution	-0.088	0.173	0.024
Cumulative Progressivity Index	-0.140	0.309	0.115
Total Progressivity Index (Ω_{25}^i)	-0.121	0.362	0.081
<i>50 Years</i>			
Affordability Effect Contribution	0.073	-0.106	0.052
Tear-down Effect Contribution	-0.144	0.355	0.063
Vintage Effects Contribution	-0.015	0.055	0.008
Cumulative Progressivity Index	-0.085	0.304	0.123
Total Progressivity Index (Ω_{50}^i)	-0.091	0.337	0.080

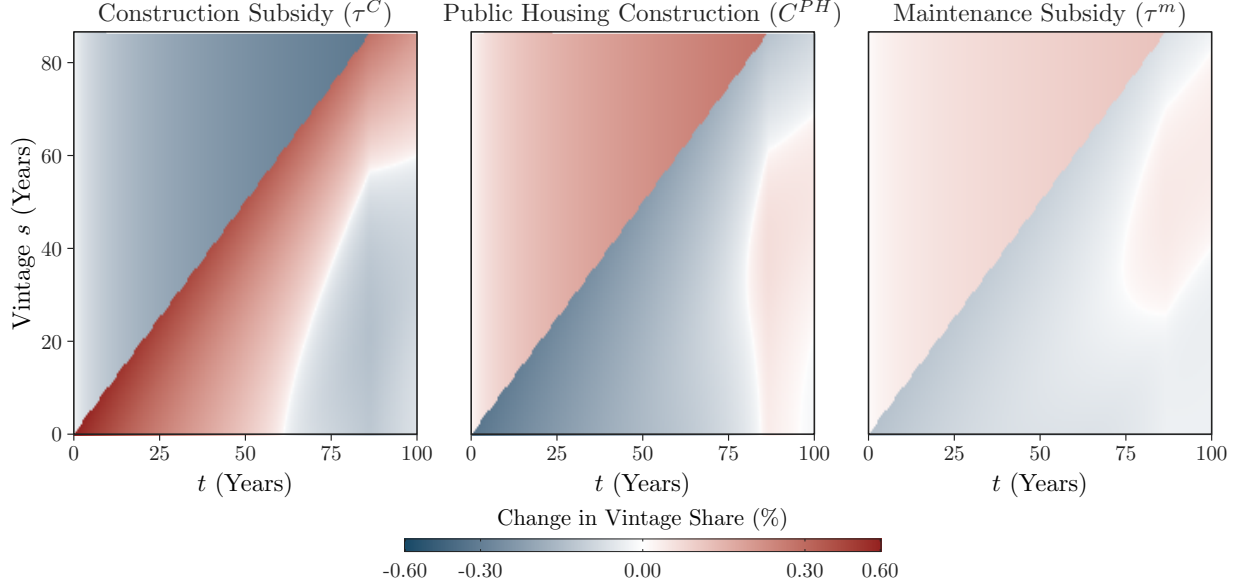
Note: Decomposition of Ω_t^i between affordability (row 1), tear-down (row 2), and vintage (row 3) effects at time horizons $t \in \{5, 25, 50\}$. The cumulative progressivity index (row 4) sums contributions of affordability, tear-down, and vintage effects. Public housing tenants are omitted.

construction subsidy (left panel) indicates a construction peak, consistent with Proposition 5. The construction peak is regressive, because the resulting imbalance between new (high-quality) units and old (low-quality) units pushes prices up at the bottom of the housing ladder.

Figure 12 shows that public housing construction (center panel) causes a construction trough, as it crowds out private housing construction. Consistent with Proposition 5, the maintenance subsidy (right panel) also causes a construction trough, which contributes positively to its short-run progressivity.

The second stark pattern from Table 6 is that tear-down effects build up slowly over time. This is because in our calibrated model, landlords do not build at the lowest housing qualities in the economy. Supply of low-quality housing is thus fully dependent on filtering. Policy can build up the stock of housing at the bottom of the ladder by incentivizing maintenance of old units, but cannot do so faster than is permitted by filtering. Thus, the combination of a progressive tear-down effect with progressive affordability and vintage effects is what ensures the gains from a maintenance subsidy materialize quickly for low-income households,

Figure 12: Vintage Share Changes and Construction Peaks and Troughs.



Note: Left panel: vintage share changes from τ^C along the transition. Center panel: vintage share changes from C^{PH} . Right panel: vintage share changes from τ^m . The x -axis is time t and the y -axis is vintages s . Shading shows magnitude of vintage share increases (red) and decreases (blue).

resulting in progressivity at all time horizons.

Additional Experiments. Can a construction subsidy targeted at affordable housing (e.g. Low-Income Housing Tax Credit) achieve progressivity? We contrast our baseline construction subsidy with an equally costly construction subsidy restricted to below-median construction. Welfare gains are higher for low-income households: 10th-percentile households gain 1.5 consumption-equivalent units per unit spent by the government on the targeted construction subsidy, compared to 0.9 for the untargeted construction subsidy. Nevertheless, 10th-percentile households prefer the maintenance subsidy. Using our progressivity index Ω_t^i , Table A2 shows that the targeted construction subsidy is more progressive than its untargeted counterpart. However, it remains regressive. The reason is that landlords do not build very-low quality housing. Thus, even a subsidy targeted to low-quality construction does not expand supply at the very bottom of the housing ladder. This supply is instead *depressed* by a regressive tear-down effect.

Next, we reproduce our main three policy counterfactuals in the model without filtering from Appendix D. We find *regressive* welfare gains from the maintenance subsidy. This is because the model without filtering fails to capture that incentivizing maintenance of old units disproportionately expands supply at the bottom of the housing ladder. Thus, it generates similar distributional implications from construction and maintenance subsidies.

This failure highlights the importance of filtering to study the progressivity of housing policy.

Finally, we assess the progressivity of additional instruments $i \in \mathcal{I}$. We consider a rent control expansion.²⁹ The expansion improves welfare for households allocated a rent-controlled unit, but generates welfare losses for households who remain in MR housing, similarly to public housing construction. Welfare losses are especially large for the lowest-income households in MR housing due to a strong affordability effect: to make-up for revenue lost at qualities $q < q^{RC}$, landlords raise prices higher up the ladder, pushing demand down and driving up the gap between $p(q^{RC})$ and the controlled rent p^{RC} .

We consider a reduction in land use regulation, captured by an unanticipated and permanent wedge $\tau^{P_0} < 0$. We set τ^{P_0} such that the implicit cost of the wedge corresponds to the cost of our main three policy counterfactuals. Table A3 shows that the resulting discounted lifetime welfare gains are progressive. While reducing the cost of land use regulation lowers prices along the ladder, which incentivizes landlords to tear-down old units, it also lowers the opportunity cost of old housing units, disincentivizing tear downs. Thus, reducing land use regulation induces a mild tear-down effect which is outweighed by a progressive affordability effect (see Table A4). However, we find the reduction in land use regulation to be less effective than the maintenance subsidy at lowering rental prices, resulting in lower welfare gains across the income distribution.

5 Conclusion

Low-income households have been hardest hit by declining housing affordability in the US. For policymakers attempting to tackle rising prices, understanding the progressivity of housing policy is therefore critical. The starting point of this paper is a simple observation: low-income households live in old, not new units. From there, we build a theory of the progressivity of housing policy. The theory sheds light on the channels through which housing policies generate regressive and progressive forces. Taking our model to the data lends credibility to our theory and enables quantitative comparisons of specific policies.

In our theoretical and quantitative results, maintenance subsidies emerge as a policy instrument more progressive than construction subsidies and more effective than public housing construction. While maintenance subsidies take up little (to no) space in the housing economics literature, they are gaining traction among policymakers. The Pennsylvania Whole-Home Repairs program, which subsidizes maintenance of housing occupied by low-income

²⁹We consider a perturbation to the minimal MR housing quality, $dq^{RC} > 0$, starting from $q^{RC} = q_{min}$. We set the controlled rent to $p^{RC} = p(q_{min})$.

households, was launched in 2022 with an initial \$125 million appropriation.³⁰ It has inspired a federal pilot program through the Whole-Home Repairs Act, a provision of the 2026 ROAD Act. Combined, our results on maintenance subsidies and the interest these subsidies have sparked in the policy space point to empirical studies of programs currently being deployed as an important avenue for future research.

References

- Abramson, B. and T. Landvoigt (2025). Curbing Rising Housing Costs: A Model-Based Policy Comparison. *Journal of Economic Perspectives* 39(3), 27–44.
- Arnott, R. J. and R. M. Braid (1997). A Filtering Model with Steady-State Housing. *Regional Science and Urban Economics* 27(4–5), 515–546.
- Baum-Snow, N. and G. Duranton (2025). Housing Supply and Housing Affordability. Technical report, National Bureau of Economic Research.
- Baum-Snow, N. and L. Han (2024). The Microgeography of Housing Supply. *Journal of Political Economy* 132(6), 1897–1946.
- Braid, R. M. (1984). The Effects of Government Housing Policies in a Vintage Filtering Model. *Journal of Urban Economics* 16(3), 272–296.
- Bratu, C., O. Harjunen, and T. Saarimaa (2023). City-Wide Effects of New Housing Supply: Evidence from Moving Chains. *Journal of Urban Economics* 133.
- Davis, M. A., W. D. Larson, S. D. Oliner, and J. Shui (2021). The Price of Residential Land for Counties, ZIP Codes, and Census Tracts in the United States. *Journal of Monetary Economics* 118, 413–431.
- Diamond, R. and W. F. Diamond (2024). Racial Differences in the Total Rate of Return on Owner-Occupied Housing. Technical report, National Bureau of Economic Research.
- Diamond, R., Z. Huang, and T. J. McQuade (2025). The Unequal Effects of Up-Zoning: Evidence from Cook County. Working paper.
- Emrath, P. (2016). Government Regulation in the Price of a New Home. Special study for housing economics, National Association of Home Builders, Washington, DC.

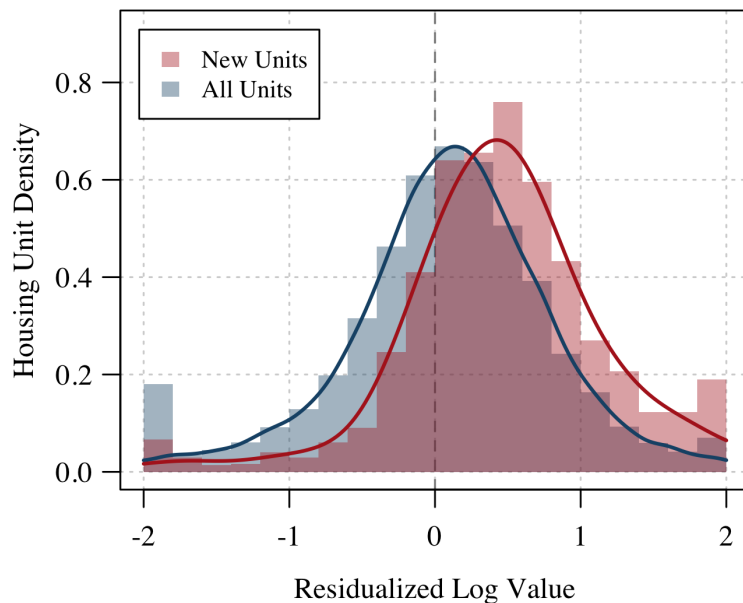
³⁰Source: Pennsylvania Senate Democrats, <https://pasenate.com/pas-celebrated-whole-home-repairs-program-becomes-national-law-with-passage-of-federal-21st-century-road-to-housing-act/>.

- Favilukis, J., P. Mabilie, and S. Van Nieuwerburgh (2023). Affordable Housing and City Welfare. *Review of Economic Studies* 90(1), 293–330.
- Glaeser, E. L. and J. Gyourko (2018). The Economic Implications of Housing Supply. *Journal of Economic Perspectives* 32(1), 3–30.
- Gyourko, J. and A. Saiz (2004). Reinvestment in the Housing Stock: The Role of Construction Costs and the Supply Side. *Journal of Urban Economics* 55(2), 238–256.
- Hauck, L. and F. Kluser (2025). Country-Wide Effects of New Housing Supply: Evidence from Moving Chains. Working paper.
- Heathcote, J., K. Storesletten, and G. L. Violante (2017). Optimal Tax Progressivity: An Analytical Framework. *Quarterly Journal of Economics* 132(4), 1693–1754.
- Hong, G. and J. Macek (2025). Redevelopment and Gentrification in General Equilibrium. Working paper.
- Kakwani, N. C. (1977). Applications of Lorenz Curves in Economic Analysis. *Econometrica* 45(3), 719–727.
- Landvoigt, T., M. Piazzesi, and M. Schneider (2015). The Housing Market(s) of San Diego. *American Economic Review* 105(4), 1371–1407.
- Liu, L., D. McManus, and E. Yannopoulos (2022). Geographic and Temporal Variation in Housing Filtering Rates. *Regional Science and Urban Economics* 93, 103758.
- Ma, L. (2025). Build What and for Whom? The Distributional Effects of Housing Supply. Working paper.
- Määtänen, N. and M. Terviö (2014). Income Distribution and Housing Prices: An Assignment Model Approach. *Journal of Economic Theory* 151, 381–410.
- Malpezzi, S. and R. K. Green (1996). What Has Happened to the Bottom of the US Housing Market? *Urban Studies* 33(10), 1807–1820.
- Mast, E. (2023). The Effect of New Market-Rate Housing Construction on the Low-Income Housing Market. *Journal of Urban Economics* 133.
- Mense, A. (2025). The Impact of New Housing Supply on the Distribution of Rents. *Journal of Political Economy: Macroeconomics* 3(1), 1–42.

- Molloy, R. (2020). The Effect of Housing Supply Regulation on Housing Affordability: A Review. *Regional Science and Urban Economics* 80, 1–5.
- Nathanson, C. G. (2025). Trickle-Down Housing Economics. Working paper.
- Nikolakoudis, G. (2025). The Economics of Segmented Housing Markets. Working paper.
- O’Flaherty, B. (1995). An Economic Theory of Homelessness and Housing. *Journal of Housing Economics* 4(1), 13–49.
- Ohls, J. C. (1975). Public Policy toward Low-Income Housing and Filtering in Housing Markets. *Journal of Urban Economics* 2(2), 144–171.
- Rosenthal, S. S. (2014). Are Private Markets and Filtering a Viable Source of Low-Income Housing? Estimates from a “Repeat Income” Model. *American Economic Review* 104(2), 687–706.
- Spader, J. (2025). Has Housing Filtering Stalled? Heterogeneous Outcomes in the American Housing Survey, 1985–2021. *Housing Policy Debate* 35(1), 3–25.
- Sweeney, J. L. (1974). A Commodity Hierarchy Model of the Rental Housing Market. *Journal of Urban Economics* 1(3), 288–323.

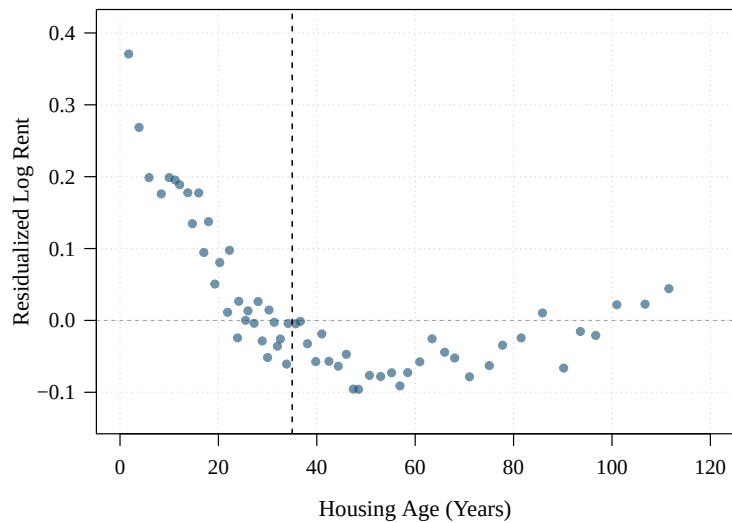
A Additional Figures and Tables

Figure A1: Owner-Occupied Housing Values for Old vs. New Housing.



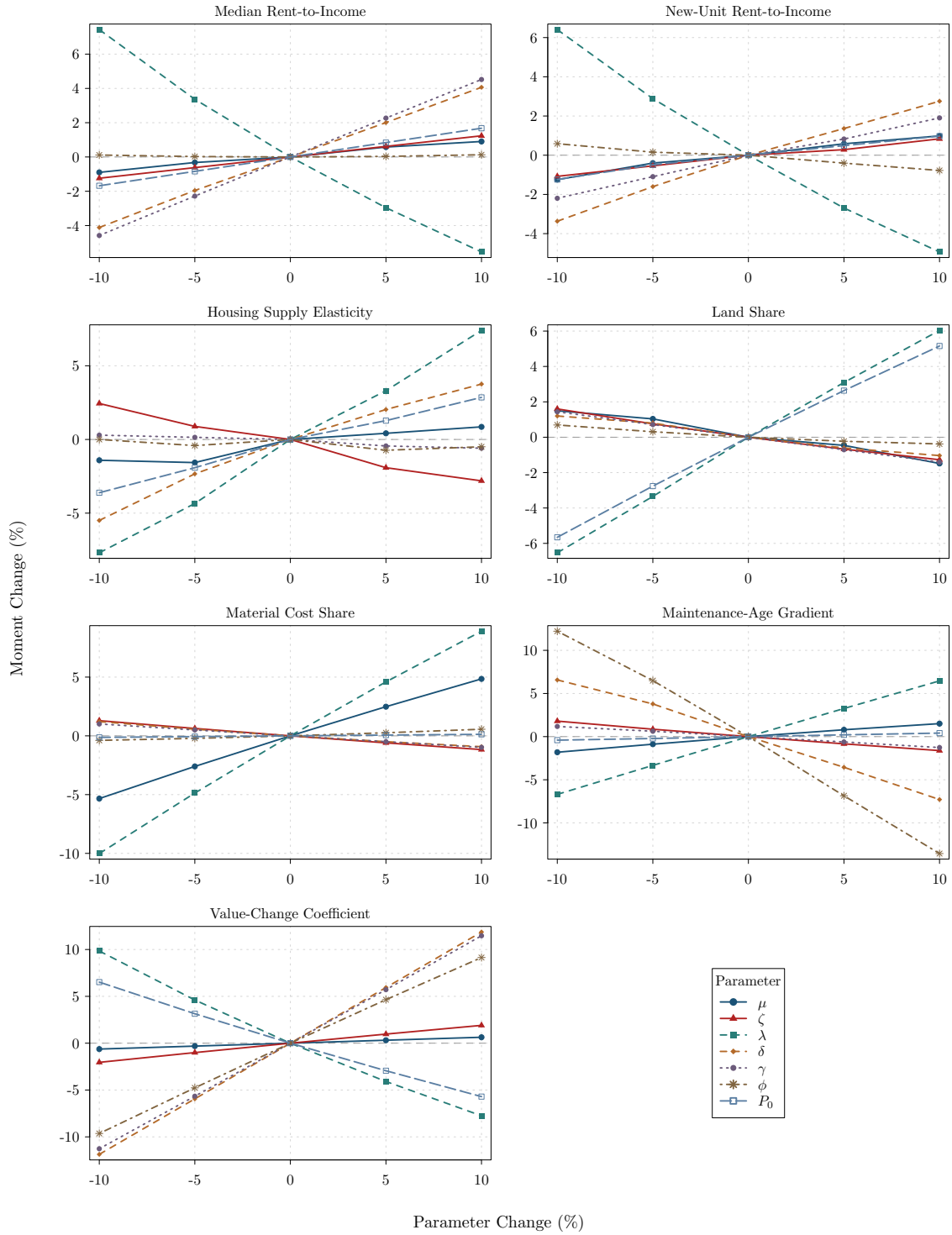
Note: Log housing value distributions of all owner-occupied units and new owner-occupied units (age < 2), residualized from MSA $\times$ year fixed effects. Source: 2001–2023 American Housing Survey (AHS).

Figure A2: Rental Prices and Housing Age.



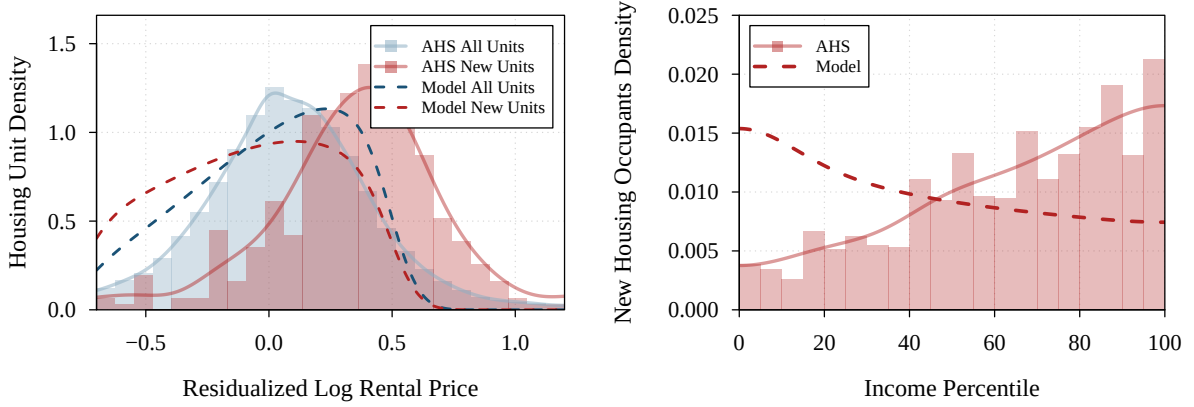
Note: Bin scatter relationship between log rental prices and housing age. Rental prices are residualized from MSA $\times$ year fixed effects. The dashed vertical line corresponds to age 35. Source: 2001–2023 AHS.

Figure A3: Calibration Sensitivity Analyses.



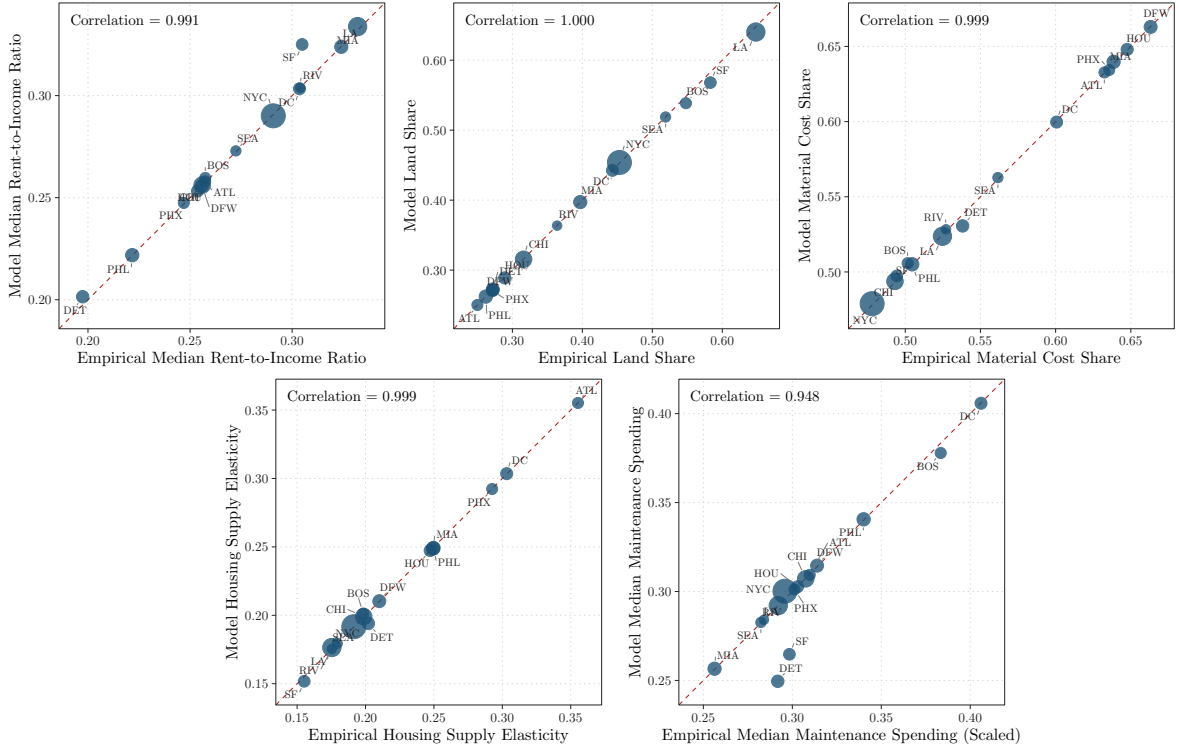
Note: Effect of perturbations to calibrated parameter values on model moment values.

Figure A4: Prices and Sorting into Old vs. New Housing: Model Without Filtering.



Note: Left panel: log rental price distributions of all units and new units in the model without filtering and in the data. Right panel: income percentile distribution of new housing occupants in the model without filtering and in the data. Rental prices are residualized from $MSA \times year$ and square footage fixed effects. Source: AHS.

Figure A5: MSA-Specific Calibration Targeted Moments.



Note: From left to right: median rent-to-income ratios, median land-to-value ratios, shares of materials in construction costs, housing supply elasticities, and maintenance spending in model and data. Markers are scaled by MSA population. Source: AHS.

Table A1: Lifetime Welfare Changes from τ^C , C^{PH} , and τ^m ($\rho = 0.02$).

Income Percentile	Construction Subsidy (τ^C)	Public Housing Construction (C^{PH})	Maintenance Subsidy (τ^m)
0.2nd	1.033	130.538	1.586
10th	1.351	-0.079	1.492
35th	1.687	-0.493	1.313
65th	1.965	-0.902	1.151
90th	2.197	-1.228	1.059

Note: Summary of discounted lifetime welfare changes from equally costly τ^C , C^{PH} , and τ^m policies. Welfare changes are expressed in consumption-equivalent terms, normalized by government spending, and discounted at the rate $\rho = 0.02$. The 0.2nd percentile of the income distribution corresponds to the median income assigned a public housing unit in the long run under the C^{PH} policy.

Table A2: Progressivity Index: Baseline vs. Targeted τ^C .

	Baseline Construction Subsidy	Targeted Construction Subsidy
<i>5 Years</i>		
Affordability Effect Contribution	0.108	0.075
Tear-Down Effect Contribution	−0.125	−0.138
Vintage Effects Contribution	−0.168	−0.181
Cumulative Progressivity Index	−0.185	−0.244
Total Progressivity Index (Ω_5^i)	−0.156	−0.094
<i>25 Years</i>		
Affordability Effect Contribution	0.083	0.061
Tear-Down Effect Contribution	−0.135	−0.156
Vintage Effects Contribution	−0.088	−0.102
Cumulative Progressivity Index	−0.140	−0.197
Total Progressivity Index (Ω_{25}^i)	−0.121	−0.049
<i>50 Years</i>		
Affordability Effect Contribution	0.073	0.055
Tear-Down Effect Contribution	−0.144	−0.170
Vintage Effects Contribution	−0.015	−0.031
Cumulative Progressivity Index	−0.085	−0.146
Total Progressivity Index (Ω_{50}^i)	−0.091	−0.021
<i>Long Run</i>		
Affordability Effect Contribution	0.075	0.055
Tear-Down Effect Contribution	−0.149	−0.171
Cumulative Progressivity Index	−0.074	−0.116
Total Progressivity Index (Ω^i)	−0.083	−0.018

Note: Decomposition of Ω_t^i between affordability, tear-down, and vintage effects at time horizons $t \in \{5, 25, 50\}$ and in the long run for baseline and targeted construction subsidies. The targeted construction subsidy is restricted to new units below the median rental price of new units. The cumulative progressivity index sums contributions of affordability, tear-down, and vintage effects.

Table A3: Lifetime Welfare Changes from $\tau^{P_0} < 0$.

Income Percentile	Land Value Subsidy (τ^L)
0.2nd	1.006
10th	0.980
35th	0.922
65th	0.860
90th	0.809

Note: Summary of discounted lifetime welfare changes from $\tau^{P_0} < 0$ policy equally costly to τ^C , C^{PH} , and τ^m . Welfare changes are expressed in consumption-equivalent terms, normalized by government spending, and discounted at the rate $\rho = 0.05$. The 0.2nd percentile of the income distribution corresponds to the median income assigned a public housing unit in the long run under the C^{PH} policy.

Table A4: Progressivity Index: $\tau^{P_0} < 0$.

	Land Use Regulation Reduction ($\tau^{P_0} < 0$)
<i>5 Years</i>	
Affordability Effect Contribution	0.053
Tear-Down Effect Contribution	0.003
Vintage Effects Contribution	−0.003
Cumulative Progressivity Index	0.053
Total Progressivity Index (Ω_5^i)	0.048
<i>25 Years</i>	
Affordability Effect Contribution	0.055
Tear-Down Effect Contribution	−0.004
Vintage Effects Contribution	−0.006
Cumulative Progressivity Index	0.044
Total Progressivity Index (Ω_{25}^i)	0.039
<i>50 Years</i>	
Affordability Effect Contribution	0.055
Tear-Down Effect Contribution	−0.008
Vintage Effects Contribution	−0.002
Cumulative Progressivity Index	0.045
Total Progressivity Index (Ω_{50}^i)	0.036
<i>Long Run</i>	
Affordability Effect Contribution	0.055
Tear-Down Effect Contribution	−0.013
Cumulative Progressivity Index	0.041
Total Progressivity Index (Ω^i)	0.038

Note: Decomposition of Ω_t^i between affordability, tear-down, and vintage effects at time horizons $t \in \{5, 25, 50\}$ and in the long run for $\tau^{P_0} < 0$ policy equally costly to τ^C , C^{PH} , and τ^m . The cumulative progressivity index sums contributions of affordability, tear-down, and vintage effects.

Table A5: Model Without Filtering Calibration Summary.

Parameter	Value	Moment	Model	Data	Source
r	0.050	—	—	—	—
θ	0.027	Log Maintenance–Age Coefficient	0.027	0.027	AHS
P_0	4.11	Median Land-to-Value	0.424	0.443	Davis et al. (2021)
μ	3.35	Median Rent-to-Income	0.274	0.291	AHS
ζ	7,871.64	Share of Materials in Construction Costs	0.490	0.525	RSMMeans
λ	2.45	Housing Supply Elasticity	0.211	0.199	Baum-Snow and Han (2024)
ψ	0.139	Annual Construction Rate	0.015	0.016	Filtering model

Note: Summary of calibration of the model without filtering with parameters, calibrated parameter values, moments used to inform parameters, model moment values, data moment values, and data source. Horizontal lines group parameters with the moments used to inform them.

B Theoretical Appendix

B.1 Proof of Lemma 1

Proof. First, it is useful to establish that $p_t(q)$ is increasing on $\mathcal{Q}_t^{MR}$. Suppose that it is not. Then there exists $q_0, q_1 \in \mathcal{Q}_t^{MR}$, with q_0 in the interior of $\mathcal{Q}_t^{MR}$ and $q_0 < q_1$, such that $p_t(q_0) \geq p_t(q_1)$. In a regular equilibrium, $p_t(\cdot)$ is continuous. It follows that no household chooses housing in a neighborhood $(q_0 - \epsilon, q_0 + \epsilon)$ for some $\epsilon > 0$. But then

$$0 = \int_{w_t^{MR}}^{\bar{w}} \mathbf{1}_{q_{wt} \geq q_0 - \epsilon} \mathbf{1}_{q_{wt} \leq q_0 + \epsilon} dF(w) < \int_{q_0 - \epsilon}^{q_0 + \epsilon} h_t(q) dq,$$

violating the MR housing market clearing condition (6). By contradiction $p_t(q)$ is increasing on $\mathcal{Q}_t^{MR}$.

Having established that $p_t(q)$ is increasing, we show there must be positive assortative matching of MR units. Consider the household problem (1). Suppose a household with income w_0 prefers $q_1 > q_0$ to q_0 , with $q_0, q_1 \in \mathcal{Q}_t^{MR}$. For any $w_1 > w_0$, we have

$$\frac{w_1 - p_t(q_1)}{w_1 - p_t(q_0)} > \frac{w_0 - p_t(q_1)}{w_0 - p_t(q_0)}. \quad (\text{A1})$$

This ensures

$$\begin{aligned}
u(w_1 - p_t(q_1), q_1) &= (w_1 - p_t(q_1)) q_1 \\
&> (w_1 - p_t(q_0)) \frac{w_0 - p_t(q_1)}{w_0 - p_t(q_0)} q_1 \\
&\geq (w_1 - p_t(q_0)) \frac{w_0 - p_t(q_0)}{w_0 - p_t(q_0)} q_0 \\
&= (w_1 - p_t(q_0)) q_0 \\
&= u(w_1 - p_t(q_0), q_0),
\end{aligned}$$

where the first inequality follows from (A1) and the second inequality follows by assumption on the preferences of the household with income w_0 . This establishes positive assortative matching. This allows us to rewrite (6) as

$$\int_{\tilde{w}_t(q)}^{\bar{w}} dF(w) = \int_q^1 h_t(x) dx.$$

Integrating the LHS yields

$$1 - F(\tilde{w}_t(q)) = \int_q^1 h_t(x) dx. \quad (\text{A2})$$

Evaluating this expression at $q = \inf\{\mathcal{Q}_t^{MR}\}$ and combining with the non-MR housing market clearing condition (7) yields

$$1 = \int_{\underline{q}}^1 h_t^{PH}(x) dx + \int_{\underline{q}}^1 h_t(x) dx, \quad (\text{A3})$$

using $\tilde{w}_t(\inf\{\mathcal{Q}_t^{MR}\}) = w_t^{MR}$, which follows from positive assortative matching. Combining (A2) and (A3) yields

$$F(\tilde{w}_t(q)) = \int_{\underline{q}}^1 h_t^{PH}(x) dx + \int_{\underline{q}}^q h_t(x) dx,$$

which we invert to complete the proof. □

B.2 Proof of Lemma 2

Proof. First, it is useful to establish that $\tilde{w}_t(q)$ is continuous. Starting from Lemma 1 and differentiating yields

$$\tilde{w}'_t(q) = \frac{1}{f(\tilde{w}_t(q))} h_t(q),$$

which is well defined. As a differentiable function, $\tilde{w}_t(q)$ is continuous.

We have established in Appendix B.1 that $p_t(q)$ is increasing on $\mathcal{Q}_t^{MR}$. Next we show that it is differentiable and characterize its slope. Consider any q in the interior of $\mathcal{Q}_t^{MR}$. For small enough $\epsilon > 0$, the household problem (1) requires that

$$u(\tilde{w}_t(q + \epsilon) - p_t(q + \epsilon), q + \epsilon) \geq u(\tilde{w}_t(q + \epsilon) - p_t(q), q),$$

which rewrites as

$$p_t(q + \epsilon)(q + \epsilon) - p_t(q)q \leq \tilde{w}_t(q + \epsilon)\epsilon,$$

and as

$$\frac{p_t(q + \epsilon) - p_t(q)}{\epsilon} \leq \frac{\tilde{w}_t(q + \epsilon) - p_t(q)}{q + \epsilon}.$$

Similarly, we have

$$u(\tilde{w}_t(q) - p_t(q), q) \geq u(\tilde{w}_t(q) - p_t(q + \epsilon), q + \epsilon),$$

which rewrites as

$$\frac{p_t(q + \epsilon) - p_t(q)}{\epsilon} \geq \frac{\tilde{w}_t(q) - p_t(q)}{q + \epsilon}.$$

This implies

$$\lim_{\epsilon \rightarrow 0^+} \frac{p_t(q + \epsilon) - p_t(q)}{\epsilon} = \frac{\tilde{w}_t(q) - p_t(q)}{q},$$

where we use the continuity of $\tilde{w}_t(q)$. Because this is also true for $\epsilon < 0$, we have

$$\lim_{\epsilon \rightarrow 0} \frac{p_t(q) - p_t(q - \epsilon)}{\epsilon} = \frac{\tilde{w}_t(q) - p_t(q)}{q}.$$

This establishes the differentiability of $p_t(q)$ and characterizes its slope as

$$p'_t(q) = \frac{u_q(\tilde{w}_t(q) - p_t(q), q)}{u_c(\tilde{w}_t(q) - p_t(q), q)} = \frac{\tilde{w}_t(q) - p_t(q)}{q}. \quad (\text{A4})$$

Note that we could have derived (A4) much more directly from the first-order conditions of (1), but that this would have assumed differentiability of $p_t(q)$ instead of proving it. This completes the proof. $\square$

B.3 Proof of Lemma 3

Proof. It follows directly from (3) that $m(q) \in \{0, \mathcal{M}(q)\}$, with landlords maintaining only as much as necessary. Consider the decision between maintaining a housing unit for an additional instant $\Delta > 0$ before tearing it down and tearing it down immediately. To first order, it is optimal to maintain if and only if

$$\Delta(p_t(q) - (1 - \tau^m)\mathcal{M}(q)) + e^{-r\Delta}(\underline{P}_{t+\Delta} - \tau^{TD}) \geq \underline{P}_t - \tau^{TD}.$$

Using $e^{-r\Delta} \approx 1 - r\Delta$ to first order, this rewrites as

$$\Delta(p_t(q) - (1 - \tau^m)\mathcal{M}(q)) + \underline{P}_{t+\Delta} - \underline{P}_t \geq r\Delta(\underline{P}_{t+\Delta} - \tau^{TD}).$$

Taking the limit $\Delta \rightarrow 0$, a necessary and sufficient condition for optimality of $m(q) = \mathcal{M}(q)$ is therefore

$$\Phi_t(q) \equiv p_t(q) - (1 - \tau^m)\mathcal{M}(q) + r\tau^{TD} - r\underline{P}_t + \dot{\underline{P}}_t \geq 0.$$

The free-entry condition (4) implies that $\underline{P}_t = (1 + \tau^{P_0})P_0$, so that $\dot{\underline{P}}_t = 0$. We recover (9). The maintenance policy is then characterized by

$$m_t(q) = \begin{cases} \mathcal{M}(q) & \text{for } q : \Phi_t(q) \geq 0 \\ 0 & \text{for } q : \Phi_t(q) < 0 \end{cases}.$$

We note that $\Phi_t(q)$ is an increasing function. Indeed $p_t(q)$ is constant for $q < q^{RC}$, potentially jumps up at q^{RC} , and is increasing for $q \geq q^{RC}$ as established by Lemma 2, while $(1 - \tau^m)\mathcal{M}(q)$ is decreasing. It follows that there exists a tear-down cutoff $q_{min,t}$ defined as

$$q_{min,t} = \inf\{q : \Phi_t(q) \geq 0\}$$

satisfying

$$q \geq q_{min,t} \implies \Phi_t(q) \geq 0, \quad q < q_{min,t} \implies \Phi_t(q) < 0.$$

This completes the proof. $\square$

B.4 Proof of Theorem 1

Proof. We start by deriving necessary conditions for a steady-state regular equilibrium. We then show these necessary conditions are sufficient for equilibrium. We build a function G mapping $\hat{p}$, the price of the lowest MR unit, to an equilibrium residual. We use G to establish existence and uniqueness.

Necessary Conditions. First, we derive steady-state assignment of MR units. We start by computing steady-state housing mass functions and construction. We note from the KFE (5) that in steady state, the housing mass functions satisfy

$$\frac{\partial h(q)}{\partial q} = 0, \quad \frac{\partial h^{PH}(q)}{\partial q} = 0$$

for all q such that $m(q) \geq \mathcal{M}(q)$. The boundary conditions for $h_t(q)$ and $h_t^{PH}(q)$, together with the maintenance policy derived in Lemma 3, then imply

$$h(q) = \frac{C}{\delta}, \quad h^{PH}(q) = \frac{C^{PH}}{\delta} \tag{A5}$$

for $q \geq q_{min}$ and

$$h(q) = h^{PH}(q) = 0 \tag{A6}$$

for $q < q_{min}$. Evaluating $\tilde{w}(q)$ at $q = 1$, yields by Lemma 1

$$\bar{w} = F^{-1} \left(\int_{\underline{q}}^1 h^{PH}(x) dx + \int_{\underline{q}}^1 h(x) dx \right).$$

Inverting and plugging in for (A5) and (A6) yields

$$1 = \frac{C^{PH}}{\delta} (1 - q_{min}) + \frac{C}{\delta} (1 - q_{min}) = \frac{C + C^{PH}}{\delta} (1 - q_{min})$$

so that

$$C^A = \frac{\delta}{1 - q_{min}}. \quad (\text{A7})$$

By Lemma 1, steady-state assignment of MR units then satisfies

$$\tilde{w}(q) = F^{-1} \left(C^{PH} \frac{1 - q}{\delta} + \frac{q - q_{min}}{1 - q_{min}} \right). \quad (\text{A8})$$

Positive assortative matching holds for any q_{min} , with

$$\tilde{w}'(q) = \frac{1}{f(\tilde{w}(q))} \left(\frac{1}{1 - q_{min}} - \frac{C^{PH}}{\delta} \right) \geq \frac{1}{f(\tilde{w}(q))} \left(\frac{1}{1 - \underline{q}} - \frac{C^{PH}}{\delta} \right) > 0$$

by Assumption 1.

Second, we derive the steady-state rental price schedule for MR units. Lemma 2 defines a linear ordinary differential equation for $p(q)$. It can be expressed as

$$p'(x)x + p(x) = \tilde{w}(x).$$

Integrating from any $\hat{q}$ to q yields

$$p(q)q - p(\hat{q})\hat{q} = \int_{\hat{q}}^q \tilde{w}(x)dx,$$

which rearranges as

$$p(q) = \frac{\hat{q}}{q}p(\hat{q}) + \frac{1}{q} \int_{\hat{q}}^q \tilde{w}(x)dx. \quad (\text{A9})$$

Third, we derive necessary conditions from the steady-state maintenance policy. It is first useful to define q^* as

$$q^* = \inf \{ q : p^{RC} - (1 - \tau^m)\mathcal{M}(q) - r((1 + \tau^{P_0})P_0 - \tau^{TD}) \geq 0 \},$$

which is the quality at which a landlord paid the rent-controlled rent p^{RC} would choose to tear down.

We note that landlords never maintain units with quality $q < \bar{q}$ where we define

$$\bar{q} = \begin{cases} \underline{q} & \text{if } q^* < \underline{q} \\ q^{RC} & \text{if } q^* > q^{RC} \\ q^* & \text{otherwise} \end{cases}.$$

Hence $q_{min} \in [\bar{q}, 1]$.

We denote by $\hat{p}$ the lowest price paid for a MR unit. By construction, $\hat{p}$ is the price for housing of quality $\hat{q} = \max\{q_{min}, q^{RC}\}$. Given $\hat{p} > r((1 + \tau^{P_0})P_0 - \tau^{TD})$, the maintenance policy pins down q_{min} as

$$q_{min} = \max \left\{ \frac{(1 - \tau^m)\gamma}{(1 - \tau^m)\gamma + \hat{p} - r((1 + \tau^{P_0})P_0 - \tau^{TD})}, \bar{q} \right\}, \quad (\text{A10})$$

where the first element in the max operator uses the expression for $\mathcal{M}(q)$ in (2). To see this, consider the case where $\hat{p} \leq p^{RC}$. Then, the rent control policy implies $p(q) = \hat{p}$ for $q < q^{RC}$ and hence

$$q_{min} = \inf\{q : \Phi(q) \geq 0\}$$

is satisfied by (A10). Consider instead the case where $\hat{p} > p^{RC}$. Then either

$$q_{min} = \inf\{q : \Phi(q) \geq 0\} = \frac{(1 - \tau^m)\gamma}{(1 - \tau^m)\gamma + \hat{p} - r((1 + \tau^{P_0})P_0 - \tau^{TD})} \geq q^{RC} \geq \bar{q},$$

satisfying (A10) or $\Phi(q^{RC}) \geq 0$ in which case

$$q_{min} = \inf\{q : \Phi(q) \geq 0\} = \bar{q} \geq \frac{(1 - \tau^m)\gamma}{(1 - \tau^m)\gamma + \hat{p} - r((1 + \tau^{P_0})P_0 - \tau^{TD})},$$

again satisfying (A10).

Fourth, we derive the steady-state free-entry condition. We start by integrating the HJB (3). Fix a t and consider $s \geq 0$. For $s \leq \bar{s}_t = \inf\{s : 1 - \delta s = q_{min,t+s}\}$, we have

$$rP_{t+s}(q_{t+s}) = p_{t+s}(q_{t+s}) - (1 - \tau^m)\mathcal{M}(q_{t+s}) + \dot{P}_{t+s}(q_{t+s}) - \delta P'_{t+s}(q_{t+s}) \quad (\text{A11})$$

and the law of motion for quality $\dot{q}_{t+s} = -\delta$. (A11) rewrites as

$$\begin{aligned} & r e^{-rs} P_{t+s}(q_t - \delta s) - e^{-rs} \left(\dot{P}_{t+s}(q_t - \delta s) - \delta P'_{t+s}(q_t - \delta s) \right) \\ &= e^{-rs} (p_{t+s}(q_t - \delta s) - (1 - \tau^m) \mathcal{M}(q_t - \delta s)) \end{aligned}$$

which we can integrate from 0 to $\bar{s}_t$ to obtain

$$P_t(q_t) - e^{-rT} P_{t+T}(q_{min,t+T}) = \int_0^{\bar{s}_t} e^{-rs} (p_{t+s}(q_t - \delta s) - (1 - \tau^m) \mathcal{M}(q_t - \delta s)) ds.$$

Plugging in for

$$P_{t+T}(q_{min,t+T}) = (1 + \tau^{P_0}) P_0 - \tau^{TD}$$

and evaluating at $q_t = 1$ delivers the first equality in (10). In steady state, we have

$$P(q) = \int_0^{\frac{q-q_{min}}{\delta}} e^{-rs} (p(q - \delta s) - (1 - \tau^m) \mathcal{M}(q - \delta s)) ds + e^{-r\frac{q-q_{min}}{\delta}} ((1 + \tau^{P_0}) P_0 - \tau^{TD}).$$

A change of variable from s to $x = q - \delta s$ then implies

$$P(q) = (1 + \tau^{P_0}) P_0 - \tau^{TD} + \int_{q_{min}}^q e^{-r\frac{q-x}{\delta}} \frac{p(x) - (1 - \tau^m) \mathcal{M}(x) - r((1 + \tau^{P_0}) P_0 - \tau^{TD})}{\delta} dx. \quad (\text{A12})$$

In a steady state, the free-entry condition must hold with equality. To see this, suppose otherwise, that there is no private construction. This violates market clearing. Indeed, this suggests

$$\bar{w} = F^{-1} \left(\frac{C^{PH}}{\delta} (1 - q_{min}) \right) < F^{-1} \left(\frac{C^{PH}}{\delta} (1 - \bar{q}) \right) < F^{-1}(1) = \bar{w},$$

where the first inequality follows from $q_{min} \geq \bar{q}$ and the second from Assumption 1.

Thus, the free-entry condition is given by

$$(1 - \tau^C) \kappa(C^A) + \tau^{TD} = \frac{1}{\delta} \int_{q_{min}}^1 e^{-r\frac{1-q}{\delta}} \left[p(q) - (1 - \tau^m) \mathcal{M}(q) - r((1 + \tau^{P_0}) P_0 - \tau^{TD}) \right] dq.$$

Fifth, we impose that rental prices leave households with non-negative consumption.

Specifically, we only impose this for the lowest-income household renting MR housing as

$$\tilde{w}(\hat{q}) \geq \hat{p},$$

and show that this is sufficient to guarantee non-negative consumption for all households when we establish sufficiency of our necessary conditions.

Sixth and last, we note that market clearing requires $q_{min} < 1$, as clear from (A6). This in turn requires

$$\hat{p} > \underline{p} \equiv r((1 + \tau^{P_0})P_0 - \tau^{TD}).$$

Hence, we have derived the necessary conditions

$$\begin{aligned} p(\hat{q}) &= \hat{p}, \\ \hat{q} &= \max\{q_{min}, q^{RC}\}, \\ \tilde{w}(q) &= F^{-1}\left(C^{PH}\frac{1-q}{\delta} + \frac{q-q_{min}}{1-q_{min}}\right) \text{ for } q \geq \hat{q}, \\ p(q) &= \min\{\hat{p}, p^{RC}\} \text{ for } q_{min} \leq q < \hat{q}, \\ p(q) &= \frac{\hat{q}}{q}\hat{p} + \frac{1}{q}\int_{\hat{q}}^q \tilde{w}(x)dx \text{ for } q \geq \hat{q}, \\ q_{min} &= \max\left\{\frac{(1-\tau^m)\gamma}{(1-\tau^m)\gamma + \hat{p} - r((1+\tau^{P_0})P_0 - \tau^{TD})}, \bar{q}\right\}, \\ (1-\tau^C)\kappa\left(\frac{\delta}{1-q_{min}}\right) + \tau^{TD} &= \frac{1}{\delta}\int_{q_{min}}^1 e^{-r\frac{1-q}{\delta}}\left[p(q) - (1-\tau^m)\mathcal{M}(q) \right. \\ &\quad \left. - r((1+\tau^{P_0})P_0 - \tau^{TD})\right]dq, \\ \hat{p} &> \underline{p}, \\ \tilde{w}(\hat{q}) &\geq \hat{p}. \end{aligned} \tag{A13}$$

We now show that they are sufficient to characterize a steady-state regular equilibrium.

Sufficiency of Necessary Conditions. Consider any

$$\{\hat{p}, \hat{q}, q_{min}, \{\tilde{w}(q)\}_{q \geq \hat{q}}, \{p(q)\}_{q \geq q_{min}}\}$$

satisfying (A13). First, we argue it characterizes all objects of the steady-state regular equilibrium.

The maintenance policy $m(q)$ is given by q_{min} , together with Lemma 3. Aggregate construction C^A is given by q_{min} , together with (A7), and private construction is $C = C^A - C^{PH}$. The MR housing cutoff w^{MR} is given by (7), given (A5) and (A6) and private construction C . The MR housing choice set is $q \in [\hat{q}, 1]$. Housing mass functions are given by (A5) and (A6) and construction C . The housing value function is given by $p(q)$ and q_{min} , together with (A12). Finally, $\tilde{w}(q)$ and $p(q)$ for $q \geq \hat{q}$ characterize allocations $\{(c_w, q_w)\}_w$ for households with $w \geq w^{MR}$.

Second, we show that the necessary conditions are sufficient to satisfy all equilibrium conditions. Our necessary conditions satisfy

$$p(q) = \frac{\hat{q}}{q} \hat{p} + \frac{1}{q} \int_{\hat{q}}^q \tilde{w}(x) dx \text{ for } q \geq \hat{q}$$

and hence the first-order condition of the MR housing household problem

$$p'(q) = \frac{u_q}{u_c} = \frac{\tilde{w}(q) - p(q)}{q}. \quad (\text{A14})$$

To show they are sufficient for household optimality, we first show the problem is strictly concave. The objective is

$$(w - p(q))q.$$

The second derivative with respect to q is then given by

$$-p'(q) - p''(q)q - p'(q),$$

provided $p''(q)$ is well defined.

Differentiating (A14) yields

$$p''(q) = \frac{q(\tilde{w}'(q) - p'(q)) - (\tilde{w}(q) - p(q))}{q^2},$$

which is well defined. Therefore, the second derivative of the household objective is

$$\begin{aligned} & -p'(q) - p''(q)q - p'(q) \\ &= -2p'(q) - \tilde{w}'(q) + p'(q) + \frac{\tilde{w}(q) - p(q)}{q} \\ &= -\tilde{w}'(q) < 0, \end{aligned}$$

where the strict inequality follows from positive assortative matching. Hence the problem is strictly concave. To show our necessary conditions are sufficient for household optimality, we must also show that $c_w \geq 0$ for all $w \geq w^{MR}$, or equivalently that $\tilde{w}(q) - p(q) \geq 0$ for all $q \geq \hat{q}$. We argue that this must be the case whenever $w^{MR} = \tilde{w}(\hat{q}) \geq \hat{p}$. Consider any $q \geq \hat{q}$. We have

$$\begin{aligned}
\tilde{w}(q) - p(q) &= \tilde{w}(q) - \frac{\hat{q}}{q}\hat{p} - \frac{1}{q} \int_{\hat{q}}^q \tilde{w}(x) dx \\
&\geq \tilde{w}(q) - \frac{\hat{q}}{q}\tilde{w}(\hat{q}) - \frac{1}{q} \int_{\hat{q}}^q \tilde{w}(x) dx \\
&= \tilde{w}(q) - \tilde{w}(\hat{q}) - \frac{1}{q} \int_{\hat{q}}^q (\tilde{w}(x) - \tilde{w}(\hat{q})) dx \\
&\geq \tilde{w}(q) - \tilde{w}(\hat{q}) - \frac{1}{q} \int_{\hat{q}}^q (\tilde{w}(q) - \tilde{w}(\hat{q})) dx \\
&= (\tilde{w}(q) - \tilde{w}(\hat{q})) \left(\frac{\hat{q}}{q} \right) \\
&\geq 0
\end{aligned}$$

where the first inequality holds by assumption and the second and third follow from positive assortative matching. Our necessary conditions are thus sufficient for household optimality. By construction, our necessary conditions are sufficient for all other equilibrium conditions. We now establish that there exists a unique

$$\{\hat{p}, \hat{q}, q_{min}, \{\tilde{w}(q)\}_{q \geq \hat{q}}, \{p(q)\}_{q \geq q_{min}}\}$$

which satisfies (A13).

Existence and Uniqueness. To show existence and uniqueness, we define a function G mapping candidate values of $\hat{p}$ to a residual equal to 0 in equilibrium. We define $G : (\underline{p}, \infty) \rightarrow$

$\mathbb{R}$ as

$$\begin{aligned}
G(\hat{p}) = & \frac{1}{\delta} \mathbf{1}_{q_{\min}(\hat{p}) < q^{RC}} \int_{q_{\min}(\hat{p})}^{q^{RC}} e^{-r \frac{1-q}{\delta}} \left[\min\{\hat{p}, p^{RC}\} - (1 - \tau^m) \mathcal{M}(q) \right. \\
& \left. - r \left((1 + \tau^{P_0}) P_0 - \tau^{TD} \right) \right] dq \\
& + \frac{1}{\delta} \int_{\hat{q}(\hat{p})}^1 e^{-r \frac{1-q}{\delta}} \left[\frac{\hat{q}(\hat{p})}{q} \hat{p} + \frac{1}{q} \int_{\hat{q}(\hat{p})}^q \tilde{w}(x; \hat{p}) dx \right. \\
& \left. - (1 - \tau^m) \mathcal{M}(q) - r \left((1 + \tau^{P_0}) P_0 - \tau^{TD} \right) \right] dq \\
& - (1 - \tau^C) \kappa \left(\frac{\delta}{1 - q_{\min}(\hat{p})} \right) - \tau^{TD},
\end{aligned}$$

with

$$\begin{aligned}
q_{\min}(\hat{p}) &= \max \left\{ \frac{(1 - \tau^m) \gamma}{(1 - \tau^m) \gamma + \hat{p} - r((1 + \tau^{P_0}) P_0 - \tau^{TD})}, \bar{q} \right\}, \\
\hat{q}(\hat{p}) &= \max\{q_{\min}(\hat{p}), q^{RC}\}, \\
\tilde{w}(q; \hat{p}) &= F^{-1} \left(C^{PH} \frac{1 - q}{\delta} + \frac{q - q_{\min}(\hat{p})}{1 - q_{\min}(\hat{p})} \right).
\end{aligned}$$

By construction,

$$\{\hat{p}, \hat{q}, q_{\min}, \{\tilde{w}(q)\}_{q \geq \hat{q}}, \{p(q)\}_{q \geq q_{\min}}\}$$

characterizes an equilibrium if and only if $G(\hat{p}) = 0$ and $\tilde{w}(\hat{q}(\hat{p}); \hat{p}) \geq \hat{p}$.

We now argue that $G(\cdot)$ is continuous, that

$$\lim_{\hat{p} \rightarrow \underline{p}} G(\hat{p}) < 0, \tag{A15}$$

and that $G(\underline{w}) > 0$. Assumption 2 ensures $\underline{w} > \underline{p}$. Because $\tilde{w}(q; \hat{p}) \geq \underline{w}$ for all q , we have $\tilde{w}(\hat{q}(\hat{p}); \hat{p}) \geq \hat{p}$ for any $\hat{p} \leq \underline{w}$, and this establishes existence of an equilibrium with $\hat{p} \in (\underline{p}, \underline{w})$. We then turn to proving uniqueness of the equilibrium.

To see that $G(\cdot)$ is continuous, we simply observe that $G(\hat{p})$ is a continuous composition of continuous functions of $\hat{p}$.

To see that (A15) holds, observe that

$$\lim_{\hat{p} \rightarrow \underline{p}} G(\hat{p}) = \lim_{q_{\min} \rightarrow 1} -(1 - \tau^C) \kappa \left(\frac{\delta}{1 - q_{\min}} \right) - \tau^{TD} < 0,$$

where we use that the first two terms of $G(\hat{p})$ vanish as $q_{\min}(\hat{p}) \rightarrow 1$.

To see that $G(\underline{w}) > 0$, we rely on Assumption 2 and on Assumption 3. We have

$$\underline{w} - (1 - \tau^m)\mathcal{M}(q^{RC}) - r((1 + \tau^{P_0})P_0 - \tau^{TD}) > 0$$

guaranteeing $\Phi(q^{RC}) \geq 0$ and hence $\hat{q}(\underline{w}) = q^{RC}$ by Assumption 2. We then have

$$\begin{aligned} G(\underline{w}) &\geq \frac{1}{\delta} \int_{q^{RC}}^1 e^{-r\frac{1-q}{\delta}} \left[\frac{q^{RC}}{q} \underline{w} + \frac{1}{q} \int_{q^{RC}}^q \tilde{w}(x; \underline{w}) dx \right. \\ &\quad \left. - (1 - \tau^m)\mathcal{M}(q) - r((1 + \tau^{P_0})P_0 - \tau^{TD}) \right] dq \\ &\quad - (1 - \tau^C)\kappa \left(\frac{\delta}{1 - q_{min}(\underline{w})} \right) - \tau^{TD}, \end{aligned}$$

because the first term of $G(\underline{w})$ is non-negative. This rewrites as

$$\begin{aligned} G(\underline{w}) &\geq \frac{1}{\delta} \int_{q^{RC}}^1 e^{-r\frac{1-q}{\delta}} \left[\underline{w} + \frac{1}{q} \int_{q^{RC}}^q (\tilde{w}(x; \underline{w}) - \underline{w}) dx \right. \\ &\quad \left. - (1 - \tau^m)\mathcal{M}(q) - r((1 + \tau^{P_0})P_0 - \tau^{TD}) \right] dq \\ &\quad - (1 - \tau^C)\kappa \left(\frac{\delta}{1 - q_{min}(\underline{w})} \right) - \tau^{TD}. \end{aligned}$$

Using Assumption 2 and $\mathcal{M}(q)$ decreasing, we then have

$$G(\underline{w}) > \frac{1}{\delta} \int_{q^{RC}}^1 e^{-r\frac{1-q}{\delta}} \frac{1}{q} \int_{q^{RC}}^q (\tilde{w}(x; \underline{w}) - \underline{w}) dx dq - (1 - \tau^C)\kappa \left(\frac{\delta}{1 - q_{min}(\underline{w})} \right) - \tau^{TD}.$$

Observing that $q^{RC} \geq q_{min}(\underline{w})$ and that $q \leq 1$, we have, using our expression for $\tilde{w}(x; \underline{w})$,

$$G(\underline{w}) > \frac{1}{\delta} \int_{q^{RC}}^1 e^{-r\frac{1-q}{\delta}} \int_{q^{RC}}^q \left(F^{-1} \left(\frac{x - q^{RC}}{1 - q^{RC}} \right) - \underline{w} \right) dx dq - (1 - \tau^C)\kappa \left(\frac{\delta}{1 - q^{RC}} \right) - \tau^{TD}.$$

We apply a change of variables from q to $t = (1 - q)/\delta$ in the outer integral to obtain

$$\begin{aligned} G(\underline{w}) &> \int_0^{\frac{1-q^{RC}}{\delta}} e^{-rt} \int_{q^{RC}}^{1-\delta t} \left(F^{-1} \left(\frac{x - q^{RC}}{1 - q^{RC}} \right) - \underline{w} \right) dx dt - (1 - \tau^C)\kappa \left(\frac{\delta}{1 - q^{RC}} \right) - \tau^{TD} \\ &> 0, \end{aligned}$$

where the last inequality follows from Assumption 3. This establishes $G(\underline{w}) > 0$ and existence.

We now show uniqueness. Suppose, towards a contradiction, that there exist two equilib-

ria. Necessarily, the equilibria must feature $\hat{p}_1 \neq \hat{p}_0$, as $\hat{p}$ pins down all equilibrium objects through (A13). Without loss, assume that $\hat{p}_1 > \hat{p}_0$. We first observe that both $q_{min}(\hat{p})$ and $\hat{q}(\hat{p})$ are non-increasing in $\hat{p}$, and that $\tilde{w}(q; \hat{p})$ is non-decreasing in $\hat{p}$. Next, we consider all three terms of $G(\cdot)$ one by one.

Consider the first term,

$$G_1(\hat{p}) \equiv \frac{1}{\delta} \mathbf{1}_{q_{min}(\hat{p}) < q^{RC}} \int_{q_{min}(\hat{p})}^{q^{RC}} e^{-r \frac{1-q}{\delta}} \left[\min\{\hat{p}, p^{RC}\} - (1 - \tau^m) \mathcal{M}(q) - r \left((1 + \tau^{P_0}) P_0 - \tau^{TD} \right) \right] dq.$$

The indicator is non-decreasing in $\hat{p}$. The integrand is non-decreasing in $\hat{p}$ and non-negative by construction because $\Phi(q) \geq 0$ for $q \geq q_{min}$. The lower bound of the integral is decreasing in $\hat{p}$ and hence, $G_1(\hat{p}_1) \geq G_1(\hat{p}_0)$.

Consider the second term,

$$G_2(\hat{p}) \equiv \frac{1}{\delta} \int_{\hat{q}(\hat{p})}^1 e^{-r \frac{1-q}{\delta}} \left[\frac{\hat{q}(\hat{p})}{q} \hat{p} + \frac{1}{q} \int_{\hat{q}(\hat{p})}^q \tilde{w}(x; \hat{p}) dx - (1 - \tau^m) \mathcal{M}(q) - r \left((1 + \tau^{P_0}) P_0 - \tau^{TD} \right) \right] dq.$$

We have

$$\begin{aligned} G_2(\hat{p}_1) - G_2(\hat{p}_0) &= \frac{1}{\delta} \int_{\hat{q}(\hat{p}_1)}^{\hat{q}(\hat{p}_0)} e^{-r \frac{1-q}{\delta}} \left[\frac{\hat{q}(\hat{p}_1)}{q} \hat{p}_1 + \frac{1}{q} \int_{\hat{q}(\hat{p}_1)}^q \tilde{w}(x; \hat{p}_1) dx \right. \\ &\quad \left. - (1 - \tau^m) \mathcal{M}(q) - r \left((1 + \tau^{P_0}) P_0 - \tau^{TD} \right) \right] dq \\ &\quad + \frac{1}{\delta} \int_{\hat{q}(\hat{p}_0)}^1 e^{-r \frac{1-q}{\delta}} \left[\frac{\hat{q}(\hat{p}_1)}{q} \hat{p}_1 + \frac{1}{q} \int_{\hat{q}(\hat{p}_1)}^q \tilde{w}(x; \hat{p}_1) dx \right. \\ &\quad \left. - \frac{\hat{q}(\hat{p}_0)}{q} \hat{p}_0 - \frac{1}{q} \int_{\hat{q}(\hat{p}_0)}^q \tilde{w}(x; \hat{p}_0) dx \right] dq. \end{aligned}$$

The first term is non-negative because $\hat{q}(\hat{p})$ is non-increasing in $\hat{p}$ and $\Phi(q) \geq 0$ for $q \geq q_{min}$. Consider the integrand of the second term for any q . It is non-negative if and only if

$$\begin{aligned} &\hat{q}(\hat{p}_1) \hat{p}_1 - \hat{q}(\hat{p}_0) \hat{p}_0 + \int_{\hat{q}(\hat{p}_1)}^{\hat{q}(\hat{p}_0)} \tilde{w}(x; \hat{p}_1) dx + \int_{\hat{q}(\hat{p}_0)}^q (\tilde{w}(x; \hat{p}_1) - \tilde{w}(x; \hat{p}_0)) dx \\ &= \hat{q}(\hat{p}_0) (\hat{p}_1 - \hat{p}_0) + \int_{\hat{q}(\hat{p}_1)}^{\hat{q}(\hat{p}_0)} (\tilde{w}(x; \hat{p}_1) - \hat{p}_1) dx + \int_{\hat{q}(\hat{p}_0)}^q (\tilde{w}(x; \hat{p}_1) - \tilde{w}(x; \hat{p}_0)) dx \\ &\geq 0. \end{aligned}$$

Consider the expression following the first equality. The first term is positive. The second is non-negative because $\hat{q}(\hat{p})$ is non-increasing in $\hat{p}$ and because $\tilde{w}(x; \hat{p})$ is increasing in x , so that $\tilde{w}(x; \hat{p}_1) \geq \tilde{w}(\hat{q}(\hat{p}_1); \hat{p}_1) \geq \hat{p}_1$. The third is non-negative because $\tilde{w}(x; \hat{p})$ is non-decreasing in $\hat{p}$. Hence, the second term of $G_2(\hat{p}_1) - G_2(\hat{p}_0)$ is positive and we have $G_2(\hat{p}_1) > G_2(\hat{p}_0)$.

Consider the third term,

$$G_3(\hat{p}) \equiv -(1 - \tau^C)\kappa \left(\frac{\delta}{1 - q_{min}(\hat{p})} \right) - \tau^{TD}.$$

It is increasing in $\hat{p}$ as $q_{min}(\hat{p})$ is non-increasing in $\hat{p}$ and κ is non-decreasing in its argument. Hence we have $G_3(\hat{p}_1) \geq G_3(\hat{p}_0)$.

It follows that

$$G(\hat{p}_1) - G(\hat{p}_0) > 0,$$

a contradiction of

$$G(\hat{p}_1) = G(\hat{p}_0) = 0.$$

Hence, the equilibrium is unique. This completes the proof. $\square$

B.5 Proof of Theorem 3

Proof. We start by deriving the representation result for the price schedule and then show comparative statics with respect to policy are well posed.

Representation Result. Starting from (A9) in the proof of Theorem 1, we can set $\hat{q} = 1$ to recover

$$p(q) = \frac{1}{q}p(1) - \frac{1}{q} \int_q^1 \tilde{w}(x) dx.$$

Combining with (A8) then yields

$$p(q) = \frac{1}{q}p(1) - \frac{1}{q} \int_q^1 F^{-1} \left(C^{PH} \frac{1-x}{\delta} + \frac{x - q_{min}}{1 - q_{min}} \right) dx \equiv \tilde{p}(q; p(1), q_{min}). \quad (\text{A16})$$

Differentiating, we obtain

$$\frac{dp(q)}{di} = \frac{1}{q} \frac{dp(1)}{di} - \frac{1}{q} \int_q^1 \frac{1-x}{f(\tilde{w}(x))} dx \left[\frac{1}{\delta} \frac{dC^{PH}}{di} - \frac{1}{(1-q_{min})^2} \frac{dq_{min}}{di} \right]. \quad (A17)$$

Defining

$$\begin{aligned} \alpha^i &\equiv \frac{dp(1)}{di}, \\ \beta^i &\equiv \frac{dq_{min}}{di} - \frac{(1-q_{min})^2}{\delta} \frac{dC^{PH}}{di}, \end{aligned} \quad (A18)$$

yields the expression in Theorem 3, provided the differentiation is well posed, which we now establish.

Policy Comparative Statics. We now show that the comparative statics are well posed. We start from the necessary and sufficient conditions (A13) derived in the proof of Theorem 1. In a steady-state equilibrium without a binding minimal quality occupancy standard ($\underline{q} \rightarrow 0$) and without rent control ($q^{RC} = p^{RC} = 0$), $\hat{q} = q_{min} > \bar{q}$, and the conditions reduce to

$$\begin{aligned} \tilde{w}(q) &= F^{-1} \left(C^{PH} \frac{1-q}{\delta} + \frac{q-q_{min}}{1-q_{min}} \right), \\ p(q_{min}) &= (1-\tau^m) \mathcal{M}(q_{min}) + r \left((1+\tau^{P_0}) P_0 - \tau^{TD} \right), \\ p(q) &= \frac{q_{min}}{q} p(q_{min}) + \frac{1}{q} \int_{q_{min}}^q \tilde{w}(x) dx, \\ (1-\tau^C) \kappa \left(\frac{\delta}{1-q_{min}} \right) + \tau^{TD} &= \frac{1}{\delta} \int_{q_{min}}^1 e^{-r \frac{1-q}{\delta}} \left[p(q) - (1-\tau^m) \mathcal{M}(q) \right. \\ &\quad \left. - r \left((1+\tau^{P_0}) P_0 - \tau^{TD} \right) \right] dq. \end{aligned} \quad (A19)$$

We retain C^{PH} , which we perturb below, and we write $T = (1-q_{min})/\delta$ for the terminal age of a housing unit. It is useful to re-introduce aggregate construction C^A . We have

$$C^A = \frac{\delta}{1-q_{min}} \implies \frac{dq_{min}}{di} = \frac{(1-q_{min})^2}{\delta} \frac{dC^A}{di}. \quad (A20)$$

Differentiating the first condition in (A19) and using (A20), we obtain

$$\frac{d\tilde{w}(q)}{di} = \frac{1}{\delta} \frac{1-q}{f(\tilde{w}(q))} \left[\frac{dC^{PH}}{di} - \frac{dC^A}{di} \right]. \quad (A21)$$

The second and third conditions in (A19) determine the level and the slope of the rental price schedule. Differentiating them jointly at a fixed quality q , applying the Leibniz rule to the lower limit of the integral, and using $p'(q_{min}) = [\tilde{w}(q_{min}) - p(q_{min})]/q_{min}$ from Lemma 2, we obtain

$$\begin{aligned} \frac{dp(q_{min})}{di} &= r \left[P_0 \frac{d\tau^{P_0}}{di} - \frac{d\tau^{TD}}{di} \right] - \frac{d\tau^m}{di} \mathcal{M}(q_{min}) \\ &\quad - [p'(q_{min}) - (1 - \tau^m) \mathcal{M}'(q_{min})] \frac{(1 - q_{min})^2}{\delta} \frac{dC^A}{di}, \\ \frac{dp(q)}{di} &= \frac{q_{min}}{q} \frac{dp(q_{min})}{di} + \frac{1}{q} \int_{q_{min}}^q \frac{d\tilde{w}(x)}{di} dx, \end{aligned} \quad (\text{A22})$$

where $dp(q_{min})/di$ denotes the change in the rental price schedule evaluated at the initial tear-down cutoff. Last, we differentiate the free-entry condition in (A19), noting that the terms arising from the perturbation of q_{min} in the lower limit of the integral vanish because $\Phi(q_{min}) = 0$:

$$\begin{aligned} (1 - \tau^C) \kappa'(C^A) \frac{dC^A}{di} &= \frac{1}{\delta} \int_{q_{min}}^1 e^{-r \frac{1-q}{\delta}} \left[\frac{dp(q)}{di} + \frac{d\tau^m}{di} \mathcal{M}(q) - r P_0 \frac{d\tau^{P_0}}{di} + r \frac{d\tau^{TD}}{di} \right] dq \\ &\quad + \frac{d\tau^C}{di} \kappa(C^A) - \frac{d\tau^{TD}}{di}. \end{aligned} \quad (\text{A23})$$

Plugging the first three expressions in the last and changing variables from q to $s = (1 - q)/\delta$ yields a linear equation for dC^A/di ,

$$\begin{aligned} A \frac{dC^A}{di} &= \frac{d\tau^m}{di} \int_0^T e^{-rs} \left[\mathcal{M}(1 - \delta s) - \frac{q_{min}}{1 - \delta s} \mathcal{M}(q_{min}) \right] ds \\ &\quad + \frac{d\tau^{P_0}}{di} r P_0 \int_0^T e^{-rs} \left[\frac{q_{min}}{1 - \delta s} - 1 \right] ds \\ &\quad - \frac{d\tau^{TD}}{di} \left[r \int_0^T e^{-rs} \left[\frac{q_{min}}{1 - \delta s} - 1 \right] ds + 1 \right] \\ &\quad + \frac{d\tau^C}{di} \kappa(C^A) \\ &\quad + \frac{dC^{PH}}{di} \frac{1}{\delta} \int_0^T \frac{e^{-rs}}{1 - \delta s} \int_{q_{min}}^{1 - \delta s} \frac{1 - x}{f(\tilde{w}(x))} dx ds, \end{aligned} \quad (\text{A24})$$

where

$$\begin{aligned}
A &\equiv (1 - \tau^C) \kappa' (C^A) \\
&+ \frac{(1 - q_{min})^2}{\delta} \int_0^T \frac{e^{-rs} q_{min}}{1 - \delta s} [p'(q_{min}) - (1 - \tau^m) \mathcal{M}'(q_{min})] ds \\
&+ \frac{1}{\delta} \int_0^T \frac{e^{-rs}}{1 - \delta s} \int_{q_{min}}^{1-\delta s} \frac{1-x}{f(\tilde{w}(x))} dx ds \\
&> 0,
\end{aligned} \tag{A25}$$

where the inequality follows from $p'(q_{min}) > 0$ by Lemma 2 and from $\mathcal{M}'(q_{min}) < 0$. Given a specific policy perturbation di and its exogenous effect on $C^{PH}, \tau^m, \tau^{P_0}, \tau^{TD}$, and τ^C , there is hence a unique dC^A/di which solves the system. This allows us to recover $dp(q)/di$ for all q , ensuring our comparative statics are well posed. We make two comments. First, instead of the change dC^A/di , we could have solved for the change dq_{min}/di . We chose to reintroduce C^A as a first step towards solving for the dynamic path dp_t/di , which depends on the construction history. Second, note that perturbations to q^{RC}, p^{RC} , and $\underline{q}$ have no first-order effects, starting from a steady-state equilibrium without a binding minimal quality occupancy standard ($\underline{q} \rightarrow 0$) and without rent control ($q^{RC} = p^{RC} = 0$), which is why those objects do not enter the above equations. This completes the proof. $\square$

B.6 Proof of Proposition 1

Proof. From (A16) and (A17) in the proof of Theorem 3, we have

$$\frac{\partial \tilde{p}(q)}{\partial p(1)} = \frac{1}{q} > 0$$

and

$$\frac{d}{dq} \left(\frac{\partial \tilde{p}(q)}{\partial p(1)} \right) = \frac{-1}{q^2} < 0.$$

We also have

$$\frac{\partial \tilde{p}(q)}{\partial q_{min}} = \frac{1}{q} \int_q^1 \frac{1-x}{f(\tilde{w}(x))} dx \frac{1}{(1-q_{min})^2} \geq 0$$

and

$$\frac{d}{dq} \left(\frac{\partial \tilde{p}(q)}{\partial q_{min}} \right) = \frac{-1}{(1-q_{min})^2} \left[\frac{1}{q^2} \int_q^1 \frac{1-x}{f(\tilde{w}(x))} dx + \frac{1}{q} \frac{1-q}{f(\tilde{w}(q))} \right] < 0.$$

This completes the proof. $\square$

B.7 Proof of Proposition 2

Proof. We start from (A24) and (A25) in the proof of Theorem 3. Because $A > 0$, the sign of dC^A/di is the sign of the loading on the perturbed instrument. The loading on $d\tau^C/di$ is $\kappa(C^A) > 0$ and the loading on dC^{PH}/di is positive, while the loading on $d\tau^m/di$ is negative, given

$$\mathcal{M}(q) - \frac{q_{min}}{q} \mathcal{M}(q_{min}) = \gamma \left[\frac{1-q}{q} - \frac{1-q_{min}}{q} \right] < 0 \quad (\text{A26})$$

for $q > q_{min}$. This implies $dC^A/d\tau^C > 0$, $dC^A/dC^{PH} > 0$, and $dC^A/d\tau^m < 0$. From (A20),

$$\frac{dq_{min}}{di} = \frac{(1-q_{min})^2}{\delta} \frac{dC^A}{di},$$

implying $dq_{min}/d\tau^C > 0$, $dq_{min}/dC^{PH} > 0$, and $dq_{min}/d\tau^m < 0$.

Starting with the policy $di = d\tau^C$, we have, from (A21) and (A22) evaluated at $q = 1$,

$$\begin{aligned} \frac{dp(1)}{d\tau^C} &= -q_{min} [p'(q_{min}) - (1-\tau^m)\mathcal{M}'(q_{min})] \frac{(1-q_{min})^2}{\delta} \frac{dC^A}{d\tau^C} \\ &\quad - \frac{1}{\delta} \int_{q_{min}}^1 \frac{1-x}{f(\tilde{w}(x))} dx \frac{dC^A}{d\tau^C} \\ &< 0. \end{aligned}$$

This entails, using (A18),

$$\alpha^{\tau^C} = \frac{dp(1)}{d\tau^C} < 0, \quad \beta^{\tau^C} = \frac{dq_{min}}{d\tau^C} > 0.$$

Next, for the policy $di = dC^{PH}$, we have from (A23)

$$0 < \delta(1-\tau^C)\kappa'(C^A) \frac{dC^A}{dC^{PH}} = \int_{q_{min}}^1 e^{-r\frac{1-q}{\delta}} \frac{dp(q)}{dC^{PH}} dq \quad (\text{A27})$$

and, from (A21) and (A22),

$$\begin{aligned} q \frac{dp(q)}{dC^{PH}} &= -q_{min} [p'(q_{min}) - (1-\tau^m)\mathcal{M}'(q_{min})] \frac{dq_{min}}{dC^{PH}} \\ &\quad + \frac{1}{\delta} \int_{q_{min}}^q \frac{1-x}{f(\tilde{w}(x))} \left[1 - \frac{dC^A}{dC^{PH}} \right] dx. \end{aligned} \quad (\text{A28})$$

Suppose that $dC^A/dC^{PH} \geq 1$. Then both terms in (A28) are non-positive and the first is strictly negative, so that $dp(q)/dC^{PH} < 0$ for all q , violating (A27). It must therefore be that $dC^A/dC^{PH} < 1$ and hence $dC/dC^{PH} < 0$. The right-hand side of (A28) is then increasing in q . If $qdp(q)/dC^{PH}$ were non-positive for all q , so would be $dp(q)/dC^{PH}$, again violating (A27). It is therefore strictly positive for some q and, being increasing, strictly positive at $q = 1$. This entails, using (A18),

$$\alpha^{C^{PH}} = \frac{dp(1)}{dC^{PH}} > 0, \quad \beta^{C^{PH}} = \frac{(1 - q_{min})^2}{\delta} \left[\frac{dC^A}{dC^{PH}} - 1 \right] < 0.$$

Last, for the policy $di = d\tau^m$, using (A18), we have

$$\beta^{\tau^m} = \frac{dq_{min}}{d\tau^m} < 0.$$

The affordability effect is ambiguous. Indeed, from (A21) and (A22) evaluated at $q = 1$,

$$\begin{aligned} \alpha^{\tau^m} = & -q_{min}\mathcal{M}(q_{min}) - q_{min} [p'(q_{min}) - (1 - \tau^m)\mathcal{M}'(q_{min})] \frac{(1 - q_{min})^2}{\delta} \frac{dC^A}{d\tau^m} \\ & - \frac{1}{\delta} \int_{q_{min}}^1 \frac{1 - x}{f(\tilde{w}(x))} dx \frac{dC^A}{d\tau^m}, \end{aligned}$$

where the first term is negative and the last two are positive because $dC^A/d\tau^m < 0$. This completes the proof. $\square$

B.8 Proof of Theorem 4

Proof. First, we derive the conditions for a dynamic equilibrium and state the representation result. Second, we differentiate those conditions and show the dynamic response to a permanent perturbation to housing policy is characterized by the path of changes in construction. Third, we invoke the Banach fixed-point theorem to establish a unique such path. Last, we show that this path converges to the comparative statics change in aggregate construction, implying that the time-dependent policy loadings converge to the comparative statics policy loadings.

Dynamic Equilibrium Conditions and Representation Result. Starting from a steady-state equilibrium without a binding minimal quality occupancy standard ($\underline{q} \rightarrow 0$) and without rent control ($q^{RC} = p^{RC} = 0$), so that $\hat{q} = q_{min} > \bar{q}$, dynamic conditions for

equilibrium are

$$\begin{aligned}
\tilde{w}_t(q) &= F^{-1} \left(\int_{\underline{q}}^1 h_t^{PH}(x) dx + \int_{\underline{q}}^q h_t(x) dx \right), \\
p_t(q_{min,t}) &= (1 - \tau^m) \mathcal{M}(q_{min,t}) + r \left((1 + \tau^{P_0}) P_0 - \tau^{TD} \right), \\
p_t(q) &= \frac{q_{min,t}}{q} p_t(q_{min,t}) + \frac{1}{q} \int_{q_{min,t}}^q \tilde{w}_t(x) dx, \\
(1 - \tau^C) \kappa(C_t^A) + \tau^{TD} &= \int_0^{\bar{s}_t} e^{-rs} \left[p_{t+s}(1 - \delta s) - (1 - \tau^m) \mathcal{M}(1 - \delta s) \right. \\
&\quad \left. - r \left((1 + \tau^{P_0}) P_0 - \tau^{TD} \right) \right] ds,
\end{aligned} \tag{A29}$$

where $\bar{s}_t = \inf\{s : s = T_{t+s}\}$ is the terminal age of a unit built at time t . The first condition follows from Lemma 1, the second condition follows from the fact that $q_{min,t} > \bar{q}$ implies that $\Phi_t(q_{min,t}) = 0$, the third follows from integrating the rental price schedule in Lemma 2, and the fourth follows because free entry must hold with equality in the neighborhood of a steady-state equilibrium. In the initial steady-state, those conditions are necessary and sufficient, as evident from comparing them to (A19) in the proof of Theorem 3.

Towards establishing the representation result, we re-introduce housing construction in the dynamic equilibrium conditions by noting that

$$\begin{aligned}
\int_{\underline{q}}^1 h_t^{PH}(x) dx + \int_{\underline{q}}^q h_t(x) dx &= \int_{q_{min,t}}^1 \frac{C_{t-\frac{1-x}{\delta}}^{PH}}{\delta} dx + \int_{q_{min,t}}^q \frac{C_{t-\frac{1-x}{\delta}}}{\delta} dx \\
&= \int_0^{T_t} C_{t-s}^{PH} ds + \int_{\frac{1-q}{\delta}}^{T_t} C_{t-s} ds,
\end{aligned}$$

where we introduce a time subscript for public housing, towards studying changes in the policy instrument C^{PH} . We note that from market clearing,

$$\begin{aligned}
\hat{C}_t &= \frac{\int_{q_{min,t}}^1 \frac{C_{t-\frac{1-q}{\delta}}}{\delta} dq}{T_t} = \frac{\int_{q_{min,t}}^1 h_t(q) dq}{T_t} = \frac{1 - \int_{q_{min,t}}^1 h_t^{PH}(q) dq}{T_t} \\
&= \frac{\delta}{1 - q_{min,t}} \left[1 - \int_0^{T_t} C_{t-s}^{PH} ds \right].
\end{aligned} \tag{A30}$$

Hence, we can write

$$\int_{\frac{1-q}{\delta}}^{T_t} C_{t-s} ds = \frac{1 - \int_0^{T_t} C_{t-s}^{PH} ds}{1 - q_{min,t}} \int_{\frac{1-q}{\delta}}^{T_t} \delta V_{s,t} ds = \frac{1 - \int_0^{T_t} C_{t-s}^{PH} ds}{1 - q_{min,t}} \int_{q_{min,t}}^q V_{\frac{1-x}{\delta},t} dx.$$

Therefore, we have

$$p_t(q) = \frac{1}{q} p_t(1) - \frac{1}{q} \int_q^1 F^{-1} \left(\int_0^{T_t} C_{t-s}^{PH} ds + \frac{1 - \int_0^{T_t} C_{t-s}^{PH} ds}{1 - q_{min,t}} \int_{q_{min,t}}^x V_{\frac{1-y}{\delta},t} dy \right) dx.$$

Now we can consider a perturbation of housing policy. We obtain

$$\begin{aligned} \frac{dp_t(q)}{di} &= \frac{1}{q} \frac{dp_t(1)}{di} \\ &\quad - \frac{1}{q} \int_q^1 \frac{1}{f(\tilde{w}_t(x))} \left[1 - \frac{\int_{q_{min,t}}^x V_{\frac{1-y}{\delta},t} dy}{1 - q_{min,t}} \right] dx \int_0^{T_t} \frac{dC_{t-s}^{PH}}{di} ds \\ &\quad + \frac{1}{q} \int_q^1 \frac{1}{f(\tilde{w}_t(x))} \left[\left(1 - \int_0^{T_t} C_{t-s}^{PH} ds \right) \frac{V_{T,t}(1 - q_{min,t}) - \int_{q_{min,t}}^x V_{\frac{1-y}{\delta},t} dy}{(1 - q_{min,t})^2} \right. \\ &\quad \quad \left. + \frac{C_{t-T_t}^{PH}}{\delta} \left[1 - \frac{\int_{q_{min,t}}^x V_{\frac{1-y}{\delta},t} dy}{1 - q_{min,t}} \right] \right] dx \frac{dq_{min,t}}{di} \\ &\quad - \frac{1}{q} \int_q^1 \frac{1}{f(\tilde{w}_t(x))} \frac{1 - \int_0^{T_t} C_{t-s}^{PH} ds}{1 - q_{min,t}} \int_{q_{min,t}}^x \frac{dV_{\frac{1-y}{\delta},t}}{di} dy dx. \end{aligned}$$

Starting from a steady state, $V_{s,t} = 1$ and $C_{t-s}^{PH} = C^{PH}$ for all s, t , so that $\int_{q_{min}}^x V_{\frac{1-y}{\delta},t} dy = x - q_{min}$ and $\int_0^T C_{t-s}^{PH} ds = TC^{PH}$. The two terms multiplying $dq_{min,t}/di$ then combine, since

$$[1 - TC^{PH}] \frac{1 - x}{(1 - q_{min})^2} + \frac{C^{PH}}{\delta} \frac{1 - x}{1 - q_{min}} = \frac{1 - x}{(1 - q_{min})^2},$$

and we obtain

$$\begin{aligned} \frac{dp_t(q)}{di} &= \frac{1}{q} \frac{dp_t(1)}{di} \\ &\quad - \frac{1}{q} \int_q^1 \frac{1 - x}{f(\tilde{w}(x))} dx \frac{1}{1 - q_{min}} \int_0^T \frac{dC_{t-s}^{PH}}{di} ds \\ &\quad + \frac{1}{q} \int_q^1 \frac{1 - x}{f(\tilde{w}(x))} dx \frac{1}{(1 - q_{min})^2} \frac{dq_{min,t}}{di} \\ &\quad - \frac{1}{q} \int_q^1 \frac{1}{f(\tilde{w}(x))} \frac{1 - TC^{PH}}{1 - q_{min}} \int_{q_{min}}^x \frac{dV_{\frac{1-y}{\delta},t}}{di} dy dx. \end{aligned} \tag{A31}$$

We re-write the last term as

$$\begin{aligned}
& \frac{1}{q} \int_q^1 \frac{1}{f(\tilde{w}(x))} \frac{1 - TC^{PH}}{1 - q_{min}} \int_{q_{min}}^x \frac{dV_{\frac{1-y}{\delta},t}}{di} dy dx \\
&= \frac{1}{q} \int_q^1 \frac{1}{f(\tilde{w}(x))} \frac{1 - TC^{PH}}{1 - q_{min}} \int_{q_{min}}^1 \mathbf{1}_{y < x} \frac{dV_{\frac{1-y}{\delta},t}}{di} dy dx \\
&= \int_{q_{min}}^1 \frac{1}{q} \int_q^1 \frac{1}{f(\tilde{w}(x))} \frac{1 - TC^{PH}}{1 - q_{min}} \mathbf{1}_{y < x} \frac{dV_{\frac{1-y}{\delta},t}}{di} dx dy \\
&= \int_0^T \frac{\delta}{q} \int_q^1 \frac{1}{f(\tilde{w}(x))} \frac{1 - TC^{PH}}{1 - q_{min}} \mathbf{1}_{1-\delta s < x} \frac{dV_{s,t}}{di} dx ds \\
&= \int_0^T \frac{\delta}{q} \int_{\max\{q, 1-s\delta\}}^1 \frac{1}{f(\tilde{w}(x))} dx \frac{1 - TC^{PH}}{1 - q_{min}} \frac{dV_{s,t}}{di} ds.
\end{aligned}$$

Defining

$$\begin{aligned}
\alpha_t^i &\equiv \frac{dp_t(1)}{di}, \\
\beta_t^i &\equiv \frac{dq_{min,t}}{di} - (1 - q_{min}) \int_0^T \frac{dC_{t-s}^{PH}}{di} ds, \\
\eta_{s,t}^i &\equiv \frac{dV_{s,t}}{di},
\end{aligned} \tag{A32}$$

yields the expression in Theorem 4, provided the differentiation is well posed. Comparing (A31) to (A17) in the proof of Theorem 3 confirms that the affordability and tear-down perturbations are unchanged.

We note that by construction,

$$1 = \frac{1}{1 - q_{min,t}} \int_{q_{min,t}}^1 V_{\frac{1-x}{\delta},t} dx.$$

This requires

$$0 = \frac{\int_{q_{min,t}}^1 V_{\frac{1-y}{\delta},t} dy - V_{T,t}(1 - q_{min,t})}{(1 - q_{min,t})^2} \frac{dq_{min,t}}{di} + \frac{1}{1 - q_{min,t}} \int_{q_{min,t}}^1 \frac{dV_{\frac{1-y}{\delta},t}}{di} dy.$$

When perturbing a steady state, $V_{s,t} = 1$ for all s, t , implying

$$0 = \frac{1}{\delta} \int_{q_{min}}^1 \frac{dV_{\frac{1-y}{\delta},t}}{di} dy = \int_0^T \frac{dV_{s,t}}{di} ds = \int_0^T \eta_{s,t}^i ds. \tag{A33}$$

Differentiating Dynamic Equilibrium Conditions. Evaluating the first condition in (A29) at $q = 1$ gives $\int_0^{T_t} C_{t-s}^A ds = 1$. Differentiating around the initial steady-state yields

$$0 = \frac{-1}{\delta} \frac{dq_{min,t}}{di} C^A + \int_0^T \frac{dC_{t-s}^A}{di} ds \implies \frac{dq_{min,t}}{di} = (1 - q_{min}) \int_0^T \frac{dC_{t-s}^A}{di} ds, \quad (\text{A34})$$

where we have used that in steady state $C^A = \delta/(1 - q_{min})$. Proceeding as in the proof of Theorem 3, we differentiate the remaining conditions in (A29) to obtain

$$\begin{aligned} \frac{d\tilde{w}_t(q)}{di} &= \frac{1}{f(\tilde{w}(q))} \left[\int_0^{\frac{1-q}{\delta}} \frac{dC_{t-s}^{PH}}{di} ds - \int_0^{\frac{1-q}{\delta}} \frac{dC_{t-s}^A}{di} ds \right], \\ \frac{dp_t(q_{min})}{di} &= r \left[P_0 \frac{d\tau^{P_0}}{di} - \frac{d\tau^{TD}}{di} \right] - \frac{d\tau^m}{di} \mathcal{M}(q_{min}) \\ &\quad - [p'(q_{min}) - (1 - \tau^m) \mathcal{M}'(q_{min})] (1 - q_{min}) \int_0^T \frac{dC_{t-s}^A}{di} ds, \\ \frac{dp_t(q)}{di} &= \frac{q_{min}}{q} \frac{dp_t(q_{min})}{di} + \frac{1}{q} \int_{q_{min}}^q \frac{d\tilde{w}_t(x)}{di} dx, \\ (1 - \tau^C) \kappa'(C^A) \frac{dC_t^A}{di} &= \frac{d\tau^C}{di} \kappa(C^A) - \frac{d\tau^{TD}}{di} \\ &\quad + \int_0^T e^{-rs} \left[\frac{dp_{t+s}(1 - \delta s)}{di} + \frac{d\tau^m}{di} \mathcal{M}(1 - \delta s) \right. \\ &\quad \left. - r \left(P_0 \frac{d\tau^{P_0}}{di} - \frac{d\tau^{TD}}{di} \right) \right] ds. \end{aligned} \quad (\text{A35})$$

We introduce the mapping $\mathcal{G} : X \rightarrow X$, where X is the space of bounded continuous functions on $[0, \infty)$, endowed with the supremum norm. $\mathcal{G}$ maps paths $\{\Delta_t\}$ into updated paths, with

$\Delta_t \equiv dC_t^A/di$:

$$\begin{aligned}
\mathcal{G}(\{\Delta_t\})_t = & \frac{1}{(1 - \tau^C)\kappa'(C^A)} \left[\frac{d\tau^C}{di} \kappa(C^A) \right. \\
& - \frac{d\tau^{TD}}{di} \left[1 + r \int_0^T e^{-rs} \left[\frac{q_{min}}{1 - \delta s} - 1 \right] ds \right] \\
& + \frac{d\tau^{P_0}}{di} r P_0 \int_0^T e^{-rs} \left[\frac{q_{min}}{1 - \delta s} - 1 \right] ds \\
& + \frac{d\tau^m}{di} \int_0^T e^{-rs} \left[\mathcal{M}(1 - \delta s) - \frac{q_{min}}{1 - \delta s} \mathcal{M}(q_{min}) \right] ds \\
& + \left. \int_0^T \frac{e^{-rs}}{1 - \delta s} \int_{q_{min}}^{1 - \delta s} \frac{1}{f(\tilde{w}(x))} \int_0^{\frac{1-x}{\delta}} \frac{dC_{t+s-u}^{PH}}{di} du dx ds \right] \\
& - \frac{q_{min} [p'(q_{min}) - (1 - \tau^m)\mathcal{M}'(q_{min})] (1 - q_{min})}{(1 - \tau^C)\kappa'(C^A)} \int_0^T \frac{e^{-rs}}{1 - \delta s} \int_0^T \Delta_{t+s-u} du ds \\
& - \frac{1}{(1 - \tau^C)\kappa'(C^A)} \int_0^T \frac{e^{-rs}}{1 - \delta s} \int_{q_{min}}^{1 - \delta s} \frac{1}{f(\tilde{w}(x))} \int_0^{\frac{1-x}{\delta}} \Delta_{t+s-u} du dx ds,
\end{aligned} \tag{A36}$$

where we impose $\Delta_t \equiv 0$ for $t < 0$. By construction, a path $\{\Delta_t\}$ is a solution to (A35) if and only if it solves $\mathcal{G}(\{\Delta_t\}) = \{\Delta_t\}$.

Banach Fixed-Point Theorem. As clear from (A35), a unique path $\{\Delta_t\}$ entails a unique path $\{p_t(q)\}$ for all q , our object of interest. Hence, we now show that there exists a unique fixed point to $\mathcal{G}$. By the Banach fixed-point theorem, we need to show that for any two paths $\{\Delta_t^1\}, \{\Delta_t^2\} \in X$,

$$\sup_t |\mathcal{G}(\{\Delta_t^2\})_t - \mathcal{G}(\{\Delta_t^1\})_t| < B \sup_t |\Delta_t^2 - \Delta_t^1|$$

for $B < 1$. Only the last two lines of (A36) depend on $\{\Delta_t\}$, and we bound

$$\begin{aligned}
\left| \int_0^T [\Delta_{t+s-u}^2 - \Delta_{t+s-u}^1] du \right| & \leq T \sup_t |\Delta_t^2 - \Delta_t^1|, \\
\left| \int_0^{\frac{1-x}{\delta}} [\Delta_{t+s-u}^2 - \Delta_{t+s-u}^1] du \right| & \leq \frac{1-x}{\delta} \sup_t |\Delta_t^2 - \Delta_t^1|.
\end{aligned}$$

Using $(1 - q_{min})T = (1 - q_{min})^2/\delta$, we obtain the bound with

$$B \equiv \frac{1}{(1 - \tau^C)\kappa'(C^A)} \left[\frac{(1 - q_{min})^2}{\delta} \int_0^T \frac{e^{-rs} q_{min}}{1 - \delta s} [p'(q_{min}) - (1 - \tau^m)\mathcal{M}'(q_{min})] ds \right. \\ \left. + \frac{1}{\delta} \int_0^T \frac{e^{-rs}}{1 - \delta s} \int_{q_{min}}^{1-\delta s} \frac{1-x}{f(\tilde{w}(x))} dx ds \right]. \quad (\text{A37})$$

Differentiating the free-entry condition in (A29) and proceeding as in the derivation of (A24), the bracket in (A37) is $|dP(1)/dC^A|$, so that

$$B = \frac{1}{(1 - \tau^C)\kappa'(C^A)} \left| \frac{dP(1)}{dC^A} \right|.$$

The free-entry condition in a steady state is

$$(1 - \tau^C)\kappa(C^A) = P(1) - (1 + \tau^{P_0})P_0.$$

Differentiating with respect to $P(1)$, we solve for the construction supply elasticity as

$$\varepsilon_{C^A} = \frac{d \log C^A}{d \log P(1)} = \frac{P(1)}{(1 - \tau^C)\kappa'(C^A) C^A}.$$

Hence, we have that

$$B = \frac{P(1)}{(1 - \tau^C)\kappa'(C^A) C^A} \frac{C^A}{P(1)} \left| \frac{dP(1)}{dC^A} \right| \\ = \varepsilon_{C^A} \left| \frac{d \log P(1)}{d \log C^A} \right| \\ < \frac{1}{2}$$

when perturbing a stable equilibrium. Therefore, by the Banach fixed-point theorem, there exists a unique solution to (A35), Δ_t^* , which characterizes dC_t^A/di for all t and hence $dp_t(q)/di$ for all t and all q .

Convergence to Comparative Statics. We now establish convergence of $dp_t(q)/di$ to our comparative statics $dp(q)/di$. Consider dC^A/di , the comparative static change in construction induced by di . We note that $\mathcal{G}(\{\Delta_t\})_t$ depends on $\{\Delta_t\}$ only through its values

on $[t - T, t + T]$, so that the bound above also holds locally,

$$|\mathcal{G}(\{\Delta_t^2\})_t - \mathcal{G}(\{\Delta_t^1\})_t| < B \sup_{z \in [t-T, t+T]} |\Delta_z^2 - \Delta_z^1|.$$

Furthermore, comparing (A24) from the proof of Theorem 3 and (A36), reveals that for $t > T$, $\mathcal{G}(\{dC^A/di\})_t = dC^A/di$. Hence, for $n > 1$,

$$\begin{aligned} e_n &\equiv \sup_{t > nT} \left| \Delta_t^* - \frac{dC^A}{di} \right| = \sup_{t > nT} \left| \mathcal{G}(\{\Delta_t^*\})_t - \mathcal{G}\left(\left\{\frac{dC^A}{di}\right\}\right)_t \right| \\ &< B \sup_{t > (n-1)T} \left| \Delta_t^* - \frac{dC^A}{di} \right| = B e_{n-1}. \end{aligned}$$

Hence,

$$e_n \xrightarrow{n \rightarrow \infty} 0 \implies \Delta_t^* = \frac{dC_t^A}{di} \xrightarrow{t \rightarrow \infty} \frac{dC^A}{di}.$$

It follows that

$$\begin{aligned} \alpha_t^i &= \frac{dp_t(1)}{di} \xrightarrow{t \rightarrow \infty} \frac{dp(1)}{di} = \alpha^i, \\ \beta_t^i &= \frac{dq_{min,t}}{di} - (1 - q_{min}) \int_0^T \frac{dC_{t-s}^{PH}}{di} ds \xrightarrow{t \rightarrow \infty} \frac{dq_{min}}{di} - \frac{(1 - q_{min})^2}{\delta} \frac{dC^{PH}}{di} = \beta^i, \\ \eta_{s,t}^i &= \frac{dV_{s,t}}{di} \xrightarrow{t \rightarrow \infty} 0, \end{aligned}$$

where the second limit holds since C^{PH} is constant for $t \geq 0$ and $(1 - q_{min})T = (1 - q_{min})^2/\delta$, and the last limit follows from noting that, from (13) evaluated at the initial steady-state,

$$\frac{dV_{s,t}}{di} = \frac{1}{\hat{C}} \left[\frac{dC_{t-s}}{di} - \frac{d\hat{C}_t}{di} \right] \xrightarrow{t \rightarrow \infty} 0,$$

given convergence of C_t^A and hence of C_t and, from (A30), of $\hat{C}_t$. This completes the proof. $\square$

B.9 Proof of Proposition 3

Proof. First, consider $di = d\tau^C$. From (A36) and (A37) in the proof of Theorem 4, we have

$$\Delta_t^* = \mathcal{G}(\{\Delta_t^*\})_t < \frac{\kappa(C^A)}{(1 - \tau^C)\kappa'(C^A)} + B \sup_t |\Delta_t^*|.$$

It follows that starting from a stable equilibrium,

$$\sup_t |\Delta_t^*| < \frac{\kappa(C^A)}{(1 - \tau^C)\kappa'(C^A)} + \frac{1}{2} \sup_t |\Delta_t^*| \implies \sup_t |\Delta_t^*| < \frac{2\kappa(C^A)}{(1 - \tau^C)\kappa'(C^A)}$$

so that

$$\frac{dC_t^A}{d\tau^C} = \Delta_t^* > \frac{\kappa(C^A)}{(1 - \tau^C)\kappa'(C^A)} - \frac{1}{2} \sup_t |\Delta_t^*| > 0.$$

From (A34) and (A35) in the proof of Theorem 4, we then have

$$\frac{dq_{min,t}}{d\tau^C} > 0, \quad \frac{d\tilde{w}_t(q)}{d\tau^C} < 0, \quad \frac{dp_t(q_{min})}{d\tau^C} < 0$$

and hence, using (A32),

$$\alpha_t^{\tau^C} = \frac{dp_t(1)}{d\tau^C} < 0, \quad \beta_t^{\tau^C} = \frac{dq_{min,t}}{d\tau^C} > 0.$$

Second, consider $di = d\tau^m$. Following the same reasoning as for $di = d\tau^C$ and noting that the loading on $d\tau^m/di$ in (A36) is negative by (A26) in the proof of Proposition 2, we find $dC_t^A/d\tau^m < 0$ and hence

$$\beta_t^{\tau^m} = \frac{dq_{min,t}}{d\tau^m} < 0.$$

Furthermore, from (A35) in the proof of Theorem 4 evaluated at $q = 1$, we have

$$\begin{aligned} \frac{dp_t(1)}{d\tau^m} &= -\mathcal{M}(q_{min})q_{min} \\ &+ q_{min} [p'(q_{min}) - (1 - \tau^m)\mathcal{M}'(q_{min})] (1 - q_{min}) \int_0^T \frac{-dC_{t-s}^A}{d\tau^m} ds \\ &+ \int_{q_{min}}^1 \frac{1}{f(\tilde{w}(x))} \int_0^{\frac{1-x}{\delta}} \frac{-dC_{t-s}^A}{d\tau^m} ds dx, \end{aligned}$$

which is negative in a short-run window since $dC_{t-s}^A/d\tau^m = 0$ for $t - s < 0$, so that the last two terms vanish as $t \rightarrow 0$. Hence

$$\alpha_t^{\tau^m} = \frac{dp_t(1)}{d\tau^m} < 0$$

in a short-run window.

Last, consider $di = dC^{PH}$. The loading on dC^{PH}/di in (A36) is strictly smaller in a

short-run window than in the long run, because $dC_{t+s-u}^{PH}/di = 0$ for $t + s - u < 0$, so that the reasoning used for $d\tau^C$ and $d\tau^m$ does not sign dC_t^A/dC^{PH} . Both $\alpha_t^{C^{PH}}$ and $\beta_t^{C^{PH}}$ can therefore not be signed. This completes the proof. $\square$

B.10 Proof of Proposition 4

Proof. Consider a perturbation of housing policy starting from a steady state. From the proof of Theorem 4, we have

$$\frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} = \frac{-\delta}{q} \int_{\max\{q, 1-s\delta\}}^1 \frac{1}{f(\tilde{w}(x))} dx \frac{1 - TC^{PH}}{1 - q_{min}}.$$

For $q > 1 - s\delta$, we therefore have

$$\frac{d}{ds} \left(\frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} \right) = 0, \quad \frac{d^2}{dq ds} \left(\frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} \right) = 0.$$

For $q < 1 - s\delta$, we have

$$\frac{d}{ds} \left(\frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} \right) = \frac{-\delta^2}{q} \frac{1}{f(\tilde{w}(1 - \delta s))} \frac{1 - TC^{PH}}{1 - q_{min}} < 0,$$

and

$$\frac{d^2}{dq ds} \left(\frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} \right) = \frac{\delta^2}{q^2} \frac{1}{f(\tilde{w}(1 - s\delta))} \frac{1 - TC^{PH}}{1 - q_{min}} > 0,$$

where we note once more that $(1 - TC^{PH})/(1 - q_{min}) > 0$ by Assumption 1. In addition, at $s = (1 - q)/\delta$, the derivative of $\partial \tilde{p}^{TD}(q)/\partial V_s$ with respect to q jumps up by

$$\frac{\delta}{q} \frac{1}{f(\tilde{w}(q))} \frac{1 - TC^{PH}}{1 - q_{min}} > 0.$$

Because $\partial \tilde{p}^{TD}(q)/\partial V_s$ is continuous, our combined results imply that it is non-increasing in s for all s and decreasing in s for some s , and that its derivative with respect to q is non-decreasing in s for all s and increasing in s for some s . Now consider $\eta_{s,t}$ in a construction

peak. We have

$$\begin{aligned}
\int_0^T \frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} \eta_{s,t}^i ds &= \int_0^T \left(\frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} - \frac{\partial \tilde{p}^{TD}(q)}{\partial V_z} \right) \eta_{s,t}^i ds \\
&= \int_0^z \left(\frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} - \frac{\partial \tilde{p}^{TD}(q)}{\partial V_z} \right) \eta_{s,t}^i ds \\
&\quad - \int_z^T \left(\frac{\partial \tilde{p}^{TD}(q)}{\partial V_z} - \frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} \right) \eta_{s,t}^i ds \\
&\geq 0,
\end{aligned}$$

where the first equality follows from (A33) in the proof of Theorem 4 and the inequality follows because both integrands are non-negative, given $\eta_{s,t}^i \geq 0$ for $s \leq z$ and $\eta_{s,t}^i < 0$ for $s > z$ in a construction peak. We also have

$$\begin{aligned}
\frac{d}{dq} \left(\int_0^T \frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} \eta_{s,t}^i ds \right) &= \int_0^z \frac{d}{dq} \left(\frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} - \frac{\partial \tilde{p}^{TD}(q)}{\partial V_z} \right) \eta_{s,t}^i ds \\
&\quad - \int_z^T \frac{d}{dq} \left(\frac{\partial \tilde{p}^{TD}(q)}{\partial V_z} - \frac{\partial \tilde{p}^{TD}(q)}{\partial V_s} \right) \eta_{s,t}^i ds \\
&< 0,
\end{aligned}$$

where the strict inequality follows from $z \in (0, T)$, so that both integrands are strictly signed on a set of positive measure. The case of a construction trough is symmetric. This completes the proof. $\square$

B.11 Proof of Proposition 5

Proof. The result follows directly from the proof of Proposition 3. By construction, the vintage loadings sum to zero at all t , from (A33) in the proof of Theorem 4. We have $dC_t^A/d\tau^C = dC_t/d\tau^C > 0$ for all $t \geq 0$ and $dC_t/d\tau^C = 0$ for $t < 0$. It follows that in a short-run window $t \in (0, \epsilon)$ for $\epsilon > 0$ small enough, $d\tau^C$ induces $\eta_{s,t}^{\tau^C} > 0$ for $0 < s \leq t$ and $\eta_{s,t}^{\tau^C} < 0$ for $s > t$, which is the definition of a construction peak, with $z = 0$. Similarly, because $dC_t^A/d\tau^m = dC_t/d\tau^m < 0$, $d\tau^m$ induces a construction trough. Because dC_t/dC^{PH} is not signed in a short-run window, the vintage effects of dC^{PH} are ambiguous. This completes the proof. $\square$

C Quantitative Appendix

C.1 Microfoundation for Construction Technology

At each instant, builders contemplate, for any quality q , many development opportunities indexed by j . Each opportunity entails a fixed cost and an opportunity-specific cost. The fixed cost, μq , corresponds to material costs which must be paid no matter how good the opportunity is. We denote the opportunity-specific cost, which captures labor costs, by $\vartheta_{j,q}$. The development cost of quality q opportunity j is then

$$\mu q + \vartheta_{j,q}.$$

A good development opportunity (a low $\vartheta_{j,q}$ corresponding to a low labor cost) may reflect availability of a land plot well-suited for quality q housing or of a worker who has the right skills to build quality q housing. We leave such forces outside of our model and capture them in a reduced-form manner. We assume that the arrival rate of projects with opportunity-specific cost $\vartheta_{j,q} < \vartheta$ is given by $\Lambda_q(\vartheta)$, with

$$\Lambda_q(\vartheta) = \left(\frac{\vartheta}{\zeta q} \right)^{1/\lambda}.$$

A high ζ shifts opportunity-specific costs up, while a high λ also spreads them out, generating congestion.

The cost of aggregate quality q construction $C_t^A(q)$ is then given by

$$\begin{aligned} & \int_0^{C_t^A(q)} (\mu q + \Lambda_q^{-1}(u)) du \\ &= \int_0^{C_t^A(q)} [\mu q + \zeta q u^\lambda] du \\ &= \mu q C_t^A(q) + \zeta q \frac{(C_t^A(q))^{1+\lambda}}{1+\lambda}. \end{aligned}$$

Competitive builders charge the marginal cost of construction at quality q , which is given by

$$q \left(\mu + \zeta (C_t^A(q))^\lambda \right),$$

as in the main text.

C.2 Data

We use the American Housing Survey from 2001–2023 as our main data source in calibrating the model. From 2015 onwards the geography of the AHS is restricted to 15 large US MSAs. We therefore restrict our calibration to these. We harmonize old MSA codes for the waves 2013 and prior into 2013 CBSA classifications by assigning each MSA to the CBSA which has the most overlap. We construct housing unit age by taking the survey year minus the year built. We drop observations which report a rent of less than \$100 or a value of less than \$10,000 in 2000 dollars. We residualize post-tax income with respect to year and CBSA to fit the distribution. Following Heathcote et al. (2017), we measure post-tax income as $\tilde{y} = \mu y^{1-\tau}$ where y is pre-tax income, τ is 0.181, and μ is chosen to match an average tax rate of 0.2. We truncate the bottom and top 5% of the income distribution. We also restrict observations to household heads of age 25–60. In taking a user-cost view of housing we impute rents following Diamond and Diamond (2024).

We then merge in data from the following sources. First we source building cost data from RSMeans. RSMeans is a widely used building cost data source that provides standardized costs and location adjustments for materials and installation. It is typically used as a benchmark measure for building costs when researchers need a consistent estimate across locations and building types (see for example Gyourko and Saiz 2004 and Glaeser and Gyourko 2018 among others). We source, for 2015, indices across locations for the relative costs of materials and installation. We calculate that across all locations materials contribute 56% of building costs. We assume that this remains fixed over time. Given this input mix, this allows us to calculate the relative expenditure on materials by metro area as $(0.56 \times \text{materials cost index}) / (\text{total cost index})$. We use historical city cost indices which describe the level of building costs across US metros over time. We obtain data for all years except for 2001 and 2002.

We further merge into the data the share of land value over time for each CBSA from Davis et al. (2021). We take data for 2012 through to 2022. We also take supply elasticities from Baum-Snow and Han (2024). In particular we average `gamma101b_units_fmm` estimates over tracts to get MSA elasticities. We source building permits from the building permits survey as a proxy for construction.

D Model Without Filtering

This appendix presents the model without filtering used to benchmark our model with filtering in Section 4. The benchmark model is identical to our model with filtering, except

that a unit built at quality q remains at quality q until it is torn down. We describe the benchmark model's maintenance technology and its implications for the landlord problem and housing market clearing. We then describe how we calibrate the benchmark model and use it to benchmark counterfactual policy experiments.

Maintenance Technology. Because quality is constant, maintenance requirements depend on the age a of a unit rather than on its quality. We impose

$$\mathcal{D}^Q(m, a) = \begin{cases} \infty & \text{if } m < \mathcal{M}^Q(a) \\ 0 & \text{if } m \geq \mathcal{M}^Q(a) \end{cases}, \quad \mathcal{M}^Q(a) = \psi e^{\theta a},$$

with $\psi > 0$ and $\theta > 0$. Improperly maintained units ($m < \mathcal{M}^Q(a)$) lose their quality instantly. Sufficiently maintained units ($m \geq \mathcal{M}^Q(a)$) retain their quality. There is no intensive margin for maintenance.

Landlords. The value $P_t(q, a)$ of a land unit with a quality q housing unit of age a solves

$$\begin{aligned} rP_t(q, a) = \max_{m \geq 0} & \mathbf{1}_{m < \mathcal{M}^Q(a)} (r(\underline{P}_t - \tau^{TD}) - (1 - \tau^m)m) \\ & + \mathbf{1}_{m \geq \mathcal{M}^Q(a)} (p_t(q) - (1 - \tau^m)m + \partial_t P_t(q, a) + \partial_a P_t(q, a)). \end{aligned}$$

Free entry requires

$$(1 + \tau^{P_0})P_0 \geq P_t(q, 0) - (1 - \tau^C)\kappa(C_t^A(q), q)$$

for all q , with equality whenever $C_t(q) > 0$. As in Lemma 3, landlords maintain a unit if and only if

$$p_t(q) - (1 - \tau^m)\mathcal{M}^Q(a) - r((1 + \tau^{P_0})P_0 - \tau^{TD}) \geq 0.$$

Because $\mathcal{M}^Q(a)$ is increasing, this defines a quality-specific tear-down age $T_t(q)$.

Housing Stock. The mass $h_t(q, a)$ of private units of quality q and age a satisfies

$$\partial_t h_t(q, a) + \partial_a h_t(q, a) = 0 \quad \text{for } a < T_t(q),$$

with $h_t(q, 0) = C_t(q)$ and $h_t(q, a) = 0$ for $a \geq T_t(q)$.

D.1 Calibration

We calibrate the model without filtering to the same representative US city as in Section 4.2, with the same income distribution, $r = 0.05$, $\mu(q) = \mu q$, and $\zeta(q) = \zeta q$. We set θ to the coefficient from a regression of log maintenance spending on housing unit age, which θ equals exactly in the model. We calibrate P_0 , μ , ζ , λ , and ψ to match the median land-to-value ratio, the median rent-to-income ratio, the share of materials in construction costs, the housing supply elasticity, and the construction rate. Because rental prices do not vary with age, the median rent-to-income ratio replaces the ratio of median new unit rent to median income. We target the construction rate of the calibrated model with filtering, so that both models feature the same average lifespan of housing units. Table A5 summarizes our calibration of the model without filtering.

D.2 Counterfactual Policy Experiments

We consider the same three policy perturbations as in Section 4: $\tau^C = 0.01$, and equally costly increases in C^{PH} and τ^m . The government maintains public housing units at $\mathcal{M}^Q(a)$ and tears them down at the baseline tear-down age $T(q)$.